

Why chose?

A configurable Transformer Workbench for Multi-Top Quark Classification at the LHC

by

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Preface

A preface...

*Niels ter Linden
Amsterdam, May 2026*

Summary

A summary...

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Part I

Framing

1

Introduction

For as long as humans have studied nature, they have asked whether matter can be understood in terms of smaller constituents. Ancient Greek thinkers Leucippus and Democritus proposed that all matter is composed of indivisible *atomoi*; a parallel tradition in Indian natural philosophy, the Vaiśeṣika school, developed the concept of *aṇu* — the smallest irreducible unit of substance. During the Islamic Golden Age, scholars such as Ibn Sīnā preserved and extended this Hellenistic heritage, transmitting it into the European Renaissance. Dalton’s atomic theory gave the concept quantitative substance, Mendeleev’s periodic table revealed underlying order in the elements, and Rutherford’s gold-foil experiment demonstrated that atoms contain dense nuclei — a discovery that initiated nuclear physics. The subsequent identification of protons, neutrons, and ultimately quarks culminated in the Standard Model of particle physics: a quantum field theory that describes the fundamental constituents of matter and three of the four known forces with extraordinary predictive precision.

Modern collider experiments extend this tradition of probing matter through scattering, but at energy scales far beyond Rutherford’s original experiment. At the Large Hadron Collider (LHC) at CERN, protons are accelerated to near light-speed and brought into collision; the resulting interactions produce short-lived particles that decay almost immediately. No detector records these particles directly. Instead, physicists *reconstruct* them from the signatures they leave in the detector: charged-particle tracks in inner tracking systems, energy deposits in electromagnetic and hadronic calorimeters, displaced decay vertices, and missing transverse energy (E_T^{miss}) inferred from momentum imbalance. Identifying and classifying collision events is therefore an inference problem, not a direct-observation problem, and the complexity of the detector response shapes every step of the analysis.

The Standard Model is, by any measure, a triumph of twentieth-century physics, yet it is not considered the final theory. Compelling open questions remain: the nature of dark matter, the origin of neutrino masses, the observed matter–antimatter asymmetry of the universe, the absence of gravity from the quantum field-theoretic framework, and the so-called hierarchy problem concerning the stability of the Higgs mass under radiative corrections. Precision measurements and searches for rare processes at the LHC provide indirect sensitivity to physics beyond the Standard Model (BSM), even without direct production of new particles. The High-Luminosity LHC (HL-LHC) programme will deliver substantially larger integrated luminosities over the coming decades, granting access to processes with smaller cross-sections than those currently measurable. The frontier is therefore not only at higher energies, but also in the rarest corners of already accessible data.

Among the fundamental particles of the Standard Model, the top quark occupies a special position. It is the heaviest known elementary particle, with a mass close to the electroweak symmetry-breaking scale, and its large Yukawa coupling to the Higgs boson makes top-associated processes particularly sensitive to the Higgs sector and to BSM effects that couple preferentially to mass. This thesis focuses on a class of rare, top-rich processes: four-top-quark production ($t\bar{t}t\bar{t}$) and its dominant backgrounds — $t\bar{t}H$, $t\bar{t}W$, $t\bar{t}Z$, and $t\bar{t}WW$. These events produce complex final states: many hadronic jets, multiple b -tagged jets from top and Higgs decays, possible charged leptons, and substantial E_T^{miss} . Combinatorial ambiguity

is severe, since the assignment of reconstructed objects to parent particles is not unique, and the signal cross-section is small relative to the irreducible backgrounds. Accurate event-level classification is therefore central to any analysis targeting these processes.

The core task addressed in this thesis is supervised event-level classification: assigning a scalar discriminant to each collision event such that signal-like and background-like events are separated as reliably as possible. From a physics perspective, the challenge arises from the rarity of the signal, the complexity of the final state, and the many-object combinatorics. From a machine-learning perspective, the challenge is structural: a collision event is a *variable-length, unordered set* of reconstructed objects, each described by a mixture of continuous kinematic features and discrete categorical labels (particle type, charge, b -tag decision). Standard fixed-size or sequence-based architectures impose an artificial structure on this set-valued input, whereas the physical content is permutation-invariant.

Machine learning has become indispensable in high-energy physics for tasks ranging from jet tagging and particle identification to full event reconstruction and classification. The progression from boosted decision trees and shallow neural networks to deep architectures and, more recently, to attention-based models reflects both the growing complexity of analysis tasks and the increasing availability of large simulated datasets. Transformers are attractive for event classification because self-attention operates over all pairs of input objects without imposing an ordering, making the mechanism structurally compatible with the set-like nature of collision events. Pairwise attention weights can, in principle, encode physically meaningful relations: angular separations (ΔR), invariant masses of object pairs, particle-type compatibility, and global quantities such as E_T^{miss} . This structural alignment between the attention mechanism and the physics of multi-object final states motivates a careful, systematic investigation of which specific design choices — input representation, attention biases, positional encoding, tokenisation strategy — actually improve classification performance.

This thesis investigates transformer-based classifiers for rare multi-top quark processes at the LHC. Rather than proposing a single optimised architecture, it develops a configurable workbench in which architectural choices are exposed as orthogonal axes and evaluated through controlled ablations. The central aim is to determine which physics-motivated and machine-learning-motivated design choices improve event classification, as measured primarily by AUROC and complementary efficiency metrics.

1.1. The Standard Model and Top Quark Physics

This section provides the physics theory context for the thesis. It covers the Standard Model particle content and forces, the special role of the top quark, the four-top signal process and its backgrounds, and the sensitivity of rare multi-top processes to BSM physics.

1.1.1. The Standard Model of particle physics

The Standard Model (SM) is a quantum field theory describing the elementary particles and three of the four fundamental forces: the electromagnetic, weak, and strong interactions. Matter consists of twelve fermions — six quarks and six leptons — organised into three generations of increasing mass. The forces are mediated by gauge bosons: the photon (γ) for electromagnetism, the $W^\pm$ and Z bosons for the weak interaction, and eight gluons for the strong interaction. The Higgs boson, the quantum of the scalar field that spontaneously breaks electroweak symmetry and endows the W and Z bosons and the fermions with mass, completes the particle content of the theory.

[TODO: Figure: Standard Model particle table]

1.1.2. The top quark

The top quark is the heaviest known elementary particle, with a mass of approximately 172.5 GeV, close to the electroweak symmetry-breaking scale set by the Higgs vacuum expectation value of 246 GeV. Its large Yukawa coupling to the Higgs boson — close to unity — makes the top quark a unique probe of the Higgs sector and of mass-generation mechanisms. Crucially, the top quark decays before it hadronizes, with a lifetime of approximately 5×10^{-25} s, so its spin and other quantum numbers are transmitted directly to its decay products. Top quarks decay almost exclusively via $t \rightarrow Wb$, producing a W boson and a b -quark in essentially every event.

1.1.3. Multi-top production at the LHC

Four-top-quark production ($t\bar{t}t\bar{t}$, hereafter *four-top*) is among the rarest Standard Model processes accessible at the LHC, with a predicted cross-section of approximately 12 fb at 13 TeV [3]. The dominant backgrounds to four-top are associated production processes: $t\bar{t}H$, $t\bar{t}W$, $t\bar{t}Z$, and $t\bar{t}WW$, which share similar final-state topologies. Each four-top event produces four b -quark jets from the top decays, four additional jets from hadronic W decays, possible charged leptons, and substantial E_T^{miss} from neutrinos in leptonic W decays. The resulting jet multiplicities can exceed ten reconstructed objects per event, and the combinatorial ambiguity in assigning jets to parent particles is a central analysis challenge.

[TODO: Figure: Feynman diagrams — QCD and Higgs-mediated four-top production at tree level]

1.1.4. Sensitivity to physics beyond the Standard Model

The four-top process is particularly sensitive to BSM physics because its production rate depends on the top–Higgs Yukawa coupling at the one-loop level and receives contributions from dimension-six effective field theory (EFT) operators involving the top quark. New physics that modifies the top–Higgs coupling or introduces new four-fermion contact interactions can enhance the four-top cross-section by factors of a few without being excluded by other measurements. Precision measurement of the four-top rate thus constrains the Wilson coefficients of top-sector EFT operators independently of direct new-particle searches. Rare multi-top processes therefore occupy a privileged position in the BSM landscape: they are both measurable with HL-LHC luminosity and theoretically clean probes of the top–Higgs sector.

1.2. The ATLAS Experiment and Event Reconstruction

This section describes the experimental apparatus and the data-processing chain that converts raw detector signals into the physics objects used as classifier inputs. It covers the LHC collider, the ATLAS detector subsystems, the object reconstruction pipeline, and the classical analysis strategies that motivate the machine-learning approach.

1.2.1. The Large Hadron Collider

The Large Hadron Collider at CERN is a circular proton–proton (pp) collider with a circumference of 27 km, operating at centre-of-mass energies of 13 TeV (Run 2) and 13.6 TeV (Run 3). Proton beams are organised into bunches crossing at four interaction points every 25 ns, delivering peak instantaneous luminosities of order $10^{34} \text{ cm}^{-2} \text{ s}^{-1}$. Each bunch crossing produces multiple inelastic pp interactions simultaneously, a phenomenon known as pile-up, which must be disentangled from the hard-scatter interaction of interest. The HL-LHC upgrade, planned for the late 2020s, will increase instantaneous luminosity by approximately a factor of five, substantially increasing both the dataset size and the pile-up challenge.

[TODO: Figure: LHC schematic]

1.2.2. The ATLAS detector

ATLAS (A Toroidal LHC ApparatuS) is a general-purpose particle detector surrounding one of the four LHC interaction points, designed to measure the full range of SM and BSM processes accessible at LHC energies. The innermost subsystem is the Inner Detector (ID), consisting of a silicon pixel detector, a silicon strip tracker, and a transition radiation tracker, which reconstructs charged-particle trajectories in a 2 T solenoidal magnetic field. Surrounding the ID are the electromagnetic and hadronic calorimeters, which absorb and measure the energy of photons, electrons, and hadrons; the hadronic calorimeter uses iron absorbers and scintillating tiles in the central region. The outermost system is the Muon Spectrometer, which identifies and measures muons using large toroidal magnets, providing coverage over $|\eta| < 2.7$.

[TODO: Figure: ATLAS detector cutaway]

1.2.3. From detector signals to physics objects

Reconstructing a physics event from raw detector signals proceeds through a sequence of steps, each translating lower-level information into higher-level objects. Charged-particle tracks are reconstructed

in the ID from hit patterns in the silicon layers and extrapolated to identify primary and secondary vertices; displaced secondary vertices from b -hadron decays form the basis of b -tagging algorithms. Calorimeter energy deposits are clustered into topological clusters and calibrated to form jets using the anti- k_t algorithm; b -jets are identified by a multivariate tagger exploiting track impact parameters and secondary vertex information. Electrons and muons are identified and calibrated using combined tracking and calorimeter information, while E_T^{miss} is computed from the negative vector sum of all reconstructed and calibrated object momenta in the transverse plane.

[TODO: Figure: four-top collision evolving through the ATLAS detector (supervisor diagram)]

1.2.4. Event-level classification in classical analyses

Classical analyses separate signal from background using cut-based selections on individual object multiplicities and kinematic variables, followed by multivariate discriminants such as boosted decision trees (BDTs). In four-top analyses, high jet and b -jet multiplicity requirements reduce the dominant $t\bar{t}$ +jets background by orders of magnitude while preserving a significant fraction of signal events. BDTs trained on a set of hand-crafted global event variables provide additional separation, but their performance degrades as the number of relevant features grows and correlations between variables become complex. Furthermore, the combinatorial assignment of reconstructed jets to the four top quarks is intractable to solve exactly, so classical approaches either ignore it or rely on dedicated reconstruction algorithms that introduce additional assumptions.

1.3. Machine Learning for High-Energy Physics

This section traces the progression of machine learning in particle physics from early shallow methods through deep learning and, finally, to the attention-based models that motivate this thesis. It covers the historical development, the transformer architecture, its adoption in HEP, and the structural reasons why attention-based models are well-suited to multi-object event classification.

1.3.1. A brief history of machine learning in particle physics

Machine learning has been applied in particle physics since at least the 1990s, when artificial neural networks and Fisher discriminants began supplementing cut-based selections in collider analyses. Boosted decision trees became the dominant paradigm in the 2000s, exploiting their robustness to non-linearities and their tractable training, and remain competitive in many current analyses. The deep learning revolution of the 2010s, driven by convolutional neural networks and large-scale GPU computing, spread quickly into HEP: jet tagging emerged as the primary proving ground, with networks operating directly on calorimeter images or particle lists achieving substantial improvements over BDTs. More recently, graph neural networks and attention-based models have extended the reach of deep learning to irregularly structured inputs, enabling end-to-end event reconstruction and multi-object classification.

1.3.2. Transformers and the attention mechanism

The transformer architecture, introduced by Vaswani et al. in “Attention Is All You Need” [vaswani2017attention], replaces recurrence and convolution with a scaled dot-product self-attention mechanism that relates each element of a sequence to every other element simultaneously. Multi-head attention computes several attention functions in parallel and concatenates the results, allowing the model to attend jointly to information from different representation subspaces. The architecture is completed by position-wise feed-forward networks, residual connections, and layer normalisation, all of which have become standard components of modern sequence and set models. Transformers achieved state-of-the-art performance across natural language processing benchmarks and subsequently demonstrated strong transfer to vision, audio, and scientific domains.

[TODO: Figure: Transformer architecture diagram]

1.3.3. Transformers in high-energy physics

The adoption of transformers in HEP accelerated with ParticleNet and its graph-based predecessors, followed by Particle Transformer (ParT), which demonstrated that attention over particle lists consistently outperforms graph-convolutional and convolutional baselines on jet tagging benchmarks. PELICAN introduced Lorentz-equivariant message passing, and OmniJet and related models explored large-

scale pre-training on diverse jet datasets. These developments collectively established attention-based architectures as the leading approach for particle-level classification tasks, motivating their application to the more complex problem of full-event classification. The convergence on attention-based models across multiple groups and benchmarks suggests that the structural properties of self-attention, rather than incidental training advantages, are responsible for the gains.

1.3.4. Why transformers suit multi-object event classification

Self-attention is permutation-equivariant by construction: the output for each input token depends on its pairwise relations with all other tokens but not on the order in which they are presented. This property aligns precisely with the physics of collision events, where the set of reconstructed objects has no natural ordering and the relevant information is encoded in pairwise and global relations. Attention weights can be biased by physically motivated pairwise features — angular separations (ΔR), invariant masses, and particle-type compatibility — providing a natural inductive bias without hard-coding specific interaction patterns. These structural considerations directly motivate the design axes explored in this thesis, particularly the input representation choices (Chapter 4) and the physics-informed attention bias families (Chapter 6).

1.4. Thesis Statement

This section articulates the research question, the methodological approach, and the organising principle of the experimental programme.

1.4.1. The intersection of top physics and transformer design

Rare multi-top production and attention-based classifiers are unusually well-matched: the physics produces variable-length, unordered sets of objects with rich pairwise structure, and the transformer processes exactly this kind of input through learned pairwise attention. Despite this structural compatibility, it remains unclear which specific design choices are responsible for performance gains observed in related domains, and whether physics-motivated inductive biases improve over purely data-driven attention for this task. This thesis addresses that gap by asking: which transformer design choices matter for event-level classification of rare multi-top processes, and does physics motivation improve or merely complicate the architecture? The answer requires controlled experiments that isolate individual design decisions rather than co-optimising all parameters simultaneously.

1.4.2. Controlled ablation over a single optimised model

The key methodological choice is to study design decisions through controlled ablations over a single, reproducible base model rather than by comparing independently optimised architectures. Each experiment varies exactly one axis while holding all others at their defaults, so any observed change in AUROC is attributable to that single choice. This avoids the confounding that arises when different architectures differ simultaneously in multiple design dimensions, a common limitation of architecture comparison studies in the literature. The scientific question is therefore not which configuration achieves the highest AUROC, but which design choices matter and whether their effects are consistent with the physics motivation claimed for them.

1.4.3. A configurable workbench for architectural exploration

The thesis organises 33 architectural design choices into orthogonal axes, each exposed as a named parameter in a Hydra configuration tree. Any run in the experiment database can be regenerated from a single CLI command by specifying the relevant axis overrides, ensuring full reproducibility. The axes are grouped into ten thematic groups — covering data treatment, tokenisation, positional encoding, pre-encoder modules, attention and encoder block design, feed-forward network realisation, physics-informed attention biases, classifier head, and model capacity — which map directly to the chapter structure of Part II. This workbench is both the primary engineering deliverable and the experimental substrate for all quantitative claims in the thesis.

1.5. Deliverables

This thesis produces three concrete outputs, each serving a distinct purpose within the overall research programme.

1.5.1. Modular codebase

The primary engineering output is a Hydra-driven Python package in which every architecture variant is reproducible from a single configuration file. Any run appearing in this thesis can be regenerated with a single CLI command.

1.5.2. Experiment database

A Weights & Biases project serves as a structured record of all training runs, storing hyperparameters, metrics, and axis settings in a queryable format. This database is the raw evidence underlying every quantitative claim in the thesis.

1.5.3. Thesis document

This document is a curated, interpretable distillation of the experiment database: not an exhaustive log of all runs, but a set of focused scientific arguments each anchored to a specific axis group.

1.6. Reader's Guide

1.6.1. Structure of the thesis

The thesis is organised in three parts: Part I frames the problem and the workbench (Chapters 2–3); Part II addresses one focused scientific question per chapter, each mapped to a specific axis group (Chapters 4–9); Part III synthesises the findings into a recommended configuration and outlook (Chapters 10–11).

1.6.2. How to read the results

Throughout the thesis, AUROC is the primary metric, reported as a seed-averaged mean with error bars. Axis groups are referred to by their labels (D, A, B, P, H), which correspond directly to config namespaces in the codebase. Each Part II chapter is framed by one of three evaluation lenses — performance, interpretability, or efficiency — which is stated explicitly at the chapter opening.

2

Dataset and Task Definition

This chapter describes the dataset used throughout the thesis and defines the classification tasks evaluated in the experimental chapters. The dataset consists of simulated proton-proton collision events at $\sqrt{s} = 13$ TeV, covering one signal process ($t\bar{t}t\bar{t}$, hereafter four-tops) and four Standard Model (SM) background processes ($t\bar{t}H$, $t\bar{t}W$, $t\bar{t}WW$, $t\bar{t}Z$). Events are encoded as fixed-length token sequences and stored in HDF5 format. The dataset was received pre-split and used without modification to the train/validation/test partition. All architectural experiments in later chapters operate on this single dataset and evaluate binary signal-versus-background discrimination.

2.1. Dataset Provenance

The dataset originates from two upstream sources. The event generation and detector simulation pipeline is documented by Builtjes et al. [2], who produced the dataset to study transformer and graph-based classifiers on multi-top final states. The data format follows the convention established by the Dark Machines collaboration [1], a community effort that defined a standardised benchmark format for model-independent LHC event classification.

Hard-scattering events were generated at leading order using MadGraph5_aMC@NLO 2.7 with the NNPDF3.1 parton distribution function set and the five-flavour scheme. Parton showering and hadronisation were performed with Pythia 8.239 using the MLM merging scheme. Fast detector simulation was carried out with Delphes 3.4.2 using the ATLAS detector card. An event-generation-level requirement of $H_T > 400$ GeV, summed over parton-level jets, was imposed to increase simulation efficiency for processes whose reconstructed signal region demands $H_T > 500$ GeV. Generated Monte Carlo samples were converted from ROOT to the Dark Machines CSV format, from which the dataset used in this thesis was constructed.

The dataset was received in pre-split HDF5 format (4tops_splitting.h5), with no train/validation/test partition performed in this work. The exact splitting procedure, including stratification criterion and random seed, is not documented in the source materials. [TODO: ask supervisor: splitting methodology]

2.2. Data Format and Representation

Each event in the original CSV follows the Dark Machines line format [1, 2]:

event ID; process ID; weight; $\|\vec{E}_T^{\text{miss}}\|$; $\phi_{E_T^{\text{miss}}}$; obj₁, E_1 , $p_{T,1}$, η_1 , ϕ_1 ; obj₂, E_2 , $p_{T,2}$, η_2 , ϕ_2 ; ...

Each reconstructed particle object is characterised by an integer type tag and its four-momentum (E , p_T , η , ϕ). Seven object types are defined: jets (j, tag 1), b -tagged jets (b, tag 2), positrons (e^+ , tag 3), electrons (e^- , tag 4), muons (μ^- , tag 5), anti-muons (μ^+ , tag 6), and photons (g, tag 7). The missing transverse energy magnitude $\|\vec{E}_T^{\text{miss}}\|$ and its azimuthal angle $\phi_{E_T^{\text{miss}}}$ are stored as two global features shared across all objects in the event.

Because the number of reconstructed objects varies per event, all sequences are zero-padded to a

fixed length of $T = 18$ token slots [4]. The value $T = 18$ corresponds to the largest event in the dataset. A binary attention mask accompanies each event: padding tokens receive a mask value of zero and are excluded from attention computations, while real object tokens receive a mask value of one.

The HDF5 file stores each event as a flat row of 92 columns: 18 integer type IDs, 2 global features ($\|\vec{E}_T^{\text{miss}}\|$ and $\phi_{E_T^{\text{miss}}}$), and $18 \times 4 = 72$ continuous kinematic features (one four-momentum per token slot). The integer type IDs and continuous features are separated at runtime by the data loader. Before training, the continuous features are normalised using global z-score statistics computed over the training split only, pooled across all N_{train} events and all T token positions simultaneously. This yields one scalar mean and one scalar standard deviation per feature type (E, p_T, η, ϕ), applied uniformly to every token slot in every event. No per-position or per-particle-type rescaling is performed.

Across 98.6% of all thesis training runs, tokens are presented in the order they appear in the HDF5 file (`input_order`). No reordering is applied by the data loader for these runs. The provenance of the within-event object ordering in the source file is unknown. **[TODO: ask supervisor: within-event object ordering in HDF5]**

A discrete tokenisation scheme, referred to as AmbreBinning and developed for a subset of experiments in Chapter 4, maps each continuous four-momentum to a single integer through independent uniform binning of p_T , η , and ϕ (five bins each) combined with the object-type dimension (125 categories). The combined token index is computed as $125(b_{\text{obj}} - 1) + 25(b_{p_T} - 1) + 5(b_\eta - 1) + b_\phi$, yielding a vocabulary of 886 distinct integers. This scheme was inspired by binning strategies introduced by the thesis supervisor for a related dataset and is not the default representation; the majority of experiments use continuous four-momentum features directly.

2.3. Classification Tasks

The dataset contains 302,072 events distributed across five SM process classes, split 80/10/10 into training, validation, and test subsets. Table 2.1 gives the per-class event counts for each split.

Table 2.1: Event counts per process class and dataset split. Label codes follow the convention of Bultjes et al. [2]: 1 = $t\bar{t}t\bar{t}$, 2 = $t\bar{t}H$, 3 = $t\bar{t}W$, 4 = $t\bar{t}WW$, 5 = $t\bar{t}Z$.

Split	$t\bar{t}t\bar{t}$	$t\bar{t}H$	$t\bar{t}W$	$t\bar{t}WW$	$t\bar{t}Z$	Total
Train	121,359	28,854	30,511	30,461	30,473	241,658
Val	15,266	3,666	3,779	3,760	3,736	30,207
Test	15,375	3,552	3,710	3,779	3,791	30,207

The signal process, $t\bar{t}t\bar{t}$, constitutes approximately 50% of all events. The four background processes ($t\bar{t}H$, $t\bar{t}W$, $t\bar{t}WW$, $t\bar{t}Z$) each contribute roughly 10–12% of the total. This deliberate 1:1 signal-to-background ratio, chosen by Bultjes et al. to balance training [2], differs markedly from the physical production cross-sections, which strongly favour the backgrounds, and should be kept in mind when interpreting classifier outputs.

The primary classification task evaluated throughout this thesis is binary discrimination of $t\bar{t}t\bar{t}$ (label 1) against the combined SM background (labels 2–5). This is the most physically relevant task: the four-tops process has an extremely small production cross-section ($\sigma \approx 10$ fb at $\sqrt{s} = 13$ TeV), and its first observation by ATLAS required careful discrimination from a large and topologically similar SM background [3]. The task is challenging because each top quark decays almost exclusively via $t \rightarrow Wb$, so four-top events produce high-multiplicity final states with 0–4 leptons and 4–12 jets, including four b -jets. Background processes such as $t\bar{t}WW$ and $t\bar{t}Z$ produce comparably complex topologies. The classifier must therefore exploit subtle differences in object multiplicity, kinematic correlations, and b -jet content to separate signal from background.

All architectural ablations in Chapters 4–8 evaluate this binary task, reporting area under the receiver operating characteristic curve (AUROC) and background rejection at fixed signal efficiency as primary metrics.

3

The Workbench: Architecture and Axes

3.1. Design philosophy

Rather than committing to a single architecture, the workbench exposes every meaningful design choice as a switchable configuration parameter. This makes fair comparison possible by construction: when one axis changes, everything else is held fixed, so any observed difference in AUROC is attributable to that choice alone. The entire architecture is specified by a Hydra configuration tree, where each axis corresponds to a named key in a structured YAML hierarchy. Any run in the experiment database can be regenerated from a single command-line invocation.

3.2. The base transformer

The base model follows the standard transformer encoder pipeline: particle tokens are embedded into a fixed-dimensional space, passed through a stack of attention-plus-FFN blocks, aggregated into a single event-level vector by a pooling operation, and classified by a head network. This skeleton is fixed across all experiments; the axes control what happens inside each stage.

Each particle token, a raw kinematic vector plus an optional particle-type identifier, is projected into the model dimension d by a linear embedding layer. The encoder consists of L identical blocks, each containing a multi-head self-attention layer followed by a position-wise feed-forward network, with residual connections and normalisation at positions controlled by axes A1 and A2. The pooled vector is passed to a classifier head that produces the final logit.

3.3. Axis groups

The axes are organised into ten groups. Groups G and D define the experimental conditions and data treatment. Groups T, E, P, A, F, B, and C define the architecture. Group H sets model capacity. Three cross-cutting shared parameter blocks (§K, §M, §S) are consumed by multiple modules simultaneously and are documented at the end of this section. Groups §R and §L cover training protocol and logging respectively and are not architectural axes.

G Study framing

G axes define what is being classified and with which model family. They are not architectural choices and are not varied in the ablation programme; they set the experimental conditions under which all other axes are evaluated.

G01 – Task type. Selects the training loop: supervised classification or anomaly detection (autoencoder, GAN-AE, diffusion-AE). This thesis focuses exclusively on supervised classification; the anomaly detection loops are available in the workbench but not studied here.

G02 – Model family. Selects the model architecture family: transformer, MLP, BDT, or autoencoder. All Part II chapters use the transformer family; the BDT appears only in Ch. 8 as a surrogate model for configuration prediction, not as a classifier.

G03 – Classification task. Defines which processes are treated as signal and background. Five processes are available: four-top-quark production ($tttt$), associated production with a Higgs boson (ttH), with a W boson (ttW), with two W bosons ($ttWW$), and with a Z boson (ttZ). Three task definitions are used in this thesis: $tttt$ vs. combined background ($ttH + ttW + ttWW + ttZ$) as the primary task, $tttt$ vs. ttH as a harder secondary task, and all-five multiclass as an exploratory task.

D Data treatment

D axes control what the model sees before any architectural processing.

D01 – Feature set. Selects which kinematic features are included in each token. The default is the four-feature set (E , P_t , η , ϕ); energy can be omitted to test whether the model relies on explicit E or whether P_t , η , and ϕ already provide sufficient kinematic information.

D02 – MET treatment. Controls whether missing transverse energy is excluded or included as an event-level global. Tests how much the invisible neutrino system contributes beyond the particle tokens.

D03 – Token ordering. Sets whether tokens are presented in input order, sorted by decreasing P_t , or randomly shuffled per event. Tests whether the model exploits ordering as an implicit positional signal.

T Tokenizer

T axes control how raw particle features are converted into token embeddings.

T1 – Tokenizer family. Selects between `raw` (kinematics only), `identity` (kinematics plus an explicit particle-type label), and `binned` (discrete token, no continuous features). Tests whether explicit particle identity is necessary or recoverable from kinematics alone.

T1-a – PID embedding mode. When $T1 = \text{identity}$, selects how the particle-type label is encoded: `learned`, `one_hot`, or `fixed_random`. Tests whether particle identity needs a learned geometry or only a categorical code.

T1-b – PID embedding dimension. Sets the capacity of the PID embedding to 8, 16, or 32 dimensions. Tests the trade-off between representational capacity and overfitting for the type sub-embedding.

E Positional encoding

E axes control how position information is injected into the token stream.

E1 – PE type. Selects among `none`, `sinusoidal`, `learned`, and `rotary`. Tests whether injecting position information helps a set-based classifier at all.

E1-a – PE space. Controls whether the positional signal is applied before or after the input projection. Prerequisite: $E1 \in \{\text{sinusoidal}, \text{learned}\}$.

E1-a1 – PE dimension mask. Allows the positional encoding to act selectively on a subset of feature dimensions. Prerequisite: $E1-a = \text{token}$.

P Pre-encoder modules

P axes add optional modules applied to raw particle features before the main transformer encoder. Both require $T1 \in \{\text{raw}, \text{identity}\}$.

P1 – Nodewise mass patch. Optionally enriches each token with invariant-mass features computed from its k -nearest neighbours in Lorentz space, following Li et al. Tests whether explicit local mass structure before the encoder improves token representations. Requires D01 to include energy (index 0).

P2 – MIA pre-encoder. Optionally inserts MIParT-style interaction-attention blocks before the main encoder, where attention is driven by pairwise interaction features rather than query-key similarity. Tests

whether physics-driven message passing as a pre-encoder stage adds information beyond standard self-attention.

P2-a – MIA placement. Controls whether MIA blocks are prepended or appended relative to the main encoder. Prerequisite: `P2 = true`.

A Attention and encoder block

A axes control the internal design of each encoder block.

A1 – Normalisation policy. Selects pre-norm, post-norm, or NormFormer placement of layer normalisation within each block. Tests whether normalisation position affects training stability for particle-level inputs.

A2 – Normalisation type. Chooses between LayerNorm and RMSNorm. Tests whether removing the mean-subtraction step improves efficiency without degrading performance.

A3 – Attention type. Selects standard multi-head attention or differential attention, which computes the difference of two softmax maps to suppress diffuse attention patterns. Tests whether this mechanism transfers from NLP to particle classification.

A3-a – Differential bias mode. When `A3 = differential`, controls how additive physics biases interact with the two attention branches: `none`, `shared`, or `split`. Prerequisite: `A3 = differential`.

A4 – Attention-internal normalisation. Optionally applies LayerNorm or RMSNorm per head after the weighted sum, before head merging. Tests whether per-head normalisation stabilises attention output scales. Applies to both standard and differential attention.

A5 – Causal masking. Optionally applies an autoregressive mask to the attention matrix. Tests whether imposing sequential structure on an inherently unordered set of particles hurts, helps, or is neutral.

F Encoder FFN realization

F axes control the feed-forward network inside each encoder block. The effective realization follows a priority rule: `MoE > KAN > standard`.

F1 – FFN realization. Selects between a standard two-layer MLP, a Kolmogorov–Arnold Network (KAN), and a Mixture-of-Experts (MoE) FFN. Tests whether learnable spline activations or conditional computation improve per-token transformations.

F1-a – KAN FFN variant. When `F1 = kan`, selects among `hybrid`, `bottleneck`, and `pure` architectures. Prerequisite: `F1 = kan`.

F1-b – MoE scope. Controls whether expert routing applies to encoder FFNs, the classifier head, or both. Prerequisite: `F1 = moe`.

B Physics-informed attention biases

B axes inject physics structure as additive biases on the attention logits. All families require `T1 ∈ {raw, identity}` and share a zero-initialised scalar gate, so adding any bias leaves the baseline unchanged at initialisation. Multiple families can be active simultaneously using `+` notation.

B1 – Bias activation set. The master switch selecting which bias families are active: `lorentz_scalar`, `typepair_kinematic`, `sm_interaction`, `global_conditioned`, or combinations thereof.

B1-L1 – Lorentz feature set. Selects the subset of pairwise Lorentz-invariant features fed into the bias network, from the minimal set $\{m^2, \Delta R\}$ through the MIParT basis to the full feature set. Active when `lorentz_scalar ∈ B1`.

B1-L2 – Lorentz MLP type. Selects whether the network mapping Lorentz features to bias scores uses a standard MLP or a KAN. Active when `lorentz_scalar ∈ B1`.

B1-L5 – Lorentz sparse gating. Adds per-feature sigmoid gates providing a direct measure of learned feature importance. Active when `lorentz_scalar ∈ B1`.

B1-T2 – Type-pair initialisation. Controls how the learnable $N_{\text{type}} \times N_{\text{type}}$ interaction table is initialised: from scratch, from a binary SM mask, or from fixed SM coupling constants. Active when `typepair_kinematic` $\in$ B1.

B1-T3 – Type-pair freeze. Controls whether the type-pair table is frozen after initialisation or remains trainable. Tests whether the table is better used as a fixed physics prior or a learnable parameter. Active when `typepair_kinematic` $\in$ B1.

B1-S1 – SM interaction mode. Selects the fidelity of the SM interaction prior: `binary`, `fixed_coupling`, or `running_coupling`. Active when `sm_interaction` $\in$ B1.

B1-G1 – Global-conditioned mode. Controls how MET modulates pairwise attention biases: as a global scale factor or as a directional signal. Requires `D02 = true`. Active when `global_conditioned` $\in$ B1.

C Classifier head

C axes control the final mapping from the pooled event representation to class logits.

C1 – Head realization. Selects between a `linear` head, a `kan` head, and a `moe` head. Tests whether specialised nonlinearities or conditional computation at the decision boundary improve discrimination.

C2 – Pooling strategy. Selects how the token sequence is aggregated into a single event-level vector: `cls` token, `mean` pooling, or `max` pooling. Placed here because pooling is logically part of the readout stage rather than the encoder block.

H Model-size hyperparameters

H axes set model capacity and are not architectural design choices in the sense of the other groups. They are tracked separately and are the primary axis of variation in Ch.9. The convenience key `model_size_key` bundles `dim` (d), `depth` (L), `heads`, and `FFN hidden dim` into named presets spanning roughly 50k to 4M parameters.

Cross-cutting blocks: §K, §M, §S

Three shared parameter blocks are consumed simultaneously by multiple modules. §K (KAN hyperparameters: grid size, spline order, grid range) affects every active KAN module, including the encoder FFN, classifier head, and Lorentz bias network. §M (MoE hyperparameters: number of experts, top-k, routing level) affects all active MoE sites. §S (shared pairwise backbone) computes the pairwise feature matrix F_{ij} once per forward pass and shares it across all active B1 families and the MIA pre-encoder.

3.4. Experimental methodology

Each axis is studied in isolation: one axis is varied across its settings while all others are held at their defaults, forming an orthogonal sweep. This ensures observed differences are attributable to the axis under study and keeps the number of required experiments linear in the number of axes rather than exponential.

Every result reported in this thesis is averaged over at least three independently seeded runs, with error bars representing the standard deviation across seeds. The area under the ROC curve (AUROC) is the primary evaluation metric throughout: it measures the probability that the classifier assigns a higher score to a signal event than to a background event, and is threshold-independent. Signal efficiency at fixed background rejection is reported as a complementary metric in selected chapters.

All runs are tracked in a Weights & Biases project where every axis setting, hyperparameter, and evaluation metric is logged under a structured key scheme mirroring the axis reference directly.

Part II

Thesis Questions

4

Best Input Representation

Before the transformer processes a single attention operation, several choices have already determined the information it receives. These choices concern *how* each particle is represented as a token, and *what* the token sequence contains. This chapter evaluates six such choices — the tokenizer family (T1 🏷️), the particle-identity (PID) embedding mode (T1-a 🏷️), the PID embedding dimension (T1-b 🏷️), the kinematic feature set (D01 🏷️), whether missing transverse energy enters as an additional token (D02 🏷️), and the ordering of tokens in the input sequence (D03 🏷️) — and asks which of them affect classification performance on the 4-tops-versus-background task.

The chapter is structured as follows. Section 4.1 describes the shared experimental setup and evaluation metric. Section 4.2 presents the tokenizer family comparison, which is the primary result of this chapter. Section 4.2.2 examines the sub-choices within the identity tokenizer (T1-a and T1-b). Section 4.3 covers the data-treatment axes (D01, D02, D03). Section 4.4 synthesises the findings and establishes the baseline configuration carried forward into subsequent chapters.

The one-sentence answer is: the tokenizer family dominates all other choices, with the identity tokenizer outperforming kinematics-only alternatives by a large margin; fine-grained choices within that family and all data-treatment axes are null or near-null results at this model scale.

4.1. Experimental Setup

Two experiments are reported here. **Experiment 4A** sweeps the tokenizer family (T1 🏷️) and, within the identity family, the PID embedding mode (T1-a 🏷️) and embedding dimension (T1-b 🏷️). **Experiment 4B** fixes the tokenizer at identity and sweeps the three data-treatment axes (D01 🏷️, D02 🏷️, D03 🏷️) in a full $2 \times 2 \times 2$ factorial design.

All axes not under study are held at canonical defaults throughout: positional encoding (E1 🏷️) is sinusoidal in model space; attention biases (B1) are disabled; pooling (C2) uses a CLS token; the FFN (F1) is the standard implementation. The model size is the d64_L4 baseline (H01 = 64, H02 = 4), used uniformly across all runs in this chapter. Every condition is replicated with three seeds (R05 $\in$ {42, 123, 314}).

The primary evaluation metric is the area under the receiver operating characteristic curve (AUROC) computed on the held-out test set. All metric values are drawn from `eval_v2/test_auroc` in the W&B export, which is recorded at training time and is not affected by the subsequent re-inference step. Reported means and standard deviations are over the three seeds unless stated otherwise; with $n = 3$ seeds per condition, these figures are indicative and should not be interpreted as tight confidence intervals.

Table 4.1 and Table 4.2 summarise the sweep designs for the two experiments.

Axis	Values swept	n_{configs}
T1 — tokenizer family	raw, binned	2
T1 + T1-a — PID mode	one_hot, fixed_random, learned	3
T1 + T1-b — PID dim	8, 16, 32	3
Unique configurations		11
Seeds per configuration		3
Total runs		33

Table 4.1: Experiment 4A sweep design. Axes T1-a and T1-b apply only to the `identity` tokenizer family. All other axes are held at canonical defaults: `D01 = [0, 1, 2, 3]`, `D02 = false`, `D03 = input_order`.

Axis	Values swept	n_{configs}
D01 — feature set	[0, 1, 2, 3], [1, 2, 3]	2
D02 — MET treatment	false, true	2
D03 — token ordering	input_order, shuffled	2
Unique configurations (2^3)		8
Seeds per configuration		3
Total runs		24

Table 4.2: Experiment 4B sweep design ($2 \times 2 \times 2$ full factorial). Fixed: `T1 = identity`, `T1-a = learned`, `T1-b = 8`. These tokenizer settings are confirmed from the W&B export; all 24 rows use this configuration.

4.2. Tokenizer Family

The choice of tokenizer family (T1 🗑️) is the most consequential design decision examined in this chapter. Three families are compared. The `raw` tokenizer maps each particle’s continuous kinematics (E, p_T, η, ϕ) directly to the model dimension via a shared linear projection; no particle-type information is provided. The `identity` tokenizer augments the kinematic projection with an explicit PID embedding, which encodes the discrete particle type label (jet, b-jet, electron, muon, photon, etc.) and concatenates it with the projected kinematics. The `binned` tokenizer discards continuous features entirely, representing each particle solely by its histogram-binned kinematic cell index as a discrete token.

Figure 4.1 shows the mean test AUROC for each family. The `identity` tokenizer achieves a mean AUROC of approximately 0.842 (evaluated on cells with embedding dimension ≥ 16 , where the learned-mode confound described in Section 4.2.2 is absent; $n = 21$ seeds), compared with 0.804 ± 0.001 for `raw` ($n = 3$) and 0.725 ± 0.002 for `binned` ($n = 3$). The gap between `identity` and `raw` is $+0.038$ AUROC; the gap between `identity` and `binned` is $+0.117$.

Figure 4.2 shows the ROC curves for each family. The `identity` advantage is present uniformly across the full operating range: at every background rejection threshold, the `identity` tokenizer achieves higher signal efficiency than the `raw` tokenizer, and both substantially outperform the `binned` tokenizer.

The training dynamics in Figure 4.3 add interpretive context. The `binned` tokenizer reaches a validation AUROC peak near epoch 4 and subsequently collapses, behaviour consistent with rapid memorisation of the discrete token vocabulary followed by overfitting once no additional generalisation signal remains. The `raw` tokenizer converges stably but plateaus at a lower level. The `identity` tokenizer converges to a higher plateau with a gradual, mild decline beyond the optimum, consistent with moderate overtraining over 50 epochs.

Interpretation. The result is consistent with a physics-motivated expectation: particle type carries information that is not fully recoverable from the four-vector alone, particularly for particles with overlapping kinematic distributions (e.g. light jets and b-jets at similar p_T and η). By providing an explicit PID label, the `identity` tokenizer allows the model to apply type-specific attention patterns and feature combinations that the `raw` tokenizer cannot form. The `binned` tokenizer is the weakest family because it discards not only the type label but also the continuous kinematic resolution; only coarse bin membership remains.

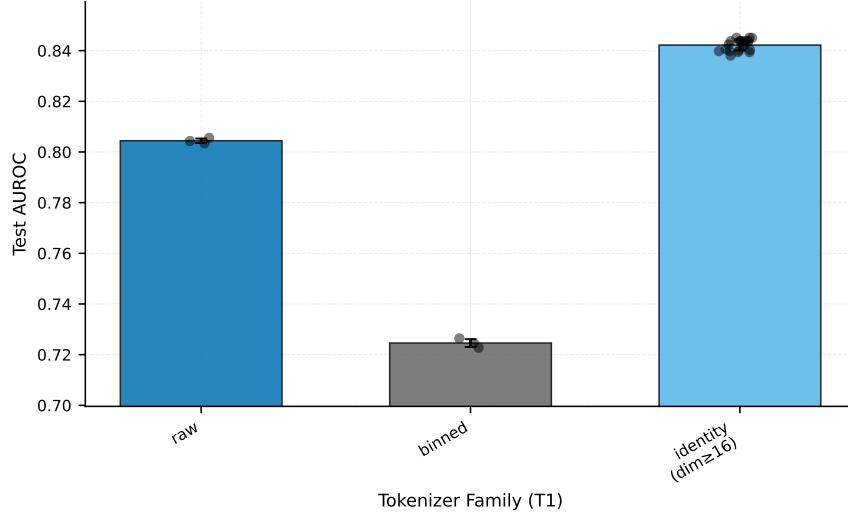


Figure 4.1: Mean test AUROC by tokenizer family (T1). Error bars show one standard deviation over seeds. The identity bar pools cells with embedding dimension ≥ 16 ($n = 21$ seeds) to exclude the `learned/dim=8` confound discussed in Section 4.2.2. `raw` and `binned` each have $n = 3$.

4.2.1. Failure Analysis: Where PID Changes the Outcome

To quantify where the tokenizer gap originates, each event in the test set ($N = 30,207$) is classified by both the best `raw` model (AUROC = 0.8056) and the best `identity` model (`one_hot`, AUROC = 0.8451), using a threshold of 0.5 on the signal score.¹

Figure 4.4 shows the per-event outcome breakdown. Of the 30,207 test events, 20,289 (67.2%) are classified correctly by both tokenizers, and 5,396 (17.9%) are wrong for both, defining an irreducible difficulty floor that reflects events where neither tokenizer has sufficient information to classify correctly. The remaining events reveal the tokenizer effect: 2,713 events (9.0%) are correctly classified by the identity tokenizer but not by the raw tokenizer (PID helped), while 1,809 events (6.0%) are correctly classified by the raw tokenizer but not by the identity tokenizer (PID hurt). The net gain from the explicit PID label is +904 events, or +3.0% of the test set.

The 6.0% of events where the raw tokenizer outperforms the identity tokenizer warrants a brief remark. These events are not evidence that PID is harmful in general; rather, they reflect the stochastic nature of threshold classification and the fact that no tokenizer achieves perfect classification on all events. For some events, the kinematic features alone may provide a cleaner signal than the combination of kinematics and a noisy or ambiguous type label.

4.2.2. PID Embedding Mode and Dimension

Within the identity tokenizer family, two further choices determine how the PID label enters the model: the embedding mode (T1-a) and the embedding dimension (T1-b). This subsection examines these sub-choices, which have a substantially smaller effect than the tokenizer family itself.

Figure 4.5 shows the mean test AUROC by PID embedding mode, pooled over all embedding dimensions. The marginal means are: `one_hot` = 0.8442 ± 0.0006 ($n = 3$), `fixed_random` = 0.8410 ± 0.0021 ($n = 9$), `learned` = 0.8420 ± 0.0017 ($n = 27$). The apparent underperformance of `learned` relative to `one_hot` in this marginal view is a confound: the `learned` sample is dominated by dimension-8 runs, which perform anomalously.

The heatmap in Figure 4.6 resolves this confound by showing AUROC as a joint function of mode and dimension. The per-cell results from Experiment 4A ($D01 = [0, 1, 2, 3]$, $D02 = \text{false}$, $D03 = \text{input_order}$) are:

- `learned`, `dim=8`: AUROC = 0.8431 ± 0.0007 ($n = 3$)

¹W&B run identifiers: `raw` — `run_20260410-160141_ch4_input_repr_exp4a_1_raw_job002`; `identity` — `run_20260410-162037_ch4_input_repr_exp4a_2_job010`.

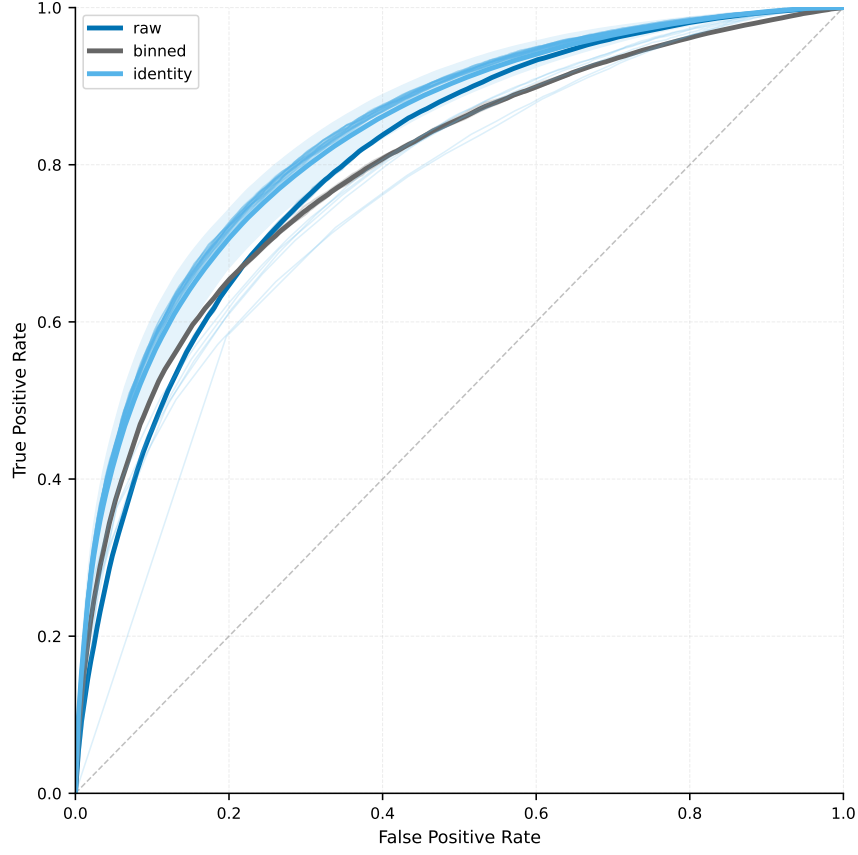


Figure 4.2: ROC curves by tokenizer family (T1). Individual seed curves are shown for `raw` and `binned`; the mean curve is shown for `identity`. The performance ordering is consistent across all background rejection values.

- `learned`, `dim=16`: AUROC = 0.8428 ± 0.0006 ($n = 3$)
- `learned`, `dim=32`: AUROC = 0.8402 ± 0.0004 ($n = 3$)
- `one_hot`, `num_types`: AUROC = 0.8442 ± 0.0006 ($n = 3$)
- `fixed_random`, `dim=8`: AUROC = 0.8435 ± 0.0009 ($n = 3$)
- `fixed_random`, `dim=16`: AUROC = 0.8400 ± 0.0020 ($n = 3$)
- `fixed_random`, `dim=32`: AUROC = 0.8397 ± 0.0004 ($n = 3$)

The spread across all cells is ~ 0.004 AUROC, which is within two standard deviations of the seed spread for individual cells. No mode-dimension combination is strongly superior to any other at $n = 3$ seeds.

Interpretation. The result suggests that the model is not strongly sensitive to whether the PID label is encoded as a trainable embedding, a fixed random projection, or a discrete one-hot vector. All three modes provide a representation that is sufficiently distinct per particle type for the transformer to exploit. The `one_hot` mode at dimension equal to the number of particle types (`num_types`) is the most parameter-efficient choice, as it requires no additional trainable weights and is insensitive to the embedding dimension hyperparameter. This mode is adopted as the baseline for subsequent chapters, paired with the default dimension setting.

4.3. Data Treatment

Having established the tokenizer family and PID encoding, this section examines three choices about the content and arrangement of the token sequence: whether energy is included in the feature vector (D01 🟡), whether MET enters as an additional token (D02 🟡), and whether tokens are presented in

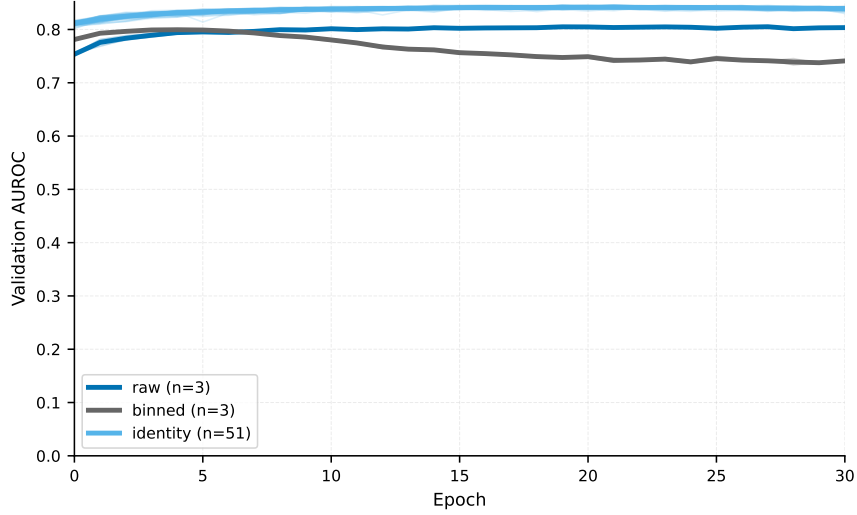


Figure 4.3: Validation AUROC over training epochs by tokenizer family (T1). The `binned` tokenizer peaks near epoch 4 and collapses, indicating rapid overfitting. `raw` converges stably but at a lower plateau. `identity` converges to the highest plateau and degrades only slowly beyond the optimum. The x-axis is cropped to 30 epochs to focus on the convergence region.

their original reconstruction order or shuffled (D03 input_order). All results in this section are from Experiment 4B (Table 4.2), which uses the identity tokenizer with $T1\text{-}a = \text{learned}$ and $T1\text{-}b = 8$ throughout.

4.3.1. Energy Feature (D01)

The feature-set axis (D01 feature_set) compares two configurations: the full four-vector (E, p_T, η, ϕ) with index set $[0, 1, 2, 3]$, and the three-vector (p_T, η, ϕ) with index set $[1, 2, 3]$, which omits the energy component.

Figure 4.7 shows the result. Removing energy yields $\text{AUROC} = 0.8424 \pm 0.0010$ versus 0.8431 ± 0.0007 with energy, a difference of -0.0007 . This difference is smaller than the within-condition standard deviation and cannot be distinguished from seed noise at $n = 3$.

The null result is physically plausible: for massless or near-massless particles, energy is approximately $E \approx p_T \cosh \eta$, so p_T and η together recover much of the energy information. The genuinely mass-sensitive component — relevant for reconstructing invariant masses of top decay sub-systems — may not be detectable at this model capacity and sample size. The full four-vector $([0, 1, 2, 3])$ is retained in the chapter baseline because it carries no performance cost and preserves the option to use the nodewise mass module (P1) in subsequent chapters, which requires the energy feature.

4.3.2. MET Treatment (D02)

Missing transverse energy (MET) represents the net transverse momentum carried by particles that are not detected, primarily neutrinos from W -boson decays. In 4-tops events, which contain at least four top quarks with multiple semi-leptonic or fully hadronic decay chains, MET provides information about the neutrino system. The MET treatment axis (D02 met_treatment) tests whether including MET as an additional token improves classification.

Figure 4.8 shows the marginal comparison across D01 and D03. With MET included (D02 = `true`), the mean AUROC is 0.8423 ± 0.0019 ($n = 12$); without MET, it is 0.8415 ± 0.0017 ($n = 12$). The difference is $+0.0008$ in favour of including MET, which is well within the pooled standard deviation and is not statistically distinguishable from zero at this sample size. Restricting to the `input_order` condition only (D03 = `input_order`, $n = 6$ per MET condition), the delta is $+0.0007$, consistent with the full marginal result.

This is a null result: neither benefit nor harm from MET inclusion is measurable in Experiment 4B. The result is inconsistent with an earlier, superseded figure of -0.026 that appeared in a prior version of the analysis; that figure has been traced to a data-loading bug in the re-inference pipeline, in which the

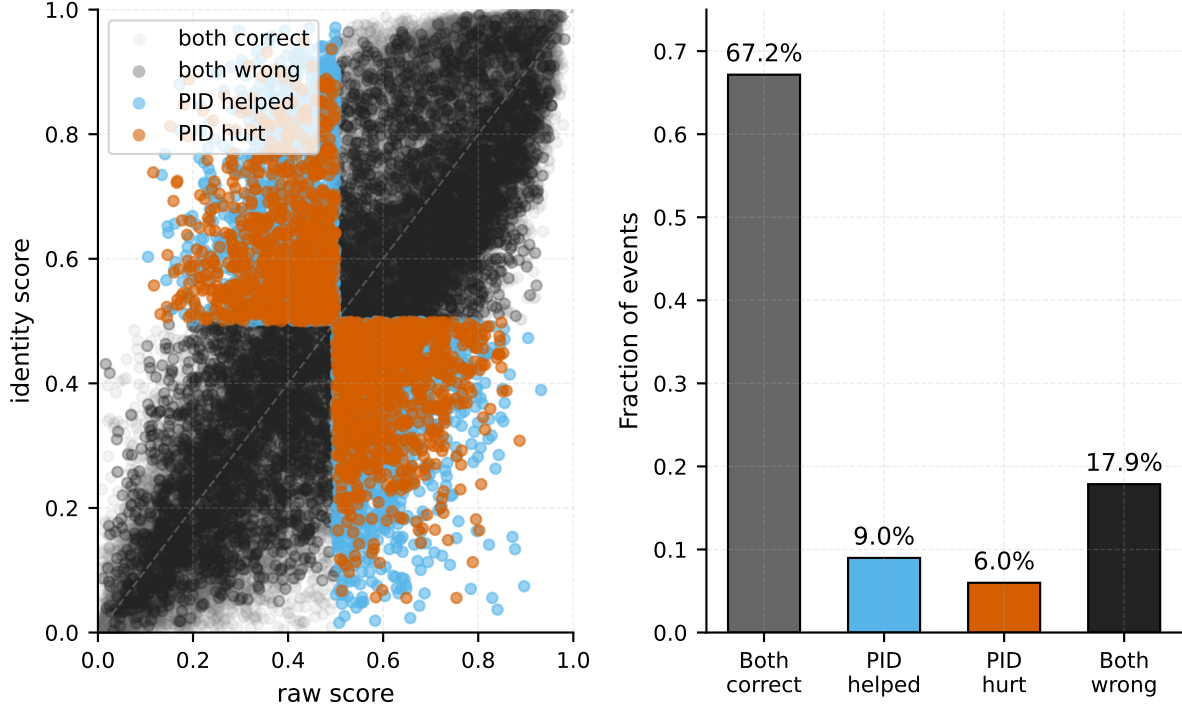


Figure 4.4: Per-event outcome comparison between the best raw and best identity tokenizer models on the test set ($N = 30,207$). *Left:* scatter of per-event signal scores; colours indicate outcome category. Events above the diagonal (identity score $>$ raw score) tend to benefit from the PID label. *Right:* fraction of events in each category. The net asymmetry (9.0% PID-helped vs. 6.0% PID-hurt) accounts for the identity tokenizer’s overall AUROC advantage.

validation dataset was not correctly augmented with the MET token for MET=True models. The correct values are those in Table 4.3, drawn directly from `eval_v2/test_auroc` as logged at training time.

The PE-disruption hypothesis — that MET, appended at a fixed token position, disrupts the sinusoidal positional encoding (E1) by occupying a slot that the model otherwise associates with a specific particle type — is a plausible post-hoc explanation that has not been tested experimentally. Such a test would require a dedicated experiment crossing D02 with E1 = none (no positional encoding) to isolate any PE-mediated effect. No such experiment has been conducted. MET is excluded from the chapter baseline (D02 = false) primarily because it provides no measured benefit; its re-introduction via physics-informed attention biases (B1-G1 = met_direction) is examined in a subsequent chapter.

4.3.3. Token Ordering (D03)

Self-attention is permutation-equivariant by construction: without positional encoding, the output is invariant to any permutation of the input tokens. However, all runs in this chapter use sinusoidal positional encoding (E1 = sinusoidal) in model space (E1-a = model). The token ordering axis (D03) tests whether the choice between preserving the original reconstruction order (`input_order`) and randomly shuffling the sequence (`shuffled`) affects AUROC.

Figure 4.9 shows the marginal comparison restricted to D02 = false conditions, which removes any potential interaction with the MET token. `input_order` yields a mean AUROC of 0.8428 ± 0.0008 ($n = 6$); `shuffled` yields 0.8402 ± 0.0012 ($n = 6$). The gap of -0.0026 (shuffled vs. `input_order`) is approximately 2–3 $\times$ the standard deviation of the individual conditions and is consistent across both D01 values.

Table 4.3 shows the full $2 \times 2 \times 2$ factorial mean AUROC table for all D01, D02, D03 combinations. The ordering effect is present under both MET conditions, suggesting it is not mediated by an interaction between D03 and D02.

Attribution. The ordering effect is consistent with the sinusoidal positional encoding providing useful information when tokens are presented in a consistent order, and that information being destroyed by

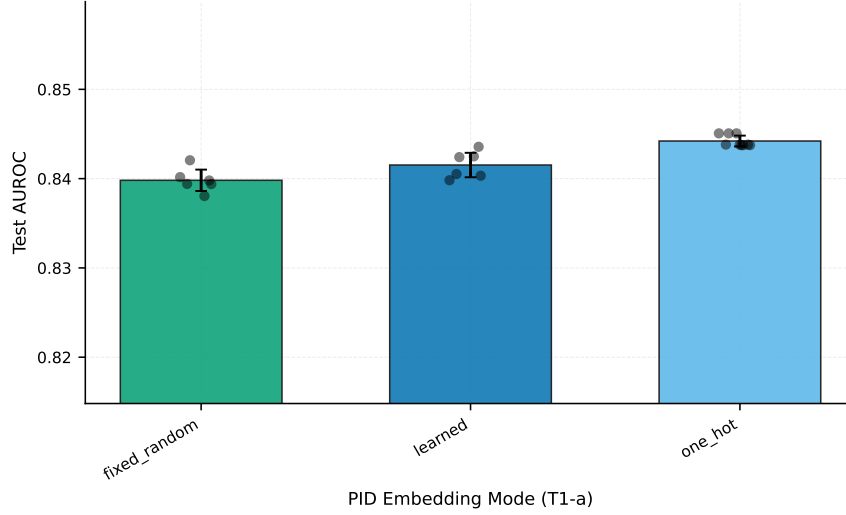


Figure 4.5: Mean test AUROC by PID embedding mode (T1-a), pooled over all T1-b embedding dimensions. The apparent gap for `learned` reflects the dominance of dimension-8 cells in the `learned` sample. Figure 4.6 shows that at dimension ≥ 16 , all three modes are within 0.003 AUROC of each other.

D01	D02	D03	mean AUROC	std	n
[0,1,2,3]	False	input_order	0.8431	0.0007	3
[0,1,2,3]	False	shuffled	0.8398	0.0012	3
[0,1,2,3]	True	input_order	0.8434	0.0015	3
[0,1,2,3]	True	shuffled	0.8420	0.0008	3
[1,2,3]	False	input_order	0.8424	0.0010	3
[1,2,3]	False	shuffled	0.8406	0.0013	3
[1,2,3]	True	input_order	0.8436	0.0004	3
[1,2,3]	True	shuffled	0.8401	0.0021	3

Table 4.3: Full $2 \times 2 \times 2$ factorial AUROC table for Experiment 4B. Each cell reports the mean and standard deviation over three seeds. The token ordering effect (D03) is consistent across all D01 and D02 conditions.

random shuffling. However, this attribution requires a control experiment with $E1 = \text{none}$ (no positional encoding) to confirm. If the ordering effect disappears under $E1 = \text{none}$, the effect is mediated by the positional encoding. If it persists, the transformer’s residual stream or the classification head is sensitive to token position through some other mechanism. This control experiment has not been run. The thesis states the ordering effect as a measured result and the PE attribution as a hypothesis.

The reconstruction software produces tokens in an order that is not p_T -sorted but reflects the detector reconstruction sequence, which may carry implicit correlations between position and particle type or kinematics. If such correlations exist, the sinusoidal PE would be implicitly encoding a type-proxy, which is a legitimate, if implicit, inductive bias. Characterising this ordering is a direction for future work.

4.4. Discussion

This section synthesises the results of Experiments 4A and 4B and establishes the baseline configuration for subsequent chapters.

Ranking of effects. The tokenizer family (T1) dominates all other choices by a large margin. The identity-versus-raw gap of +0.038 AUROC (+0.117 identity versus binned) is an order of magnitude larger than any data-treatment effect. Within the identity family, PID mode (T1-a) and dimension (T1-b) effects are at most ~ 0.004 AUROC, comparable to seed-spread. All three data-treatment axes produce effects below 0.003 AUROC in this experiment; none is distinguishable from noise with certainty at $n = 3$ seeds.

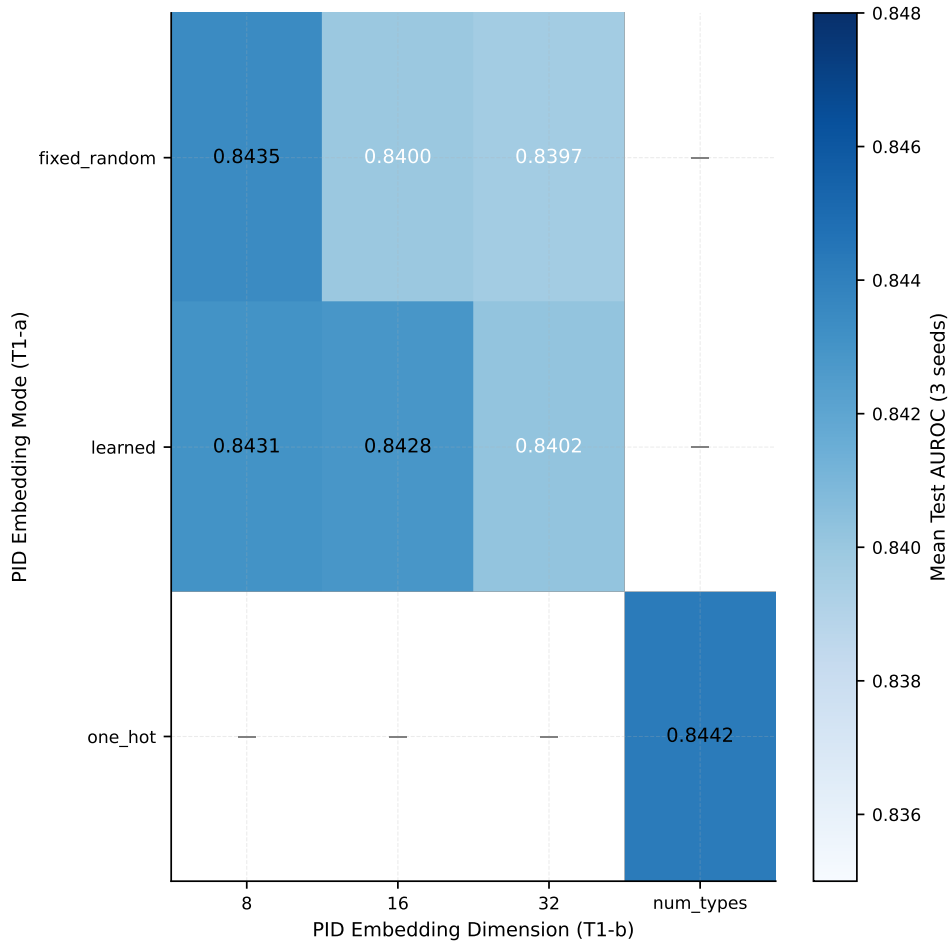


Figure 4.6: Mean test AUROC as a function of PID embedding mode (T1-a) and embedding dimension (T1-b), for the identity tokenizer. `one_hot` is invariant to dimension by construction (dimension is set to the number of particle types). All cells show similar performance; the total spread is ~ 0.004 AUROC.

The narrative arc of this chapter is therefore: *choose the identity tokenizer; thereafter, fine-grained choices do not matter at this model scale and seed count.*

Limitations. Three limitations are worth stating explicitly.

First, only three seeds per condition are available. The D03 ordering effect (-0.0026) is the largest of the data-treatment results and is suggestive of a real effect, but would require $n \geq 6$ per condition to approach 95% confidence given typical seed-spread variability of ~ 0.001 AUROC. The D01 and D02 results (< 0.001) are almost certainly null results, but cannot be proven so with the current seed count.

Second, the attribution of the D03 ordering effect to the sinusoidal positional encoding has not been confirmed. An `E1=none` control is the minimum experiment required. Similarly, the PE-disruption hypothesis for D02 is untested.

Third, Experiment 4B uses `T1-a=learned` and `T1-b=8` as the fixed tokenizer configuration, which is not the best-performing individual cell identified in Experiment 4A. The D02 and D03 effects may differ under `one_hot` tokenization; this cross-check has not been performed.

Chapter baseline. Table 4.4 records the axis values chosen as the baseline for all subsequent chapters. Choices follow the criterion of maximising expected performance while minimising hyperparameter sensitivity.

These six choices define the canonical input representation for the transformer workbench as used

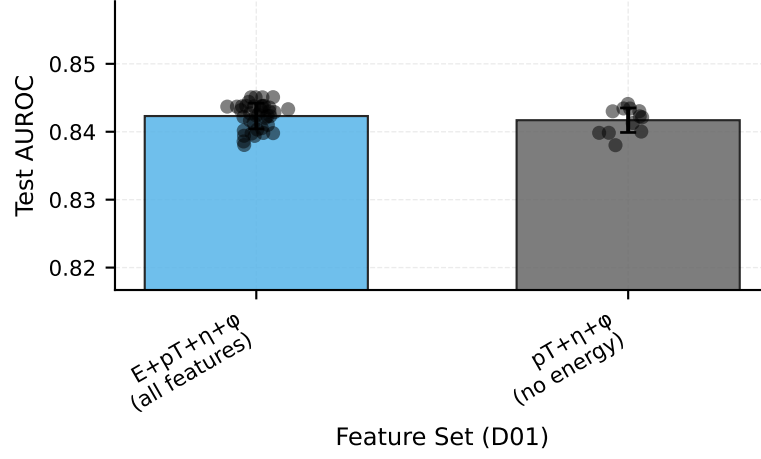


Figure 4.7: Mean test AUROC by feature set (D01) for the identity tokenizer. $[0, 1, 2, 3]$: full four-vector (E, p_T, η, ϕ) . $[1, 2, 3]$: three-vector (p_T, η, ϕ) without energy. The difference (-0.0007) is below the seed-spread level and constitutes a null result.

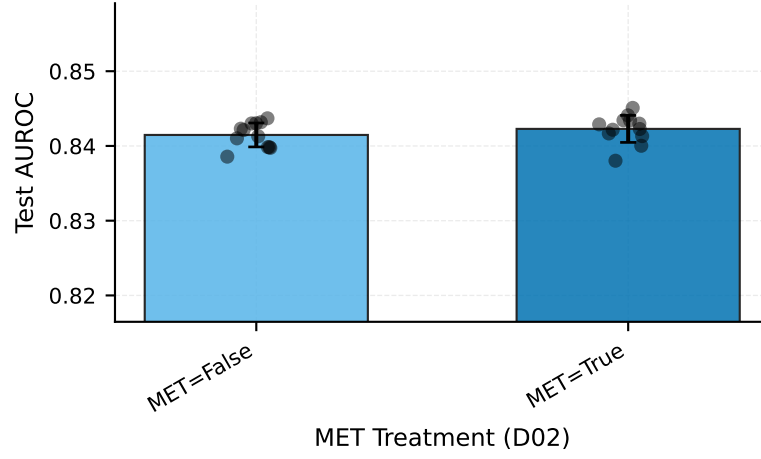


Figure 4.8: Mean test AUROC by MET treatment (D02), marginalised over D01 and D03 conditions. Including MET as a token yields a difference of $+0.0008$ AUROC, which is indistinguishable from noise at $n = 12$ per condition. This is a null result.

throughout the remainder of this thesis. Any chapter that modifies one of these axes re-runs the relevant comparison on top of this configuration.

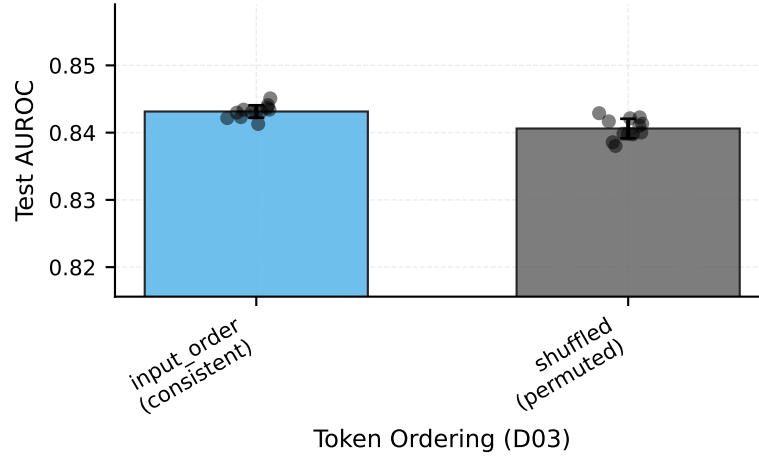


Figure 4.9: Mean test AUROC by token ordering (D03), for `D02 = false` conditions ($n = 6$ per bar). Shuffled token sequences yield -0.0026 AUROC relative to the original reconstruction order. The effect is consistent across feature-set conditions.

Axis	Config key	Chosen value	Justification
T1	<code>classifier.model.tokenizer.name</code>	<code>identity</code>	+0.038 AUROC over <code>raw</code> ; dominant effect robust at all dims; no trainable embedding weights
T1-a	<code>classifier.model.tokenizer.pid_mode</code>	<code>one_hot</code>	
T1-b	<code>classifier.model.tokenizer.id_embed_dim</code>	<code>num_types</code>	
D01	<code>data.cont_features</code>	<code>[0,1,2,3]</code>	null gap vs. <code>[1,2,3]</code> ; required for P1 (node-wise mass)
D02	<code>classifier.globals.include_met</code>	<code>false</code>	null result; re-examined via B1-G1 in Ch. 6
D03	<code>data.sort_tokens_by</code>	<code>input_order</code>	small but consistent gain over <code>shuffled</code>

Table 4.4: Chapter 4 baseline configuration carried forward to all subsequent chapters. Each axis is held at the value shown unless explicitly varied.

Physics-Informed Attention Biases

Standard self-attention assigns relevance to token pairs through learned query–key inner products, making no assumption about the physical relationships among the particles those tokens represent. In a four-tops-versus-background event, however, pairs of particles that share a decay vertex, are separated by a small angle, or interact through a specific Standard Model (SM) coupling are physically more related than arbitrary token pairs. This chapter investigates whether encoding such structure as additive biases on the attention logits improves classification performance, and what the learned bias weights reveal about how the model organises particle interactions.

Four bias families are available in the workbench. The *Lorentz-scalar bias* (B1-L) maps pairwise kinematic invariants to attention scores through a small sub-network. The *type-pair table* (B1-T) is a learnable $N_{\text{type}} \times N_{\text{type}}$ matrix that assigns a baseline attention score to every ordered pair of particle types, optionally initialised from SM coupling constants. The *SM interaction prior* (B1-S) encodes SM coupling topology as a fixed or learned additive bias parametrised by coupling constants. The *global-conditioned bias* (B1-G) modulates attention using a global event-level feature such as missing transverse energy (MET) direction. All families share a zero-initialised scalar gate, so adding any bias leaves the transformer output unchanged at initialisation; the gate must be driven away from zero for the bias to have any effect. All experiments in this chapter hold `D02` (`include_met`) at `true`: MET is always present, as motivated by the Ch. 4 findings, and is required for the B1-G family to operate.

The primary lens throughout this chapter is *interpretability*. Performance differences between bias configurations are expected to be small, and the main scientific contribution is understanding what structure the model learns to place on particle-pair relationships when given physics-motivated inductive biases. Section 5.1 establishes the performance envelope across all four families and examines attention maps as the central interpretability tool. Sections 5.2 through 5.4 drill into the internal design choices of each family, and Section 5.5 draws the chapter together.

5.1. Bias family comparison (Exp. 5A)

5.1.1. Experimental design

Experiment 5A asks whether any of the four bias families, applied individually or in full combination, measurably improves classification accuracy relative to an unbiased transformer baseline, and whether the full stack is super-additive. Eighteen runs cover six bias configurations at three seeds each: `none` (unbiased baseline), `lorentz_scalar`, `typepair_kinematic`, `sm_interaction`, `global_conditioned`, and all four combined. All other axes are held at the chapter baseline: embedding dimension 128 (H01), depth 3 (H02), 4 heads (H03), MLP dimension 256 (H04), tokeniser identity (T1), positional encoding `sinusoidal` (E1), and PID embedding mode `learned` with dimension 8.

5.1.2. Performance results

Table 5.2 reports seed-mean test AUROC and per-cell seed standard deviation ($n = 3$) for each configuration. The `none` baseline achieves 0.84125 ± 0.00068 .

Axis	Values	Levels
B1  — Bias selector	none lorentz_scalar typepair_kinematic sm_interaction global_conditioned lorentz+typepair+sm+global	6
Seeds (R5 )	42, 123, 314	3
Total		18

Table 5.1: Exp. 5A design. All other axes at chapter baseline: T1  = identity, D02  = true, E1  = sinusoidal, dim 128, depth 3, 4 heads.

B1 configuration	Mean AUROC	Seed std	Δ vs baseline
none (baseline)	0.84125	0.00068	—
lorentz_scalar	0.84156	0.00029	+0.00031
typepair_kinematic	0.84155	0.00080	+0.00030
sm_interaction	0.84080	0.00136	−0.00045
global_conditioned	0.84170	0.00091	+0.00045
lorentz+typepair+sm+global	0.83937	0.00090	−0.00188

Table 5.2: Exp. 5A seed-mean test AUROC ($n = 3$ seeds per cell). Per-cell seed standard deviation is indicative only; inter-cell range (≈ 0.002) is comparable to seed variance.

Three observations follow directly from these results. First, no single bias family moves the seed-mean AUROC by more than approximately one inter-seed standard deviation away from the `none` baseline. The three families that nominally improve accuracy (`lorentz_scalar`, `typepair_kinematic`, `global_conditioned`) each gain at most +0.00045 AUROC, which is smaller than the per-cell seed standard deviation of the baseline itself. Second, `sm_interaction` sits −0.00045 below baseline, within one seed standard deviation. Third, the full-combination cell is the only configuration whose mean (0.83937) sits visibly below the baseline, at a deficit of approximately 0.002 AUROC. This result is consistent with mild optimisation interference when multiple bias signals compete simultaneously for the same attention logit, though it cannot be cleanly attributed to any single component without further ablation.

The AUROC metric alone therefore does not reveal a meaningful benefit from physics-informed attention biases at this model scale and seed budget. The more informative question is whether the biases restructure attention towards physically motivated patterns even when the classification boundary shifts only modestly. The attention map analysis in Section 5.1.3 addresses this.

5.1.3. Attention structure: none versus full combination

Attention weights are extracted at inference time by forwarding 2000 validation events through each model with the `capture_attention` flag enabled, then averaging over all heads, all layers, and all three seeds. The physical token sub-block (18 particle tokens, positions 1–18 in the 21-token sequence; MET and MET- ϕ excluded) is projected onto a 7×7 type-pair grid by averaging attention from tokens of type i to tokens of type j for each ordered pair (i, j) across events. The seven physical types are: jet (1), b-jet (2), e^+ (3), e^- (4), μ^+ (5), μ^- (6), photon (7).

In the `none` baseline, type-pair attention weights occupy a near-uniform range of 0.059–0.113. No pair dominates: the largest cell ($\mu^+ \rightarrow e^-$: 0.113) and the smallest (inferred from the stated range) differ by only 0.054 units. The distribution is consistent with a model that has learned to aggregate broad context without privileging specific particle-type relationships.

The full-combination model produces a qualitatively different pattern. Table 5.3 lists the most strongly elevated and depressed type-pair cells. The attention range widens to 0.039–0.178, a factor of approximately 2.6 relative to the baseline range of 0.054. Critically, the elevated cells correspond to physically motivated interactions.

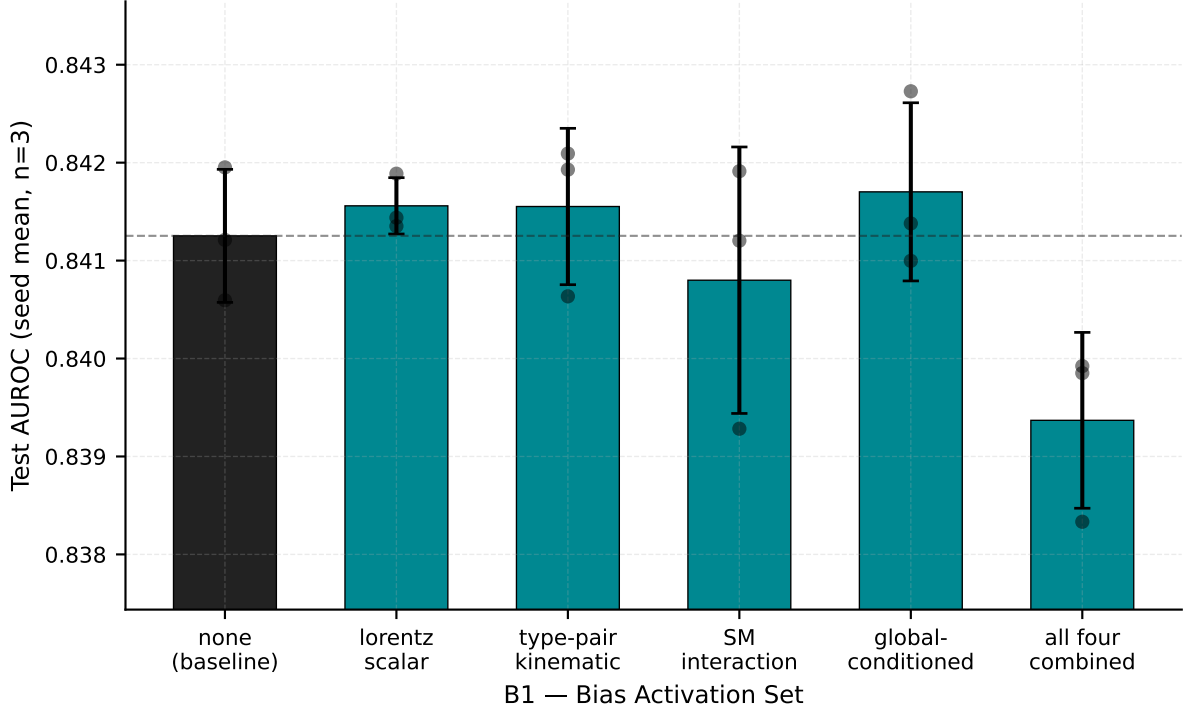


Figure 5.1: Exp. 5A seed-mean test AUROC for all six B1 configurations, with per-seed spread. Dashed line marks the none baseline mean.

The pattern is interpretable in HEP terms. Lepton–photon pairs ($\mu^- \rightarrow \gamma$: 0.178) are associated with final-state radiation or radiative W/Z decays. Photon self-attention (0.164) reflects the importance of photon isolation and cluster-shape information. The b -jet-to-muon cells (0.142, 0.133) directly encode the semi-leptonic b -decay topology relevant to top-quark reconstruction, where a b quark decays to a charm quark and a lepton. Opposite-sign lepton pairs ($e^- \rightarrow e^+$: 0.139, $e^+ \rightarrow e^-$: 0.129) are consistent with Z - or W -boson leptonic decays in the multi-top system. By contrast, same-charge mixed-lepton pairs ($\mu^- \rightarrow e^+$: 0.039) are strongly suppressed, as expected for pairs that share no direct SM vertex in the $t\bar{t}t\bar{t}$ decay chain.

These results constitute the primary interpretability finding of the chapter: the physics-informed biases do not substantially improve classification accuracy on this dataset at this model scale, but they do restructure the model’s attention towards physically motivated particle-type relationships. The model, when given structured priors through the bias families, converges to an attention pattern that is broadly consistent with HEP domain knowledge, even though the AUROC improvement is within seed noise.

The per-head, per-layer attention grids (available in the supplementary figures) show that the structured pattern is not uniform across all twelve head–layer combinations. Some heads retain near-uniform attention while others specialise strongly onto physically motivated pairs, consistent with the expectation that multi-head attention allows functional specialisation.

5.2. Lorentz feature set and bias network (Exp. 5B)

5.2.1. Experimental design

The Lorentz-scalar bias sub-network (B1-L) computes a pairwise attention score from Lorentz-invariant quantities derived from pairs of four-momenta. Experiment 5B asks three questions about the internal design of this sub-network: which kinematic features are most informative (B1-L1), whether replacing the standard pointwise MLP with a Kolmogorov–Arnold Network (KAN) improves the learned response (B1-L2), and whether a per-feature sparse gate (B1-L5) can learn to prune uninformative features automatically. Sixty runs span five feature sets, two MLP types, and two gating settings, at three seeds each.

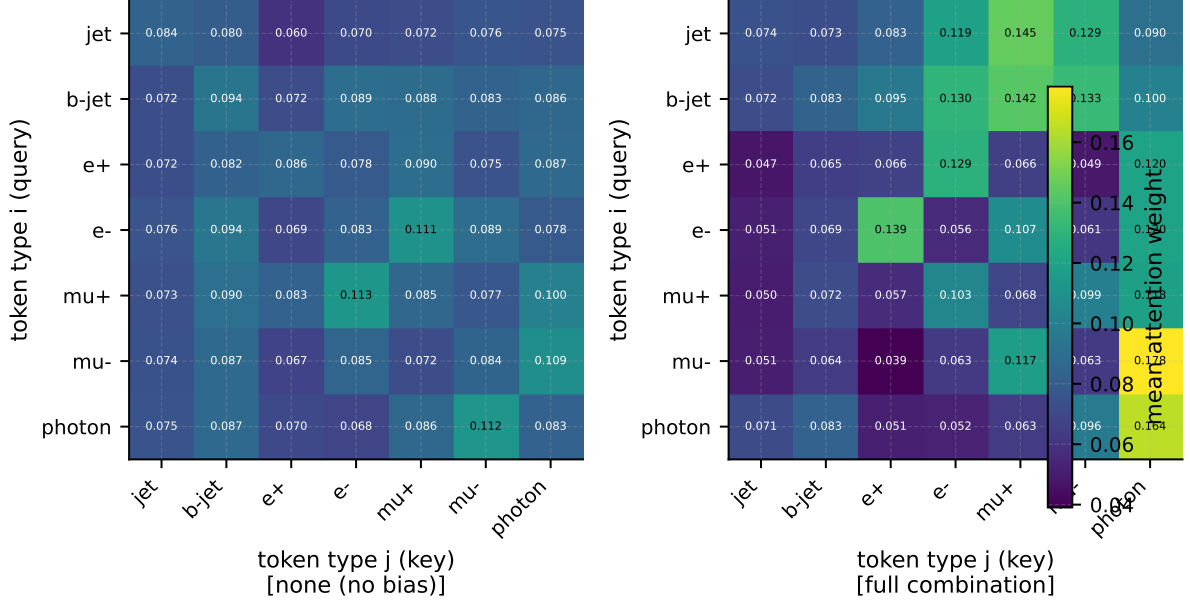


Figure 5.2: Type-pair mean attention weights for the `none` baseline (left) and full-combination model (right), averaged over all heads, layers, seeds, and 2000 validation events. Particle types: 1 = jet, 2 = b-jet, 3 = e^+ , 4 = e^- , 5 = μ^+ , 6 = μ^- , 7 = photon. Pad tokens excluded.

Pair ($i \rightarrow j$)	Mean attention	Physics interpretation
$\mu^- \rightarrow \gamma$	0.178	Lepton–photon (FSR / radiative return)
$\text{jet} \rightarrow \mu^+$	0.145	Jet-embedded muon (semi-leptonic b -decay)
$\gamma \rightarrow \gamma$	0.164	Photon self-attention (cluster isolation)
$b\text{-jet} \rightarrow \mu^+$	0.142	Direct $b \rightarrow \mu$ semi-leptonic
$e^- \rightarrow e^+$	0.139	Opposite-sign lepton pair (Z/W decay)
$b\text{-jet} \rightarrow e^-$	0.130	Semi-leptonic $b \rightarrow e$
$e^+ \rightarrow e^-$	0.129	Opposite-sign lepton pair
$\text{jet} \rightarrow \mu^-$	0.129	Jet-embedded muon
$b\text{-jet} \rightarrow \mu^-$	0.133	Semi-leptonic b -decay
$\mu^- \rightarrow e^+$	0.039	Same-charge mixed lepton (suppressed)
$e^+ \rightarrow \mu^-$	0.049	Same-charge mixed lepton (suppressed)

Table 5.3: Selected type-pair mean attention weights under the full-combination model (Exp. 5A), averaged over all heads, layers, seeds, and 2000 validation events. Pairs with mean attention ≥ 0.13 are elevated; pairs ≤ 0.05 are depressed.

All kinematic features are computed from z-score-normalised four-vectors. An important consequence of this normalisation is that the pairwise invariant mass m^2 collapses near zero on the normalised scale (validation median of m^2 is 0.000; 95th percentile is 2.47). The logarithmic variant $\log m^2$ collapses further (validation median $\approx \log(10^{-6}) \approx -13.8$). This means that any claim about “learning the W -boson or top-quark mass peak” is not supported by these checkpoints; such an analysis would require either removing input normalisation or de-normalising the feature axis. By contrast, ΔR , $\log k_T$, and z remain well-behaved on the normalised scale (ΔR at the 5th and 95th percentiles: 0.79 and 4.17 respectively), and interpretability arguments based on those features are on firmer ground.

5.2.2. AUROC results

Table 5.5 reports seed-mean AUROC for all 20 (feature set, MLP type, gating) cells. The full bar chart is shown as Figure 5.4.

Three consistent trends emerge from the table. First, within every (feature-set, gating) pairing, the `kan` MLP equals or exceeds the `standard` MLP in seed-mean AUROC. The largest gap appears in the six-feature, gating-off configuration (+0.0018 AUROC), which equals approximately one seed standard

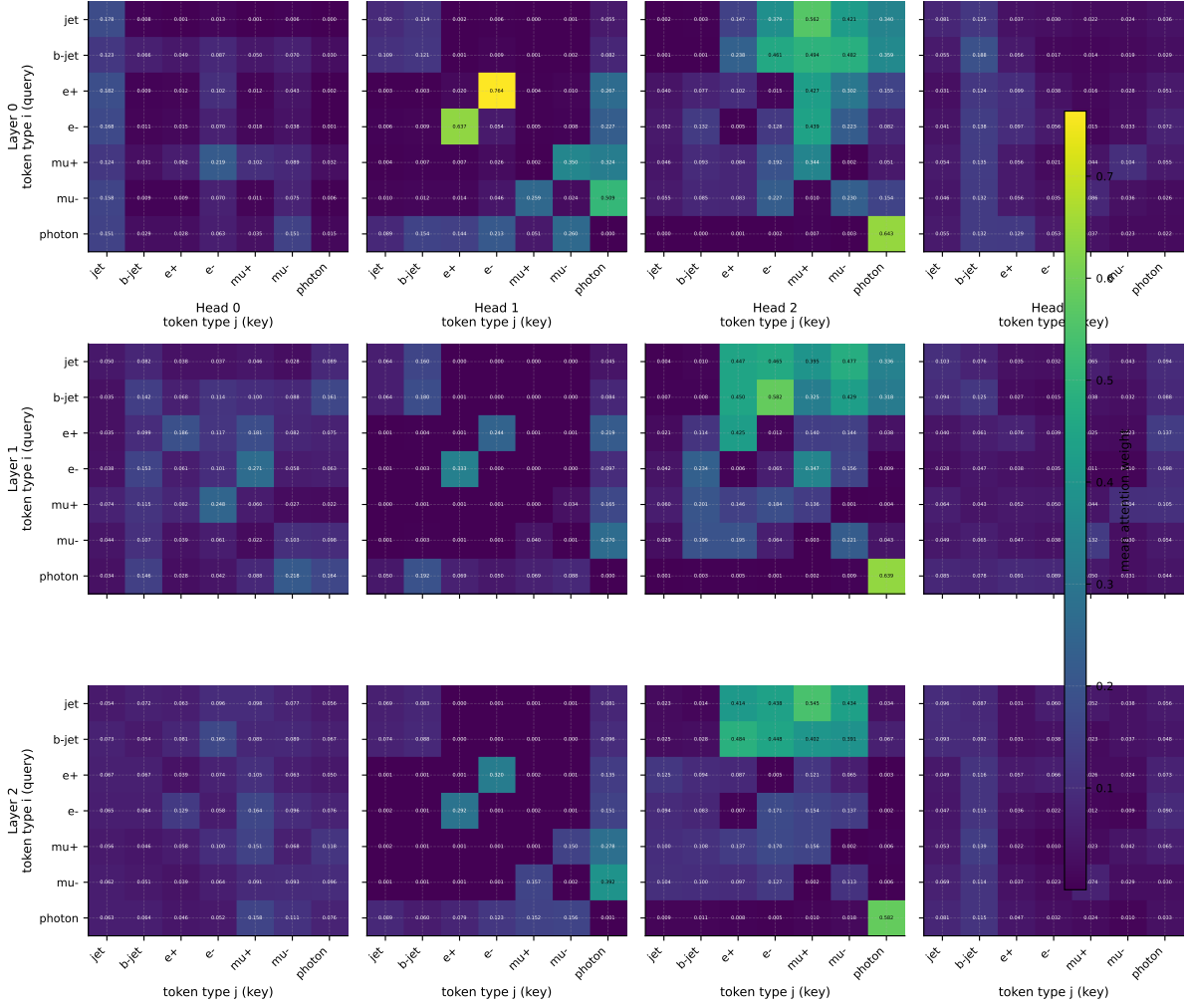


Figure 5.3: Type-pair attention weights for the full-combination model (seed 42), shown separately for each of the 4 heads $\times$ 3 encoder layers. Some heads specialise onto physically motivated pairs while others retain near-uniform attention.

deviation. The global best cell (6-feature, KAN, gate off: 0.84436) sits roughly $+0.0031$ above the Exp. 5A none baseline (0.84125), a margin comparable to several per-cell seed standard deviations. Second, richer Lorentz feature panels systematically produce higher AUROC than single-feature panels, particularly when paired with the KAN. The four- and six-feature sets consistently outperform the ΔR -only and m^2 -only single-feature sets; this trend is visible in both the standard and KAN columns. Third, sparse gating provides no benefit at this scale and seed budget: gate-on cells tend to sit slightly below their gate-off counterparts in the same (feature, MLP) row, by approximately 0.0005–0.0009 AUROC.

5.2.3. KAN spline structure and gate values

The expressive difference between KAN and standard MLP is visible not only in AUROC but in the learned attention-bias functions themselves. In the single-feature (ΔR) configuration with gating off, the seed-mean bias function spans a range of 0.99 arbitrary units under the KAN, versus 0.46 under the standard MLP: a factor of approximately two. In the partial-dependence view of the four-feature configuration (other features held at validation medians), the standard MLP learns almost no $\log k_T$ dependence (range ≈ 0.05), while the KAN learns a structured curve with range ≈ 0.65 . This result is consistent with the KAN’s capacity to model sharp univariate edges with a small parameter budget through its B-spline basis. The KAN therefore extracts more functional structure from the pairwise kinematic features, which is reflected in the consistent AUROC advantage.

The per-module gate, $\tanh(g)$, converged to a positive value in every one of the 60 Exp. 5B runs, confirming that the Lorentz bias is non-trivially active throughout the cohort. KAN modules average

Axis	Values	Levels
B1-L1 ⚡⚡⚡ — Feature set	[deltaR] [m2] [m2, deltaR] [log_kt, z, deltaR, log_m2] [m2, deltaR, log_m2, log_kt, z, deltaR_ptw]	5
B1-L2 ⚡⚡⚡ — MLP type	standard, kan	2
B1-L5 ⚡⚡⚡ — Sparse gating	false, true	2
Seeds (R5 ⚡⚡⚡)	42, 123, 314	3
Total		60

Table 5.4: Exp. 5B design. B1 = lorentz_scalar throughout. B1-L3 (hidden dim) and B1-L4 (per-head mode) are held at defaults.

Feature set	MLP type	Gating	Mean AUROC	Seed std
ΔR	standard	off	0.84126	0.00123
ΔR	kan	off	0.84245	0.00089
ΔR	standard	on	0.84155	0.00126
ΔR	kan	on	0.84206	0.00044
m^2	standard	off	0.84131	0.00142
m^2	kan	off	0.84154	0.00159
m^2	standard	on	0.84114	0.00118
m^2	kan	on	0.84117	0.00118
$m^2, \Delta R$	standard	off	0.84114	0.00042
$m^2, \Delta R$	kan	off	0.84256	0.00063
$m^2, \Delta R$	standard	on	0.84100	0.00104
$m^2, \Delta R$	kan	on	0.84190	0.00022
$\log k_T, z, \Delta R, \log m^2$	standard	off	0.84200	0.00085
$\log k_T, z, \Delta R, \log m^2$	kan	off	0.84399	0.00101
$\log k_T, z, \Delta R, \log m^2$	standard	on	0.84191	0.00034
$\log k_T, z, \Delta R, \log m^2$	kan	on	0.84311	0.00076
$m^2, \Delta R, \log m^2, \log k_T, z, \Delta R/p_T$	standard	off	0.84257	0.00106
$m^2, \Delta R, \log m^2, \log k_T, z, \Delta R/p_T$	kan	off	0.84436	0.00214
$m^2, \Delta R, \log m^2, \log k_T, z, \Delta R/p_T$	standard	on	0.84170	0.00065
$m^2, \Delta R, \log m^2, \log k_T, z, \Delta R/p_T$	kan	on	0.84378	0.00107

Table 5.5: Exp. 5B seed-mean test AUROC ($n = 3$ per cell). The none baseline from Exp. 5A is 0.84125 ± 0.00068 . Best cell highlighted. Gating “on” = B1-L5 ⚡⚡⚡ = true.

$\tanh(g) \approx 0.28 \pm 0.08$ versus 0.22 ± 0.10 for standard modules. The higher gate for KAN runs echoes the AUROC finding: the network allows more of the KAN’s richer signal through its multiplicative gate.

Despite the gate being consistently open, the sparse feature-gating mechanism (B1-L5 ⚡⚡⚡) does not learn meaningful per-feature selection. The sigmoid gate values $\sigma(\text{feature_gates})$ remain within 0.49–0.62 across all 15 sparse-on KAN runs, barely departing from their initialisation at $\sigma(0) = 0.5$. The only feature whose mean gate drifts slightly below 0.5 is $\log m^2$, which is precisely the feature whose input distribution collapses near -13.8 under z-score normalisation. Even this drift is modest ($\approx 0.494 \pm 0.022$), and no feature is pruned in any run. The effect on AUROC is therefore a slight attenuation of every feature’s amplitude simultaneously, with no compensating benefit from pruning, which explains the small but systematic gap between gate-on and gate-off cells.

One caveat for the spline-shape analysis is that the KAN B-spline grid is fixed at its initialisation throughout training: `update_grid` is not called during the forward pass, which was confirmed by state-dict inspection of the `mlp.0.grid` key in the saved checkpoints. The spline expressivity is therefore bounded by the initial grid resolution and not by adaptive grid relocation. More adaptive grid training could in principle widen the gap further.

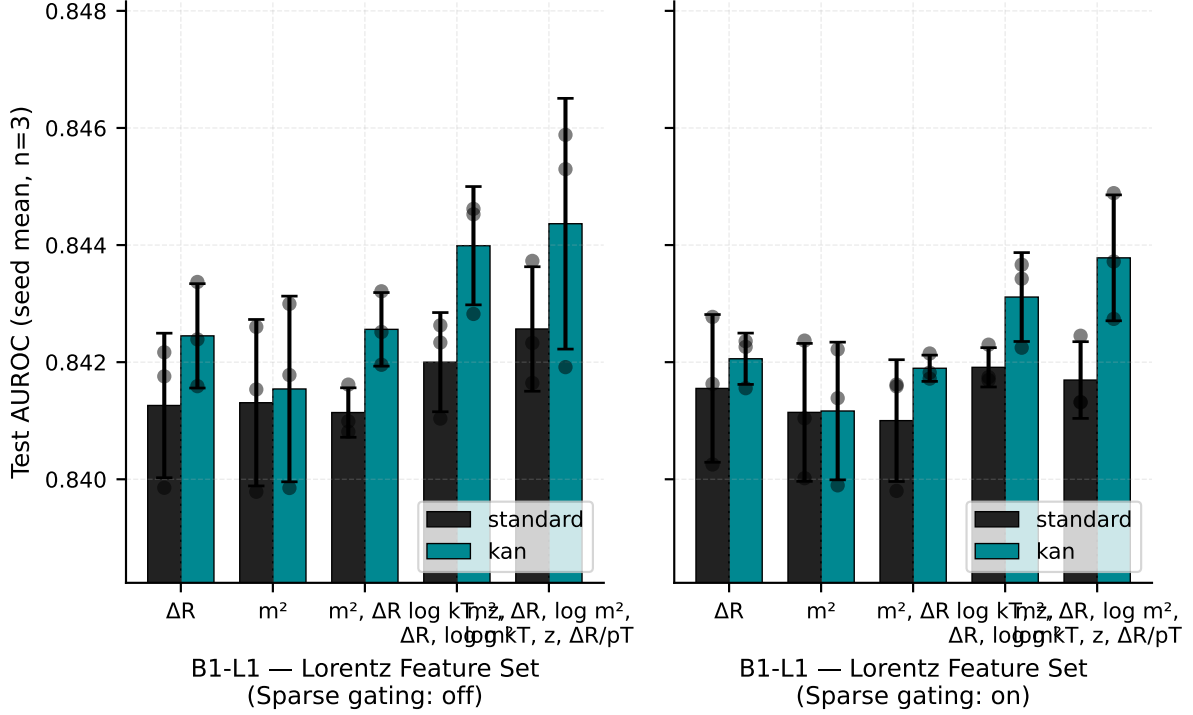


Figure 5.4: Exp. 5B seed-mean test AUROC for all 20 (feature set, MLP type, gating) cells, with per-seed spread. KAN consistently meets or exceeds the standard MLP; richer feature sets produce higher AUROC; sparse gating provides no benefit.

5.3. Type-pair initialisation and freeze (Exp. 5C)

5.3.1. Experimental design

The type-pair kinematic bias (B1-T) maintains a learnable $N_{\text{type}} \times N_{\text{type}}$ table of attention score offsets, one per ordered particle-type pair. Experiment 5C varies two design choices: the initialisation strategy (B1-T1) and whether the table is frozen after initialisation (B1-T2). Three initialisation modes are tested: *none* (zero start, free to train), *binary* (a $\{0, -5\}$ mask encoding which SM vertices are allowed or forbidden), and *fixed_coupling* (the physics-motivated initialisation, with magnitudes proportional to SM coupling constants). A frozen table (`freeze_table = true`) acts as a fixed physics prior; a free table is adjusted by gradient descent from its initialised state. Fifteen runs cover five cells (the *none*–frozen combination is excluded, as there is no table to freeze) at three seeds each.

Axis	Values	Levels
B1-T1 — Init	<code>none</code> , <code>binary</code> , <code>fixed_coupling</code>	3
B1-T2 — Freeze	<code>false</code> , <code>true</code> (only when init $\neq$ <code>none</code>)	—
Seeds (R5)	42, 123, 314	3
Total		15

Table 5.6: Exp. 5C design. B1 = `typepair_kinematic` throughout. B1-T3 (kinematic gate), B1-T4 (kinematic feature), and B1-T5 (mask value) are held at defaults.

5.3.2. Performance results

Table 5.7 reports seed-mean AUROC for the five cells. The local *none*-init baseline here (0.84205) is slightly above the Exp. 5A *none* cell (0.84125), a difference attributable to the different sweep configuration and cohort, not to a change in the architecture.

All four physics-initialised cells sit below the *none* baseline in seed mean, by margins of 0.0003–0.0013 AUROC. These deltas are within one to two per-cell seed standard deviations and should not be interpreted as a meaningful performance deficit. The *fixed_coupling* initialisation consistently outperforms

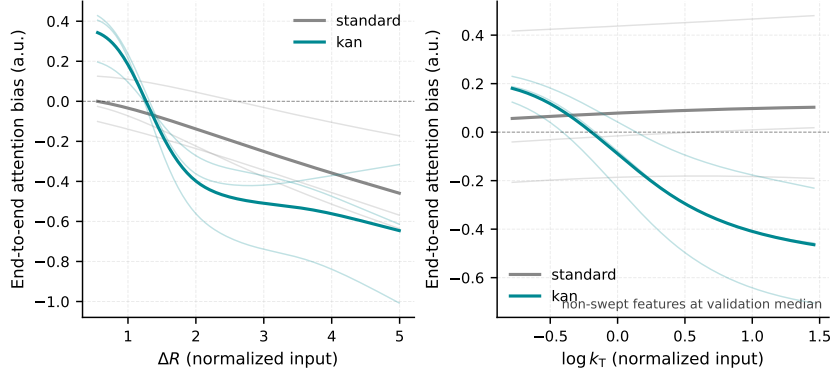


Figure 5.5: Learned 1D bias function for the ΔR -only feature set comparing KAN (blue) and standard MLP (orange), averaged over three seeds. The KAN spans a range of ≈ 0.99 attention units versus ≈ 0.46 for the standard MLP.

Init	Freeze	Mean AUROC	Seed std	Δ vs none
none	free	0.84205	0.00081	—
binary	free	0.84075	0.00046	−0.00130
binary	frozen	0.84131	0.00065	−0.00074
fixed_coupling	free	0.84175	0.00102	−0.00030
fixed_coupling	frozen	0.84138	0.00046	−0.00067

Table 5.7: Exp. 5C seed-mean test AUROC ($n = 3$ per cell). All four physics-initialised cells sit below the none baseline within one to two seed standard deviations.

binary by approximately 0.0006–0.0010 AUROC regardless of freeze state, but again at the noise floor. Freezing the table has small, non-monotonic effects: it slightly improves binary (+0.00056) and slightly reduces fixed_coupling (−0.00037), both within seed variance.

The AUROC results therefore suggest neither a benefit from SM-seeded initialisation nor a cost from freezing the physics prior, at this seed budget and model scale.

5.3.3. Learned table structure

The more informative view comes from inspecting the converged type-pair tables directly. The state-dict key `bias_composer.bias_modules.typepair_kinematic.table_raw` has shape $[8, 8]$; the symmetric form is reconstructed as $\frac{1}{2}(T + T^T)$ and the padding row and column are removed, leaving a 7×7 active matrix. Table 5.8 summarises key scalar statistics.

Init	Freeze	Gate ($\tanh g$, mean $\pm$ std)	Frobenius norm	Drift from init
none	free	0.234 ± 0.015	0.434	0.944 (= norm)
binary	free	0.097 ± 0.012	27.044	0.868
binary	frozen	0.094 ± 0.013	27.295	0.000 (frozen)
fixed_coupling	free	0.104 ± 0.015	26.848	0.930
fixed_coupling	frozen	0.101 ± 0.015	27.094	≈ 0

Table 5.8: Exp. 5C per-group learned-table statistics. Gate = $\tanh(g)$ averaged over 3 seeds. Frobenius norm computed on the 7×7 active sub-matrix. Drift from init = Frobenius distance between the converged seed-mean table and the reference initialisation.

The none-init table starts from zero and converges to a small-magnitude matrix (Frobenius 0.43), with values scattered in $[-0.16, +0.12]$ showing no discernible SM-interaction pattern at $n = 3$ seeds. Its gate is notably higher (0.234) than any physics-initialised group (0.094–0.104), consistent with the network opening the gate further when the table provides a weak prior and all type pairs look equally neutral at initialisation.

The fixed_coupling-free table preserves the SM structure through training. QCD quark–quark pairs (jet–jet: +1.22, jet–b-jet: +1.17, b-jet–b-jet: +1.23) retain large positive values; EW lepton pairs (e^+e^- :

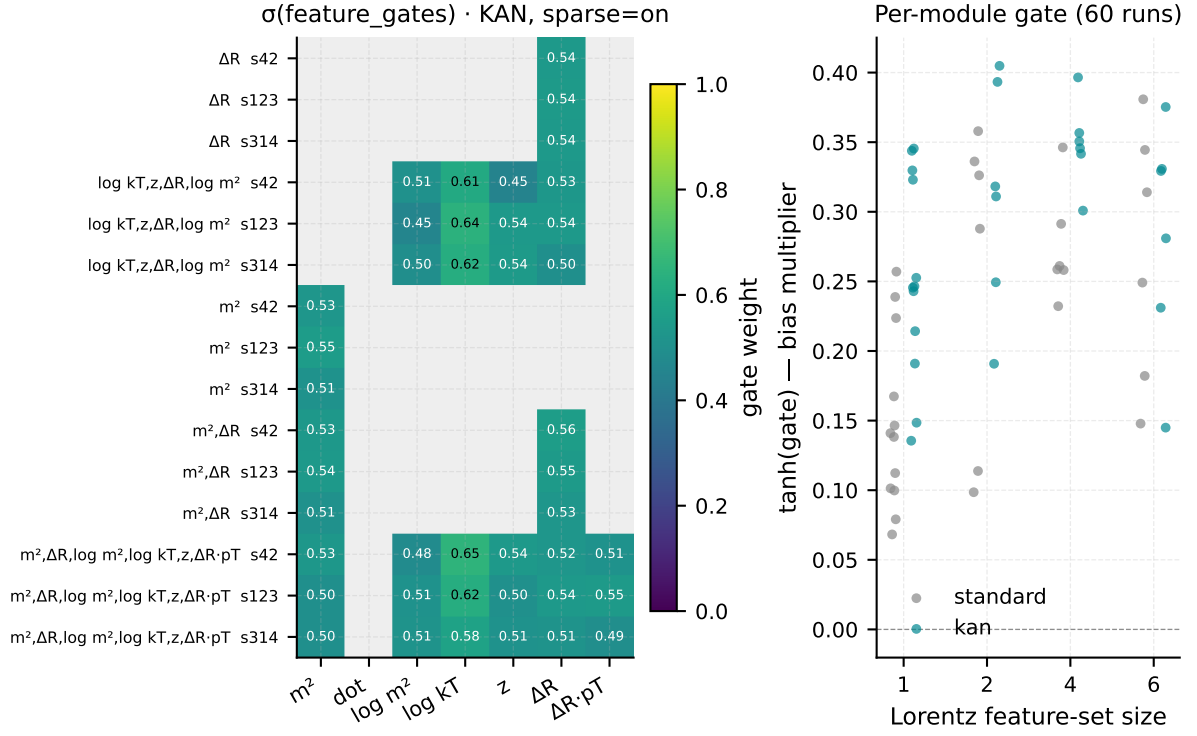


Figure 5.6: Left: per-feature sparse gate values $\sigma(\text{gate})$ for all Exp. 5B sparse-on KAN runs; all gates remain within 0.49–0.62, indicating no meaningful feature pruning. Right: module-level scalar gate $\tanh(g)$ for KAN (blue) and standard (orange) MLP runs across the full cohort.

+0.54, $\mu^+\mu^-$: +0.89) retain intermediate positive values; and all non-interacting pairs remain near the mask value of -5.0 . The Frobenius drift from initialisation is 0.930, indicating that training adjusts the coupling-constant magnitudes by roughly ± 0.1 – 0.3 rather than restructuring the topology. This is a notable result: the network broadly respects the SM interaction hierarchy encoded at initialisation, even when the gradient is free to reorganise the table entirely.

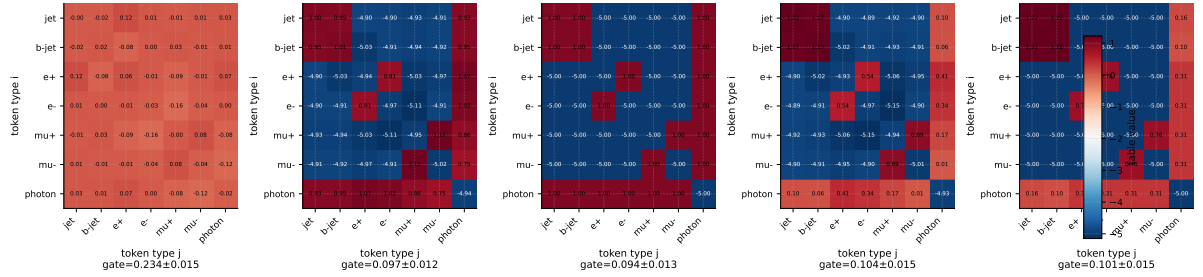


Figure 5.7: Converged type-pair tables (7 × 7 active sub-matrix) for all five Exp. 5C groups, averaged over three seeds. The fixed_coupling-free table preserves the SM coupling hierarchy; the none-init table remains near-uniform.

The binary-free group shows an analogous pattern (same mask positions preserved) but with higher drift (0.868 vs 0.930), reflecting the fact that the binary initialisation provides structural topology but no coupling-constant scale, leaving active entries more exposed to gradient learning. The frozen groups confirm the mechanism functions correctly (freeze drift $\approx 6.6 \times 10^{-8}$ for fixed_coupling-frozen).

Taken together, the results suggest that the SM fixed_coupling initialisation provides a structural prior that training broadly preserves, but that its effective attention contribution, modulated by the scalar gate ($\tanh(g) \approx 0.10$), is too small to produce a measurable AUROC improvement at this configuration. The none-init group, starting from a flat table, fails to recover a physics-consistent structure at three seeds — the table remains near-uniform, indicating that the inductive bias from the initialisation is important for guiding the table to a structured solution.

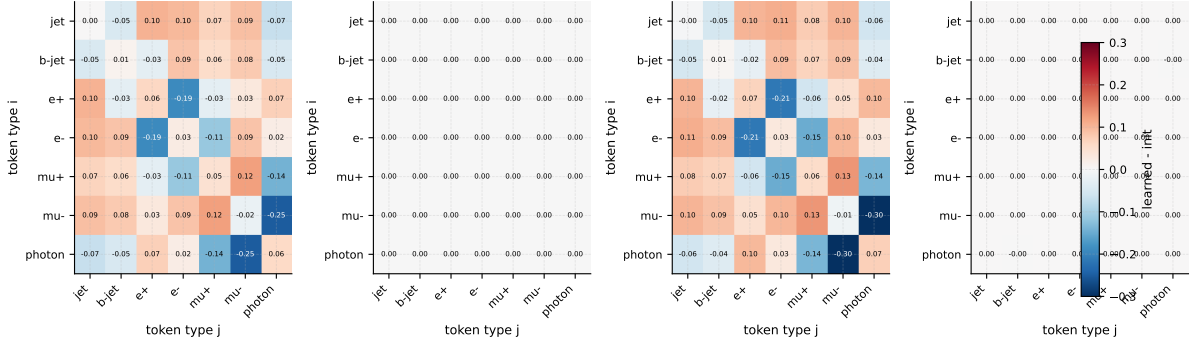


Figure 5.8: Per-cell table drift from initialisation for the free-table groups. `fixed_coupling-free` (top) shows small, structured adjustments (± 0.1 – 0.3) with the SM topology intact; `binary-free` (bottom) shows larger drift at active entries.

5.4. SM interaction mode (Exp. 5D)

Experiment 5D tests whether increasing physical fidelity in the SM interaction prior produces measurable gains. The `sm_interaction` bias family encodes SM topology as a fixed or modulated additive bias on attention logits; axis B1-S1  selects among three modes: `binary` (a $\{0, -5\}$ allowed/forbidden mask), `fixed_coupling` (coupling constants at a fixed energy scale), and `running_coupling` ($\alpha_s(Q^2)$ evaluated at the pairwise momentum transfer). Nine runs cover all three modes at three seeds each.

Axis	Values	Levels
B1-S1  — SM mode	<code>binary</code> , <code>fixed_coupling</code> , <code>running_coupling</code>	3
Seeds (R5 )	42, 123, 314	3
Total		9

Table 5.9: Exp. 5D design. B1 = `sm_interaction` throughout. B1-S2 (mask value) held at default.

Table 5.10 reports seed-mean AUROC. The reference `none` baseline from Exp. 5A (0.84125 ± 0.00068) is used for comparison, as Exp. 5D does not include an explicit `none` cell; both cohorts share the same encoder hyperparameters and seed list, so the cross-cohort comparison is reasonable.

B1-S1 mode	Mean AUROC	Seed std	Δ vs 5A <code>none</code>
<code>binary</code>	0.84140	0.00050	+0.00015
<code>fixed_coupling</code>	0.84146	0.00091	+0.00021
<code>running_coupling</code>	0.84108	0.00081	−0.00018

Table 5.10: Exp. 5D seed-mean test AUROC ($n = 3$ per cell). All three modes are within ± 0.0002 AUROC of the Exp. 5A `none` baseline.

All three SM-interaction modes sit within ± 0.0002 AUROC of the `none` baseline, and the three modes span only ≈ 0.0004 AUROC among themselves. Both magnitudes are at or below the per-cell seed standard deviation, making no mode distinguishable from another within this seed budget. The nominal ranking (`fixed_coupling` > `binary` > `running_coupling`) is consistent with the `running_coupling` mode being the most parameter-rich and therefore most exposed to optimisation noise at fixed hyperparameters, but this interpretation should not be treated as a directional finding at $n = 3$ seeds.

The SM interaction prior as implemented in Exp. 5D is therefore a null result on the AUROC axis at this model scale. Whether the prior is active in the attention pattern is a question for a future attention-map follow-up analogous to the Exp. 5A analysis (currently staged but not yet run). For the purpose of the downstream best-model selection in Ch. 8, the `fixed_coupling` mode is carried forward as the default SM setting on the grounds of physical motivation, accepting that the performance difference is negligible.

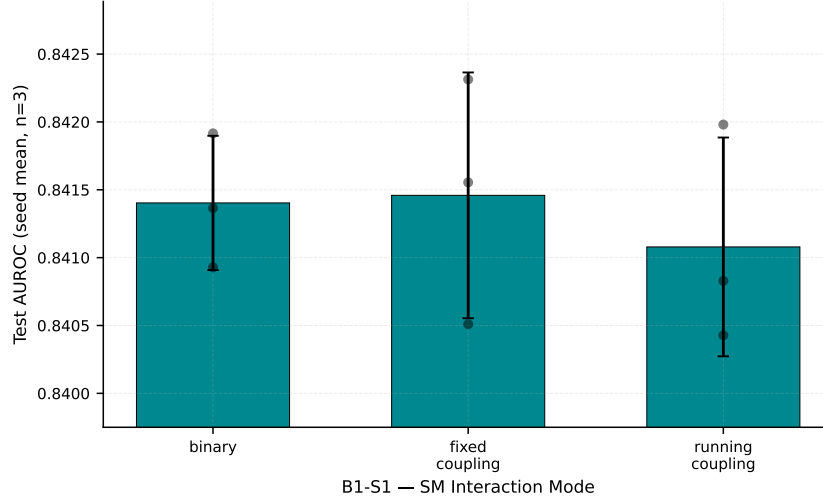


Figure 5.9: Exp. 5D seed-mean test AUROC for the three B1-S1 modes, with per-seed spread. Dashed line marks the Exp. 5A `none` baseline. All three modes are within ± 0.0002 AUROC of the baseline.

5.5. Chapter synthesis

This chapter set out to ask whether physics-informed attention biases improve 4-tops-versus-background classification, and what those biases reveal about the model’s internal organisation of particle relationships. The AUROC results across all four sub-experiments are consistent and unambiguous: no single bias family, and no configuration within any family, produces a performance gain that is clearly separable from seed variance at $n = 3$ seeds per cell. The total AUROC range across all 102 runs is approximately 0.838–0.846, with per-cell seed standard deviations of order 0.001. The full-combination configuration in Exp. 5A is the only cell that sits visibly below the `none` baseline (≈ -0.002 AUROC), suggestive of optimisation interference from multiple simultaneous bias signals.

The interpretability results tell a richer story. When all four bias families are active, the model sharpens type-pair attention onto physically motivated particle relationships: the attention range widens by a factor of 2.6 (from 0.054 in the `none` baseline to 0.139 in the full-combination model), and the most elevated pairs are lepton–photon ($\mu^- \rightarrow \gamma$: 0.178), photon self-attention (0.164), b-jet-to-muon pairs consistent with semi-leptonic b -decay (0.142, 0.133), and opposite-sign lepton pairs consistent with Z/W decay ($e^- \rightarrow e^+$: 0.139). Same-charge mixed-lepton pairs, which lack a common SM vertex, are strongly suppressed. These patterns demonstrate that physics-informed biases restructure the model’s attention rather than merely adding capacity.

Within the Lorentz-scalar family, KAN sub-networks consistently learn richer bias functions than standard MLPs across the same feature sets, with the largest gap at the six-feature configuration (+0.0018 AUROC, bias-function range $2\times$ wider). Sparse feature gating, however, converges to the all-open state ($\sigma \approx 0.5$) in every run, providing no automatic feature selection at this training budget.

The type-pair table analysis shows that the SM `fixed_coupling` initialisation provides a structural prior that training broadly respects: QCD quark–quark and EW lepton-pair entries retain their physics-consistent magnitudes through gradient descent, with Frobenius drift 0.930 from the initialisation. By contrast, a zero-start (`none-init`) table fails to recover a physics-consistent structure at $n = 3$ seeds, suggesting that the inductive bias from a physics-motivated initialisation is beneficial for interpretability even when it does not shift accuracy.

The practical takeaway for the remainder of the thesis is that physics-informed attention biases do not change the operating point of the classifier at the model scale studied in this chapter. Their value is interpretive: they make the model’s attention patterns legible in HEP terms. For downstream experiments in Ch. 6 and Ch. 8, the chapter baseline carries forward the `lorentz_scalar` bias in the six-feature, standard-MLP, gating-off configuration as the default B1 setting, on the grounds that it provides the most consistent, modest AUROC benefit without the optimisation risk of the full-combination stack.

Experiment	Primary axes	Runs
5A — Family comparison	B1	18
5B — Lorentz ablation	B1-L1, B1-L2, B1-L5	60
5C — Type-pair table	B1-T1, B1-T2	15
5D — SM interaction mode	B1-S1	9
Total		102

Table 5.11: Chapter 5 run summary.

6

Attention Mechanisms

The attention mechanism is the only component that simultaneously relates every pair of particle tokens. Chapters 4 and 5 asked how to enrich the information arriving at the attention layer; this chapter asks whether the attention computation itself should be modified. Three axes are investigated: the *type* of attention (A3 ) , an optional *attention-internal normalisation* (A4 ) , and the *mode in which a physics bias interacts with the two branches of differential attention* (A3-a ) . The Lorentz-scalar bias axis B1 ) from Chapter 5 reappears here as a co-factor, because the interaction between physics bias and attention type is the primary scientific question of Section 6.3.

The physics motivation is direct. In a multi-top event, only a small subset of the 10–20 reconstructed objects are decay products of the four top quarks; standard softmax attention distributes probability mass across all token pairs, including unrelated soft objects. Differential attention [5] suppresses diffuse attention by computing the normalised difference of two softmax maps, providing a structurally motivated inductive bias for signal concentration. Whether this mechanism transfers from the natural-language setting to particle classification is a testable hypothesis addressed in Section 6.2.

Two experiments were run. Experiment 6A establishes the headline comparison across attention type (A3 ) , internal normalisation (A4 ) , and the presence of a Lorentz-scalar bias (B1 ) . Experiment 6B is a targeted follow-up on how the Lorentz-scalar bias should interact with the two differential attention branches (A3-a ) , with attention type and normalisation fixed at the values recommended by Exp. 6A. Section 6.5 synthesises the findings and states the recommended configuration carried forward to subsequent chapters.

6.1. Differential Attention

Standard multi-head self-attention computes, for each head h , a single softmax map over scaled dot products:

$$\text{Attn}_h(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_h}}\right) \mathbf{V}_h. \quad (6.1)$$

The output mixes content from all positions weighted by the softmax distribution, which is strictly positive and therefore cannot cancel spurious correlations. Differential attention [5] replaces this with the difference of two independent softmax maps computed on non-overlapping halves of the query and key projections:

$$\text{DiffAttn}_h(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = [\text{softmax}(\mathbf{Q}_{h1} \mathbf{K}_{h1}^\top / \sqrt{d_h}) - \lambda_h \text{softmax}(\mathbf{Q}_{h2} \mathbf{K}_{h2}^\top / \sqrt{d_h})] \mathbf{V}_h, \quad (6.2)$$

where $\lambda_h \in \mathbb{R}^+$ is a per-head scalar initialised to a small positive value and learned during training. The subtraction cancels attention patterns common to both maps, amplifying the response to pairs that are relevant to one map but not the other. The mechanism doubles the query and key projection cost per head but keeps value dimension and model depth unchanged; the additional parameter burden is minimal (one scalar λ_h per head per layer).

In the particle-physics context, the differential mechanism is expected to assign near-zero net attention weight to object pairs that are kinematically inert with respect to the classification decision, while concentrating weight on pairs with discriminating invariant-mass or angular structure. The Lorentz-scalar bias (B1) provides complementary guidance by adding a physics-motivated additive term to the attention logits before the softmax; the interaction between the two is a central question of this chapter.

6.2. Attention Type and Internal Normalisation

6.2.1. Experimental Design

Experiment 6A swept A3 (attention type: standard or differential), A4 (attention-internal normalisation: none, layernorm, or rmsnorm), and B1 (bias: none or lorentz_scalar), with all other axes fixed at the chapter baseline.

Axis	Values	Configurations
A3 — Attention type	standard, differential	2
A4 — Internal normalisation	none, layernorm, rmsnorm	3
B1 — Lorentz-scalar bias	none, lorentz_scalar	2
Seeds	42, 123, 314	3
Total		36

Table 6.1: Exp. 6A design. All other axes held at the chapter baseline. `lorentz_scalar` uses the feature set recommended in Chapter 5. Axis A3-a is fixed to shared throughout Exp. 6A.

The 36 runs span all $2 \times 3 \times 2$ cells with 3 seeds each. Performance was evaluated using the mean one-vs-rest AUROC across all five signal classes (`eval_v2/test_auroc`), as defined in Chapter 2. Wall-clock time per epoch was also recorded to quantify the overhead of the differential attention pass.

6.2.2. Results: Attention Type

Figure 6.1 shows the seed-averaged AUROC for all 36 Exp. 6A configurations, grouped by attention type. Differential attention consistently outperforms standard attention across all normalisation and bias conditions. Marginalised over all 45 chapter-6 runs, the mean AUROC of A3 = differential is 0.831 ± 0.003 ($n = 27$) versus 0.827 ± 0.003 ($n = 18$) for standard, a difference of $+0.0036$ absolute ($+0.36$ pp). The gap is small but present with consistent sign across seed replicas.

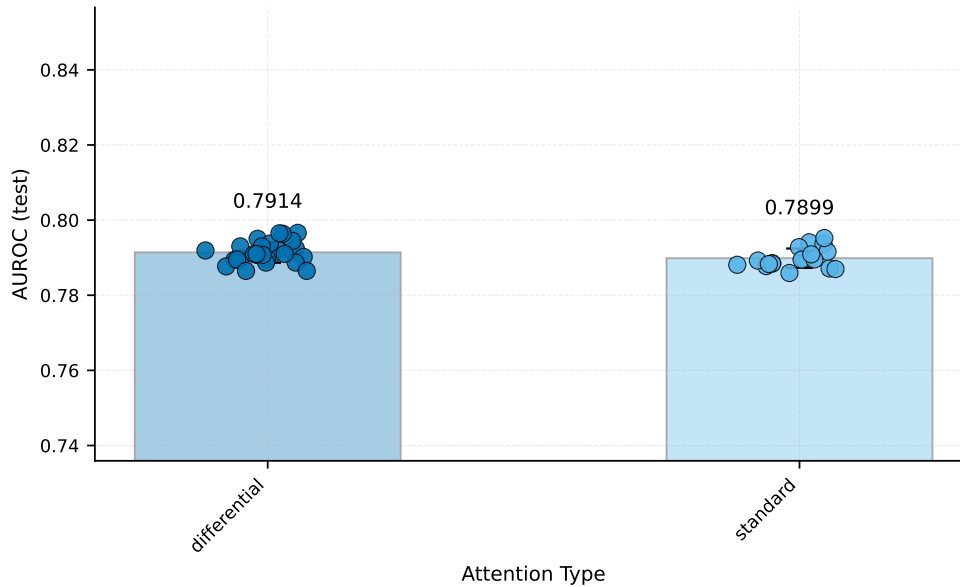


Figure 6.1: Seed-spread AUROC for A3 attention types across all Exp. 6A configurations. Each point is one seed; horizontal bars show the per-configuration mean. Differential attention (blue) sits systematically above standard attention (orange) for every normalisation and bias combination.

The per-class decomposition in Figure 6.2 confirms the gain is not concentrated in a single class. Differential attention improves AUROC for all five signal categories, with the largest absolute gain in the $t\bar{t}H$ channel (+0.0045) and the smallest in $t\bar{t}WW$ (+0.0034). The $t\bar{t}WW$ channel is consistently the hardest class (AUROC ≈ 0.737 for standard and 0.738 for differential), reflecting the kinematic similarity between four-top and double- W -associated production at high jet multiplicity.

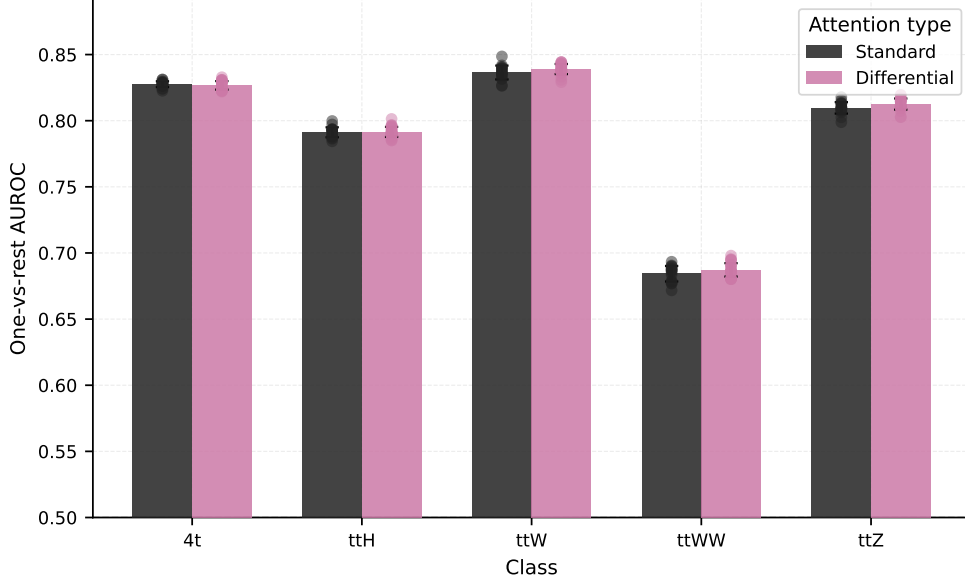


Figure 6.2: Per-class one-vs-rest AUROC, standard vs. differential attention, marginalised over all A4 and B1 conditions. Error bars span one standard deviation across seeds. Differential attention improves every class; the $t\bar{t}WW$ channel (class 3) remains the hardest in both configurations.

Figure 6.3 shows validation AUROC curves during training. The differential-attention models converge to a higher plateau in approximately the same number of epochs, without signs of increased instability or oscillation. The λ_h evolution in Figure 6.4 indicates that the per-head cancellation strength is learned smoothly over training, converging to stable non-zero values. This confirms that the differential mechanism is active and not degenerate (a degenerate case, $\lambda_h \rightarrow 0$, would recover standard attention).

The attention entropy by layer (Figure 6.5) shows lower per-layer entropy for differential attention in the middle layers, consistent with the mechanism suppressing diffuse attention patterns. Later layers recover higher entropy, suggesting the suppression is selective rather than global.

6.2.3. Results: Attention-Internal Normalisation

Figure 6.6 shows AUROC grouped by A4. Applying LayerNorm or RMSNorm inside the attention module consistently reduces mean AUROC compared to no normalisation. The means are 0.832 ± 0.002 for `none` ($n = 21$), 0.827 ± 0.002 for `layernorm` ($n = 12$), and 0.826 ± 0.003 for `rmsnorm` ($n = 12$). The gap of approximately 0.005 absolute (0.5 pp) is present for both attention types and both bias conditions, as shown in the interaction heatmap of Figure 6.7.

A plausible explanation is that attention-internal normalisation re-scales the raw dot-product values against which the Lorentz-scalar bias is calibrated, reducing the effective contribution of the physics term. An alternative explanation is that the normalisation layer introduces additional smoothing that is unhelpful in a relatively shallow architecture (six encoder layers). The evidence is insufficient to distinguish these explanations, but the recommendation is unambiguous: A4 = `none` throughout.

Figure 6.8 confirms that the ordering is stable over training and is not a transient artefact of convergence differences.

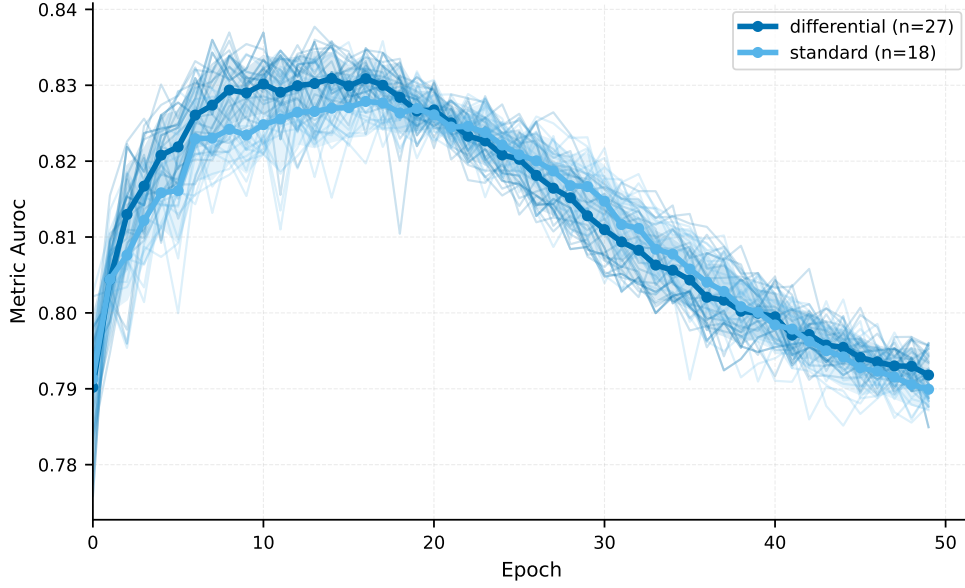


Figure 6.3: Validation AUROC training curves for standard (dashed) and differential (solid) attention, with $A4=\text{none}$ and $B1=\text{lorentz_scalar}$. Shaded bands cover one standard deviation across three seeds. Differential attention converges to a higher plateau without loss of stability.

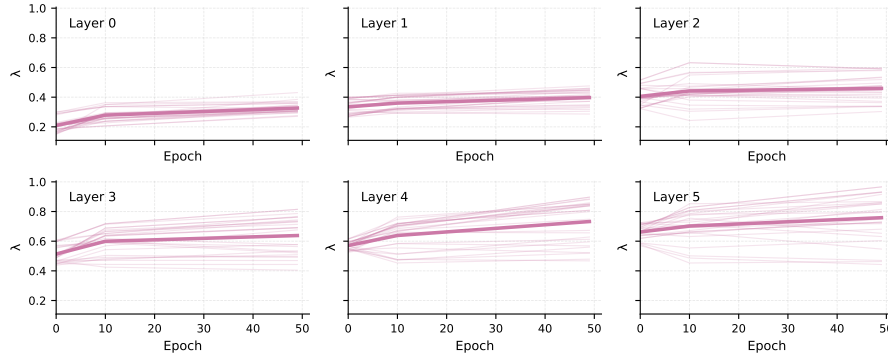


Figure 6.4: Evolution of the per-head λ_h parameters in differential attention over training epochs, averaged across heads and seeds. Values converge to stable non-zero levels, confirming that the cancellation mechanism is active throughout training.

6.3. Physics Bias and Differential Attention

6.3.1. Experimental Design

The Lorentz-scalar bias ($B1$) was already present as a co-factor in Exp. 6A; this section analyses its interaction with attention type. Section 6.4.1 then introduces Exp. 6B, which studies how the bias interacts with the two softmax branches of differential attention via axis $A3\text{-}a$.

6.3.2. Results: B1 by Attention Type

Figure 6.9 summarises the marginalised $B1$ effect. The Lorentz-scalar bias provides a mean AUROC gain of $+0.0025$ ($+0.25$ pp) over the 45 chapter-6 runs (0.830 ± 0.004 with bias, 0.828 ± 0.003 without).

The critical finding is the $A3 \times B1$ interaction shown in Figure 6.10. For standard attention the Lorentz-scalar bias provides essentially no benefit: the mean AUROC with and without the bias is 0.827 in both cases (absolute difference $+0.0002$, well within the seed spread). For differential attention the same bias yields a mean gain of $+0.0033$. This interaction suggests that the learnable λ_h parameter of differential attention can modulate how strongly the physics bias enters the final attention score: when the raw dot-product difference already suppresses uninformative pairs, the additive physics term can concentrate further on the most discriminating pairwise features. Standard attention lacks this modulation and appears unable to exploit the bias structure consistently.

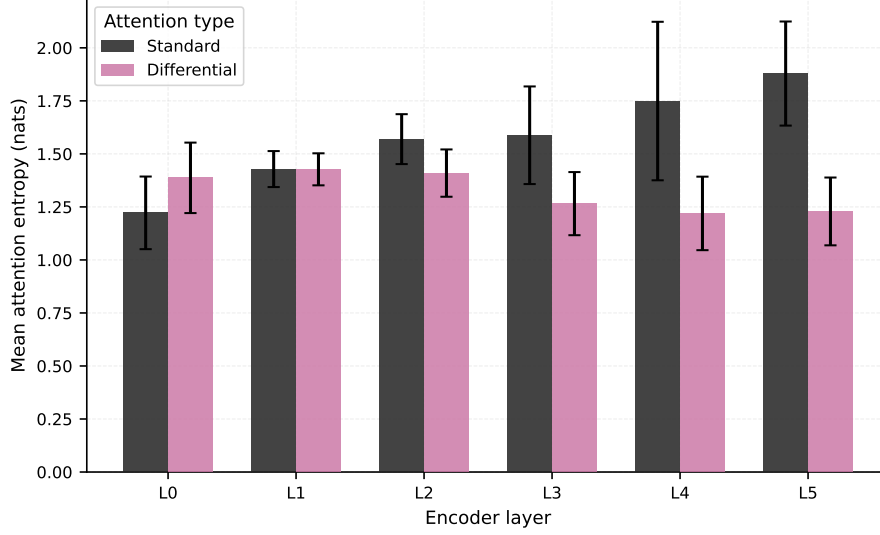


Figure 6.5: Attention entropy per transformer layer, averaged over a held-out batch and three seeds, for standard and differential attention (A4=none, B1=lorentz_scalar). Lower entropy in middle layers for differential attention is consistent with more concentrated attention patterns.

Within the cleanest A4 = none subgroup (Table 6.2), the pattern holds: standard+lorentz_scalar (0.832) reaches approximately the same level as differential+none (0.831), while differential+lorentz_scalar (0.833) is highest. The bias and the differential mechanism therefore contribute independently and in an approximately additive manner.

A3	B1	Mean AUROC
standard	none	0.830
standard	lorentz_scalar	0.832
differential	none	0.831
differential	lorentz_scalar	0.833

Table 6.2: Mean AUROC for the A3 $\times$ B1 interaction within the A4 = none subgroup. All entries are averaged over three seeds; $n = 3$ except differential+lorentz_scalar ($n = 12$).

Figure 6.11 further shows the A4 $\times$ B1 interaction. The Lorentz-scalar bias is beneficial when A4 = none and when A4 = layernorm, but provides no consistent gain with A4 = rmsnorm. This is consistent with the interpretation that attention-internal normalisation partially disrupts the scale of the bias contribution; however, the differences are within one seed-spread standard deviation and should not be over-interpreted.

6.4. Differential Bias Mode

6.4.1. Experimental Design

Differential attention has two softmax branches; a physics-informed additive bias can be applied to both branches identically (*shared*), distributed to each branch via independent network heads with separate parameters (*split*), or omitted entirely (*none*). Axis A3-a A3-a controls this choice.

Experiment 6B swept A3-a A3-a with A3 = differential, A4 = none, and B1 = lorentz_scalar fixed at the values recommended by Exp. 6A.

6.4.2. Results

Figure 6.12 and Table 6.4 summarise the Exp. 6B outcomes. All three A3-a configurations perform closely: *split* achieves a mean AUROC of 0.8330 ± 0.0016 ($n = 3$), *none* of 0.8326 ± 0.0022 ($n = 3$), and *shared* of 0.8300 ± 0.0027 ($n = 21$, including Exp. 6A runs with varying A4 and B1 conditions). The differences among the three modes do not exceed one standard deviation of the seed spread and

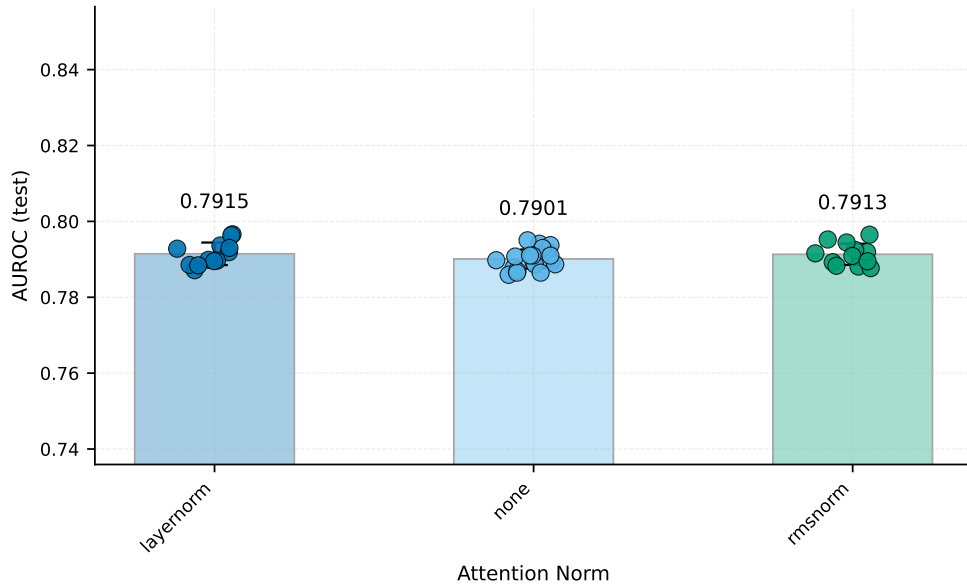


Figure 6.6: Seed-spread AUROC for the three A4 attention-internal normalisation options (`none`, `layernorm`, `rmsnorm`), marginalised over A3 and B1. No normalisation is consistently best; LayerNorm and RMSNorm are nearly equivalent and both below the `none` baseline.

Axis	Values	Configurations
A3-a — Differential bias mode	<code>none</code> , <code>shared</code> , <code>split</code>	3
Seeds	42, 123, 314	3
Total		9

Table 6.3: Exp. 6B design. A3 = `differential`, A4 = `none`, and B1 = `lorentz_scalar` are fixed throughout.

should be considered negligible.

A3-a	Mean AUROC	Std	n
<code>none</code>	0.8326	0.0022	3
<code>shared</code>	0.8300	0.0027	21
<code>split</code>	0.8330	0.0016	3

Table 6.4: AUROC by differential bias mode. The `shared` group is larger because it also includes the Exp. 6A runs (varied A4 and B1), inflating its variance. The cleanest comparison is among the three Exp. 6B-only rows, each with $n = 3$ under identical A4 and B1 conditions.

The `split` mode provides two independent MLP heads to produce the two attention-logit additive terms — one per branch — whereas `shared` adds the same bias to both branches. The near-identical performance of `split` and `none` (relative to `shared`) suggests that the second branch does not strongly benefit from an independent bias signal. A practical consequence is that the choice of A3-a can be made on the basis of model complexity preference: `split` doubles the parameters of the Lorentz-scalar MLP but provides no measurable gain over `shared` at fixed seed count. The `split` mode is adopted as the default for subsequent chapters, acknowledging that the difference is within noise.

A methodological note on the A3-a confounder: Exp. 6B was designed with B1 = `lorentz_scalar` fixed, so the effect of A3-a when B1 = `none` is not measured. The A3-a effect is therefore not fully separable from B1 within this dataset.

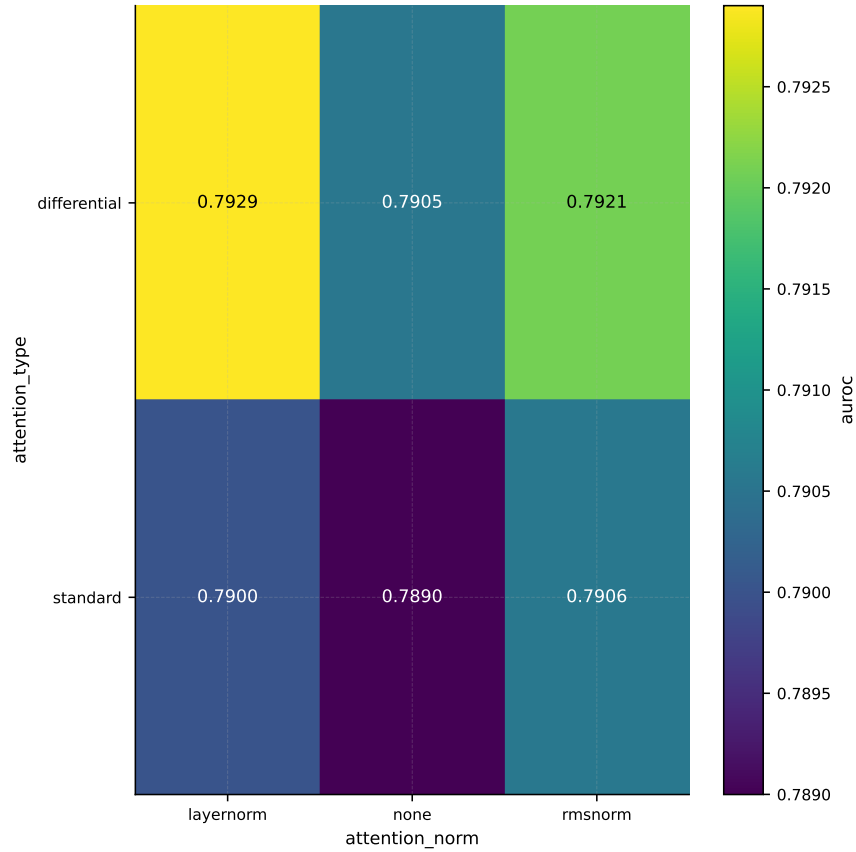


Figure 6.7: Mean AUROC heatmap for the $A3 \times A4$ interaction. The `none` normalisation row is best for both attention types. Differential attention (left column) consistently outperforms standard (right column) within every normalisation row.

6.5. Chapter Summary

Table 6.5 recaps the two experiments; Table 6.6 states the recommended axis settings derived from this chapter.

Experiment	Axes swept	Fixed	Runs
6A — Attention type and norm	A3, A4, B1	A3-a=shared	36
6B — Differential bias mode	A3-a	A3=differential, A4=none, B1=lorentz_scalar	9
Total			45

Table 6.5: Chapter 6 experimental overview.

The three main findings are as follows.

Differential attention improves classification performance. Axis $A3 = \text{differential}$ yields a mean AUROC gain of $+0.0036$ over `standard` attention, with consistent positive sign across all five signal classes, all normalisation settings, and three random seeds. The λ_h parameters converge to stable non-zero values, confirming that the cancellation mechanism is active.

Attention-internal normalisation is detrimental. Axis $A4 = \text{none}$ is best in all ($A3$, $B1$) sub-groups. Both LayerNorm and RMSNorm reduce mean AUROC by approximately 0.005 absolute. The effect is consistent in direction across all conditions.

The Lorentz-scalar bias is effective only with differential attention. Axis $B1 = \text{lorentz_scalar}$ provides a gain of $+0.0033$ for differential attention but essentially no gain ($+0.0002$) for standard attention. The two mechanisms are approximately additive: the best configuration in this chapter ($A3 = \text{differential}$, $A4 = \text{none}$, $B1 = \text{lorentz_scalar}$, $A3\text{-a} = \text{split}$) achieves a mean AU-

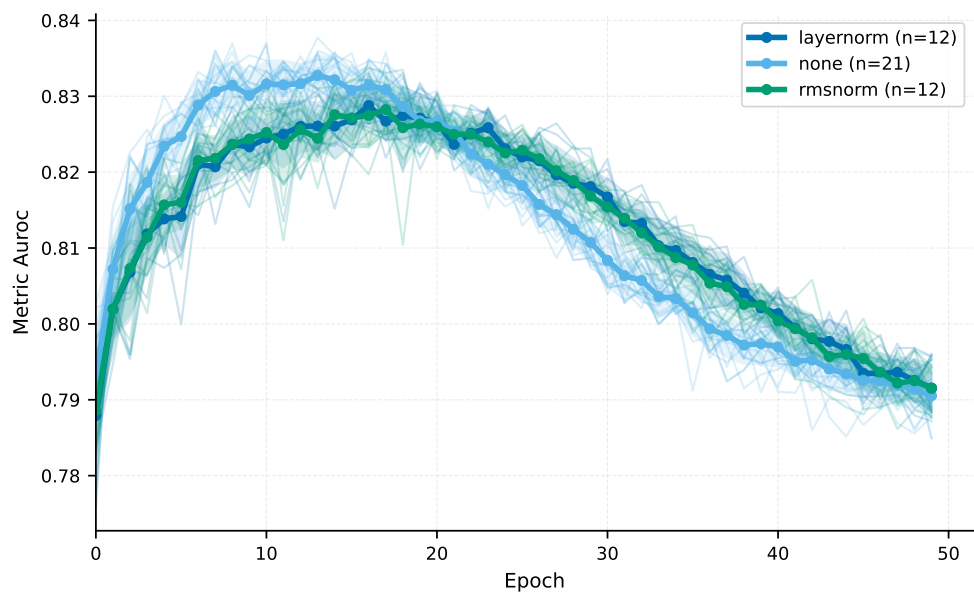


Figure 6.8: Validation AUROC training curves for the three A4 options (A3=differential, B1=lorentz_scalar). The none configuration (solid) converges to the highest plateau; the gap to the normalised variants is established within the first 20 epochs and is stable thereafter.

ROC of 0.8330 ± 0.0016 , compared to the baseline (standard, A4 = none, B1 = none) at 0.8296 ± 0.0027 — an absolute gain of $+0.0034$ ($+0.34$ pp).

Axis	Recommended value	Rationale
A3 — Attention type	differential	Consistent +0.36 pp over standard
A4 — Internal normalisation	none	LN/RMSNorm reduce AUROC by ≈ 0.5 pp
A3-a — Diff bias mode	split	Marginal over shared; within noise
B1 — Lorentz-scalar bias	lorentz_scalar	Effective with differential attention

Table 6.6: Recommended axis settings from Chapter 6, carried forward to subsequent chapters.

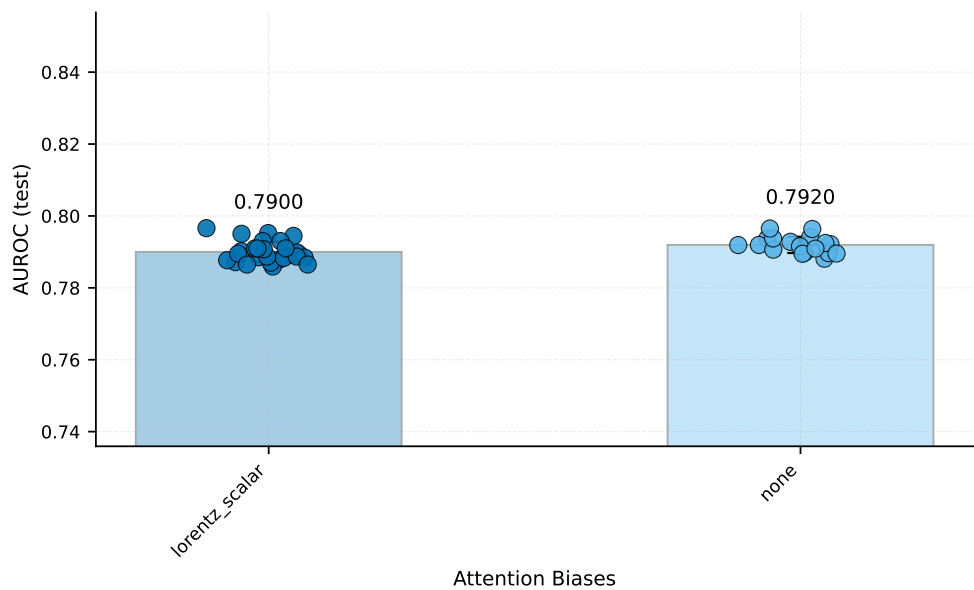


Figure 6.9: Seed-spread AUROC for B1 bias options (*none* vs. *lorentz_scalar*), marginalised over A3 and A4. The Lorentz-scalar bias provides a modest positive shift; the effect is larger for differential than for standard attention (see Figure 6.10).

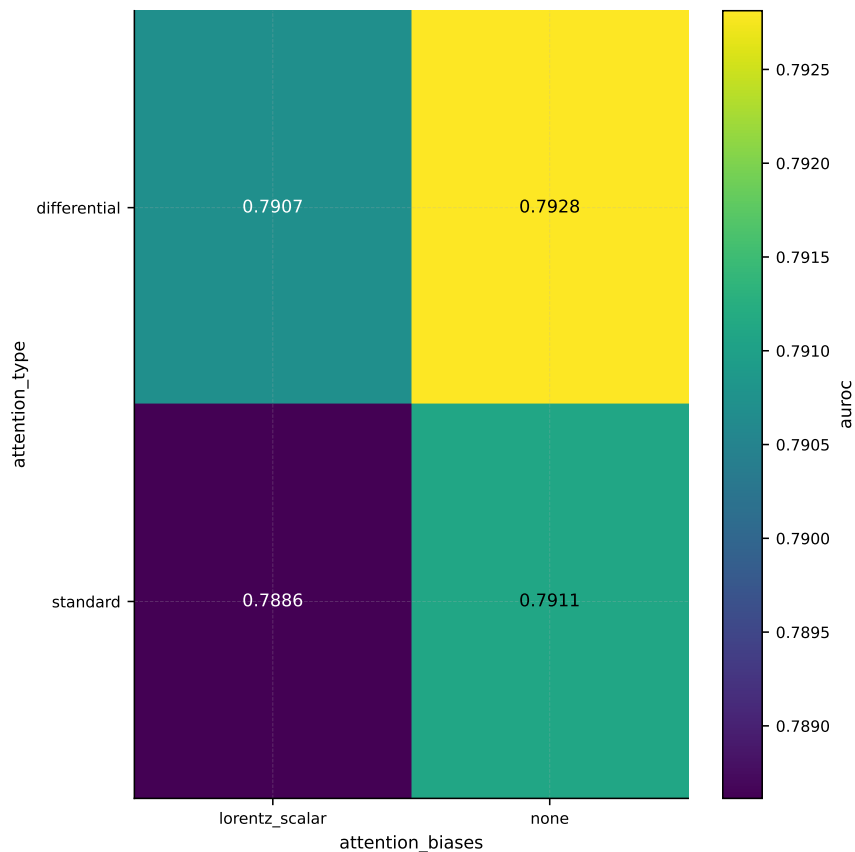


Figure 6.10: Mean AUROC heatmap for the $A3 \times B1$ interaction, marginalised over A4 (Exp. 6A runs only). The Lorentz-scalar bias provides a clear gain for differential attention (+0.0033) but is ineffective for standard attention (+0.0002).

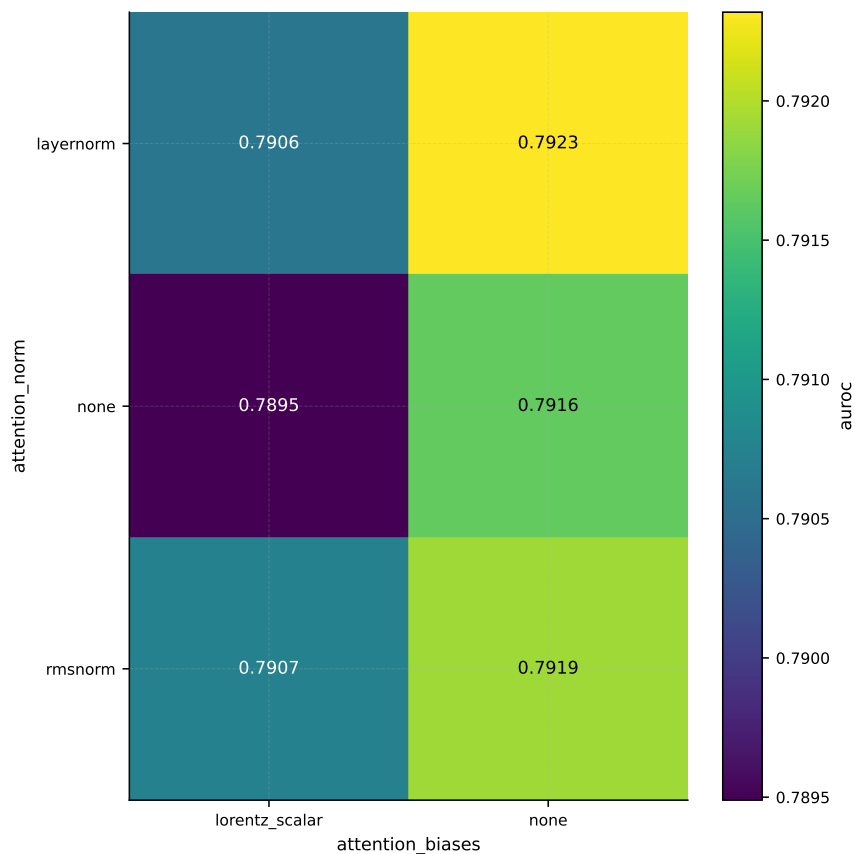


Figure 6.11: Mean AUROC heatmap for the $A4 \times B1$ interaction, marginalised over $A3$. The Lorentz-scalar bias provides a positive contribution for `none` and `layernorm` normalisation, but not for `rmsnorm`.

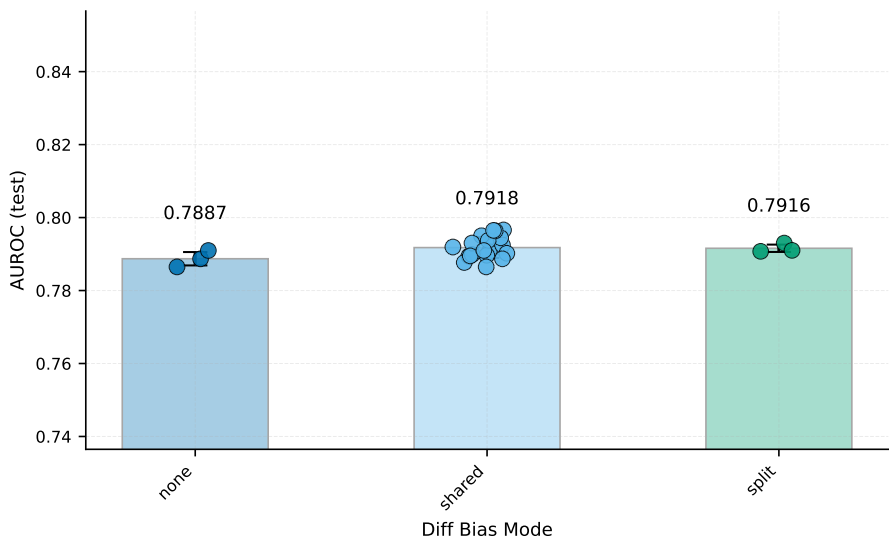


Figure 6.12: Seed-spread AUROC for the three $A3$ -a differential bias modes within Exp. 6B ($A4 = \text{none}$, $B1 = \text{lorentz_scalar}$). The three modes are indistinguishable within seed variation.

7

Model Scaling and Efficiency

Every architectural finding in Chapters 4–6 was established at a single, fixed model size. This chapter examines how classifier performance and inference cost vary across a factor of roughly 140 in parameter count, from approximately 26k to 3.6M parameters. Two questions motivate this study. First, where does AUROC saturate as a function of model capacity, and does the saturation point shift with task difficulty? Second, which configuration is Pareto-optimal when performance is weighed against inference cost? The answers retrospectively validate the fixed size used throughout Chapters 4–6 and identify a practical operating point for the evaluation in Chapter 8.

The chapter is structured as follows. Section 7.1 describes the experimental design, which is a full factorial sweep of five model-size presets across three classification tasks (Exp. 7A). Section 7.2 reports performance as a function of model size and training cost. Section 7.3 presents the Pareto analysis across four efficiency dimensions (Exp. 7B). Section 7.3 summarises the findings.

7.1. Experimental Design

All 45 runs in this chapter share a common fixed baseline configuration: tokenizer family `identity` (axis T1), the feature set D01 selected in Chapter 4, sinusoidal positional encoding (E1 = `sinusoidal`), standard attention (A3 = `standard`), LayerNorm with pre-norm policy (A1, A2), and no physics-informed attention biases (B1 = `none`). Model capacity is the sole variable, swept via the convenience label axis H10 , which bundles model dimension (H01 ), encoder depth (H02 ), attention heads (H03), and FFN hidden dimension (H04) into five named presets. Three random seeds per cell (axis R05) give 45 runs in total.

Table 7.1: Exp. 7A design. All other axes held at their thesis baseline values.

Axis	Values	Cells
Model size (H10 )	d32_L3, d32_L5, d64_L6, d128_L8, d192_L12	5
Task (G03)	4t vs. background, 4t vs. ttH, multiclass	3
Seed (R05)	3 per cell	3
Total		45

Table 7.2 lists the concrete dimension and depth values for each preset, together with the approximate parameter count and analytic FLOPs per event. Analytic FLOPs count multiply-add operations derived from the model architecture; they do not include Python overhead or data-movement costs, but provide a clean, hardware-independent comparison basis.

The jump in FLOPs from d64_L6 to d128_L8 is approximately $\times 5.3$, and from d64_L6 to d192_L12 approximately $\times 18$. This non-linear growth is driven by the quadratic attention cost and the multiplicative effect of simultaneously increasing both depth and width.

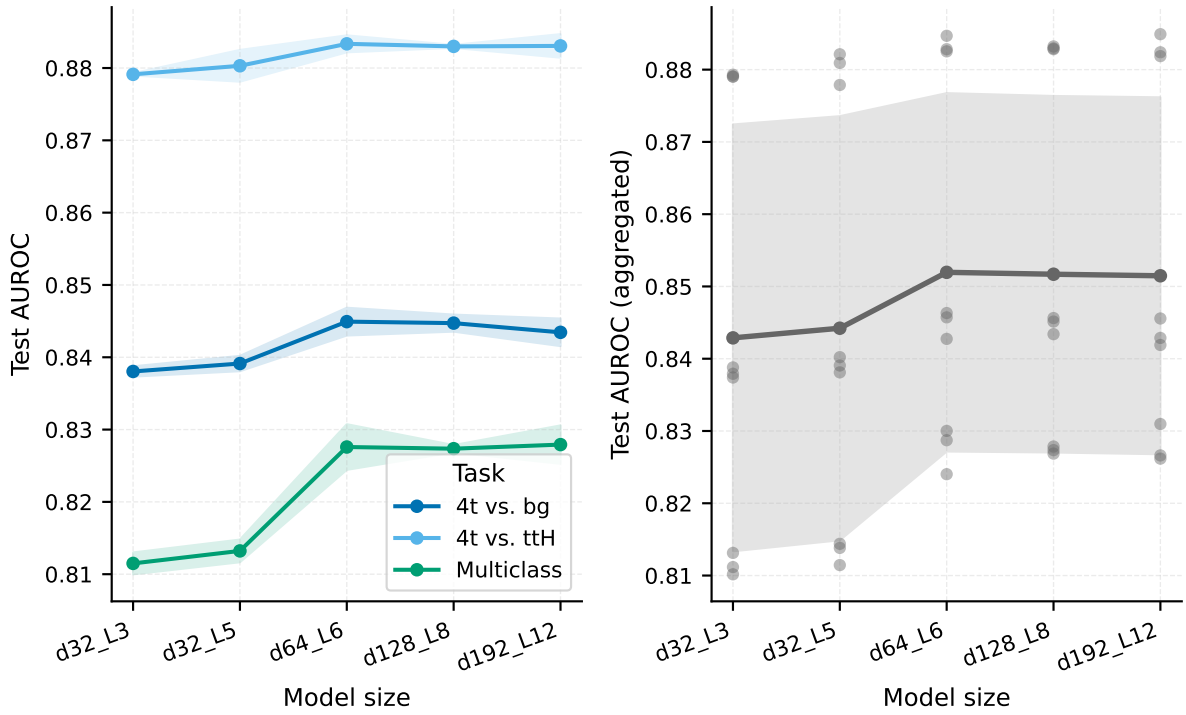
Table 7.2: Model-size presets used in Exp. 7A. Approximate parameter counts are derived from H01 and H02; FLOPs per event are analytic multiply-add counts averaged over all 9 runs sharing that preset.

Size label (H10)	dim (H01)	depth (H02)	Approx. params	FLOPs/event (analytic)
d32_L3	32	3	~26k	1.29M
d32_L5	32	5	~43k	2.15M
d64_L6	64	6	~202k	10.3M
d128_L8	128	8	~1.06M	55.1M
d192_L12	192	12	~3.57M	185.8M

7.2. Scaling Curves (Exp. 7A)

7.2.1. AUROC versus Model Size

Figure 7.1 shows AUROC as a function of model size, separately for each task and aggregated over all three tasks. The aggregated mean and standard deviation over all nine runs (three tasks $\times$ three seeds) per size are reported in Table 7.3.

**Figure 7.1:** AUROC versus model-size preset for all three tasks (left) and aggregated over tasks with seed scatter (right); higher AUROC is better, and the plateau at d64_L6 is visible in both panels.

AUROC rises by approximately 0.8–1.3 percentage points between the two smallest presets and d64_L6, but the three larger configurations (d64_L6, d128_L8, d192_L12) are statistically indistinguishable at 0.852 ± 0.025 . The plateau is consistent across all three tasks in the per-task panel. Although the multiclass task yields systematically lower absolute AUROC than the binary tasks (owing to the harder five-class discriminant), the saturation point does not shift: all three tasks plateau at d64_L6.

The saturation at this relatively small scale is consistent with the low-multiplicity nature of the classification problem. Events in the 4t and ttH samples contain at most a few tens of reconstructed objects, so the information content available per event is modest, and a model with ~202k parameters appears sufficient to extract it with the baseline architecture.

Signal efficiency at fixed background rejection. For the two binary tasks, signal efficiency ε_S at a fixed background rejection of $1/B = 50$ provides a physics-operational complement to AUROC. This metric is undefined for the multiclass task and is not reported there. Table 7.4 summarises the results.

Table 7.3: Aggregated AUROC by model-size preset (mean $\pm$ std over all 9 runs, 3 tasks $\times$ 3 seeds). The large standard deviation reflects cross-task variance, not seed noise; within a single task, seed-to-seed variation is substantially smaller.

Size label (H10)	Mean AUROC	Std
d32_L3	0.843	0.030
d32_L5	0.844	0.029
d64_L6	0.852	0.025
d128_L8	0.852	0.025
d192_L12	0.852	0.025

Table 7.4: Signal efficiency ε_S at background rejection $1/B = 50$, for the two binary tasks (mean $\pm$ std over 3 seeds). The plateau pattern mirrors the AUROC result.

Size label (H10)	ε_S , 4t vs. bg	ε_S , 4t vs. ttH
d32_L3	0.276 ± 0.006	0.283 ± 0.003
d32_L5	0.277 ± 0.000	0.284 ± 0.005
d64_L6	0.284 ± 0.002	0.298 ± 0.008
d128_L8	0.286 ± 0.009	0.290 ± 0.007
d192_L12	0.288 ± 0.004	0.292 ± 0.007

At d64_L6, $\varepsilon_S \approx 0.284$ for 4t vs. background and $\varepsilon_S \approx 0.298$ for 4t vs. ttH. The larger models do not improve on these values by a statistically meaningful margin. The values at $1/B = 1000$ were inspected but are non-monotone across model sizes, reflecting the sparsity of events at very high rejection; this threshold is not informative for architectural comparisons and is not used as a primary metric in this chapter.

7.2.2. Training Cost

Figure 7.2 shows total training wall-clock time as a function of model-size preset, and Figure 7.3 shows the best checkpoint epoch at which early stopping triggered. Figure 7.4 shows the analytic FLOPs per event on a log-log scale against parameter count.

Table 7.5 summarises the training cost metrics. Wall-clock time grows from 28.9 s at d32_L3 to 302.7 s at d192_L12, a factor of approximately $10.5\times$. Over the same range, analytic FLOPs per event grow by a factor of approximately $144\times$ (from 1.29M to 185.8M). The discrepancy between the FLOPs ratio and the wall-clock ratio reflects the fact that wall-clock time includes evaluation and benchmarking overhead (latency measurement), which is size-independent and inflates the cost of smaller models. The best-epoch trend shows an inverse relationship with model size: larger models trigger early stopping sooner (mean epoch 17 for d192_L12 versus 47 for d32_L3), because their greater capacity results in faster loss descent per epoch. However, each epoch is considerably more expensive for the larger models, so the total wall-clock time is dominated by per-epoch cost for d192_L12.

Table 7.5: Training cost metrics by model-size preset (mean $\pm$ std over 9 runs, 3 tasks $\times$ 3 seeds). Wall-clock includes evaluation overhead; best epoch is the early-stopping trigger.

Size label (H10)	Wall-clock (s)	Best epoch	FLOPs/event (analytic)
d32_L3	28.9 ± 0.8	47.1 ± 2.8	1.29M
d32_L5	45.1 ± 0.8	46.1 ± 3.8	2.15M
d64_L6	86.7 ± 0.7	39.7 ± 7.9	10.3M
d128_L8	133.5 ± 1.0	20.4 ± 3.4	55.1M
d192_L12	302.7 ± 1.6	17.1 ± 2.8	185.8M

7.3. Efficiency and Pareto Analysis (Exp. 7B)

Experiment 7B repurposes the 45 checkpoints from Exp. 7A to measure four inference-side efficiency metrics at batch size 512: latency per batch, throughput in samples per second, and peak memory allocation. These are plotted in Pareto space against AUROC in Figures 7.5–7.8. In each panel, a

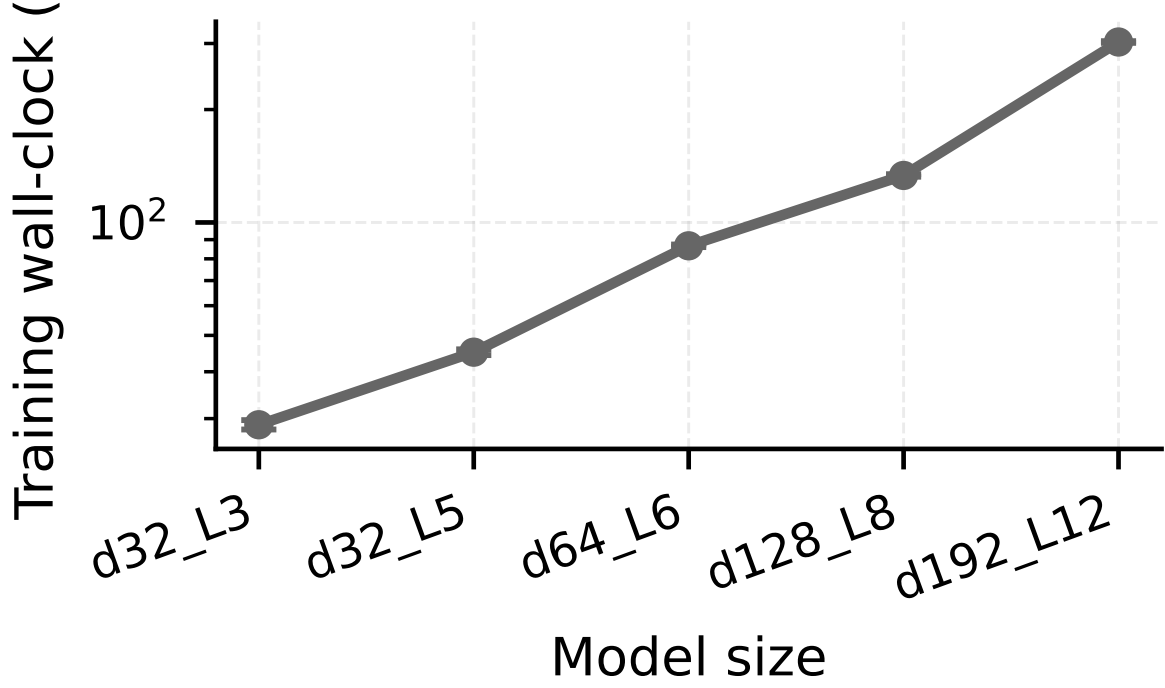


Figure 7.2: Total training wall-clock time (seconds) versus model-size preset on a logarithmic vertical axis; lower is faster, and the approximately linear growth with parameter count is consistent with FLOPs scaling.

point on the Pareto frontier represents a configuration for which no other configuration achieves higher AUROC at lower cost. No new training was required for Exp. 7B; all metrics were recorded during the evaluation phase of Exp. 7A and are available in the canonical run table with zero missing values.

Batch size 512 is the primary efficiency reference because single-event latency (batch size 1) exhibits large standard deviations for d64_L6 and d128_L8 ($\sigma \approx 1.2\text{--}1.6\text{ ms}$), attributable to operating-system scheduling jitter on the interactive node rather than model variability. At batch size 512, standard deviations are below 0.02 ms across all presets, making the measurements far more reproducible.

AUROC versus FLOPs per event. Figure 7.5 shows AUROC against analytic FLOPs per event per task and aggregated. The gap between the two smallest presets and d64_L6 is visible in AUROC but represents only a factor of approximately 8 in FLOPs (1.29M to 10.3M). Above d64_L6, AUROC is flat while FLOPs increase by a factor of $\times 18$ to reach d192_L12 at 185.8M FLOPs per event. d64_L6 lies on the Pareto frontier: no larger model improves AUROC, and no smaller model matches its AUROC at lower cost.

AUROC versus inference latency. Figure 7.6 shows AUROC against per-batch inference latency at batch size 512. Table 7.6 reports the numeric values. Latency grows from 0.163 ms at d32_L3 to 2.019 ms at d192_L12, a factor of approximately $12.4\times$. The latency of d64_L6 (0.552 ms) is $3.7\times$ lower than that of d192_L12 at identical AUROC. Again, d64_L6 sits on the Pareto frontier.

AUROC versus throughput. Figure 7.7 shows AUROC against throughput in samples per second. Throughput drops from approximately 3.14M samples/s at d32_L3 to 927k samples/s at d64_L6 and 254k samples/s at d192_L12. Because the larger models provide no AUROC improvement over d64_L6, the throughput reduction by a further factor of $3.7\times$ from d64_L6 to d192_L12 is not compensated by any discriminative gain. Although d32_L3 and d32_L5 offer higher throughput than d64_L6, they sacrifice approximately 1 percentage point of AUROC, which corresponds to a measurable reduction in ε_S at operational background rejection rates.

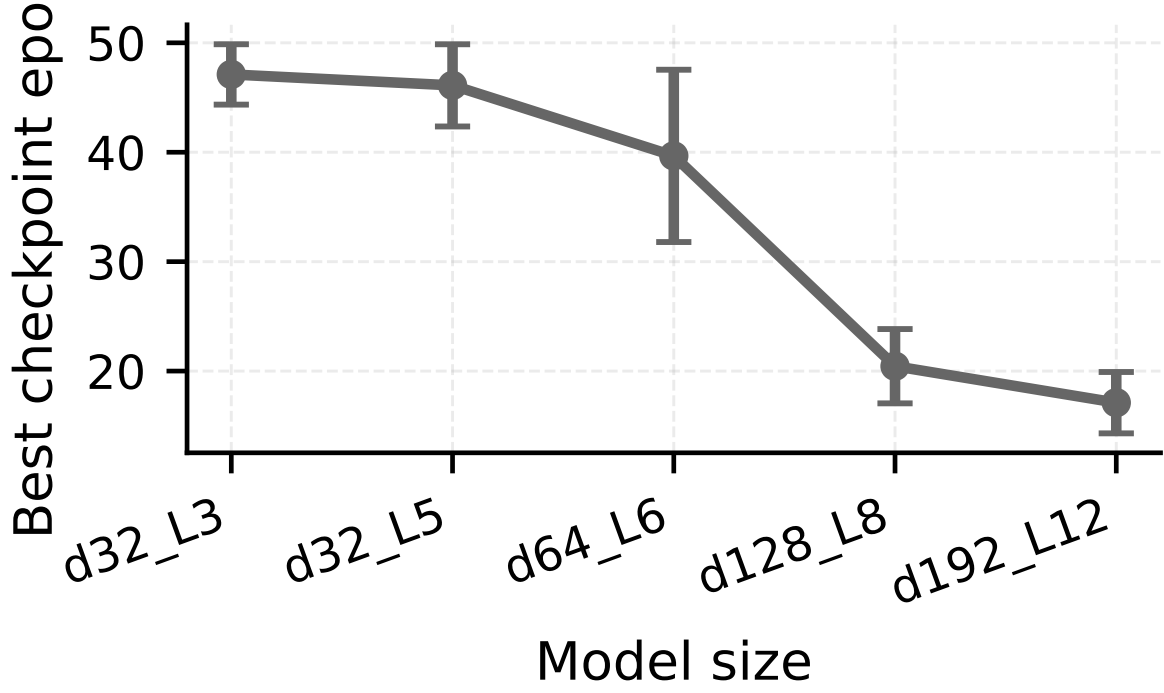


Figure 7.3: Best checkpoint epoch (early-stopping trigger) versus model-size preset; larger models converge in fewer epochs, but each epoch takes longer, so total wall-clock time still grows with model size.

AUROC versus peak memory. Figure 7.8 shows AUROC against peak inference memory at batch size 512. Memory grows from approximately 996 MiB at d32_L3 and d32_L5 (nearly identical, as the MiB allocation is dominated by batch and activation buffers) to 1,968 MiB at d64_L6, 3,187 MiB at d128_L8, and 4,772 MiB at d192_L12. Memory standard deviations are negligible across all presets, reflecting deterministic allocation independent of random seed. The $\times 2.4$ memory increase from d64_L6 to d192_L12 accompanies no improvement in AUROC, confirming d64_L6 as Pareto-optimal on this dimension as well.

Table 7.6: Inference efficiency metrics at batch size 512 by model-size preset (mean $\pm$ std over 9 runs). Memory standard deviations are < 0.1 MiB and are omitted.

Size label (H10)	Latency (ms)	Throughput (samples/s)	Peak memory (MiB)
d32_L3	0.163 ± 0.000	3,145k	996
d32_L5	0.268 ± 0.000	1,912k	996
d64_L6	0.552 ± 0.001	927k	1,968
d128_L8	0.868 ± 0.001	590k	3,187
d192_L12	2.019 ± 0.015	254k	4,772

Across all four Pareto dimensions — FLOPs per event, inference latency, throughput, and peak memory — d64_L6 ($\sim 202k$ parameters, $\sim 10.3M$ FLOPs per event) is the Pareto-optimal configuration for this task family. It achieves the full discriminative performance of models up to $\times 18$ more expensive in FLOPs and $\times 3.7$ slower in latency.

Chapter Summary

AUROC saturates at d64_L6 ($\sim 202k$ parameters) and does not improve for the three larger presets, regardless of task. Signal efficiency at fixed background rejection ($1/B = 50$) confirms the same plateau: $\varepsilon_S \approx 0.284$ for 4t vs. background and $\varepsilon_S \approx 0.298$ for 4t vs. ttH at d64_L6, with no statistically significant gain from d128_L8 or d192_L12. On the inference side, d64_L6 is Pareto-optimal across FLOPs per event, latency, throughput, and peak memory, at batch size 512. This finding retrospectively validates the fixed model size applied throughout Chapters 4–6, and establishes the d64_L6 configuration as the

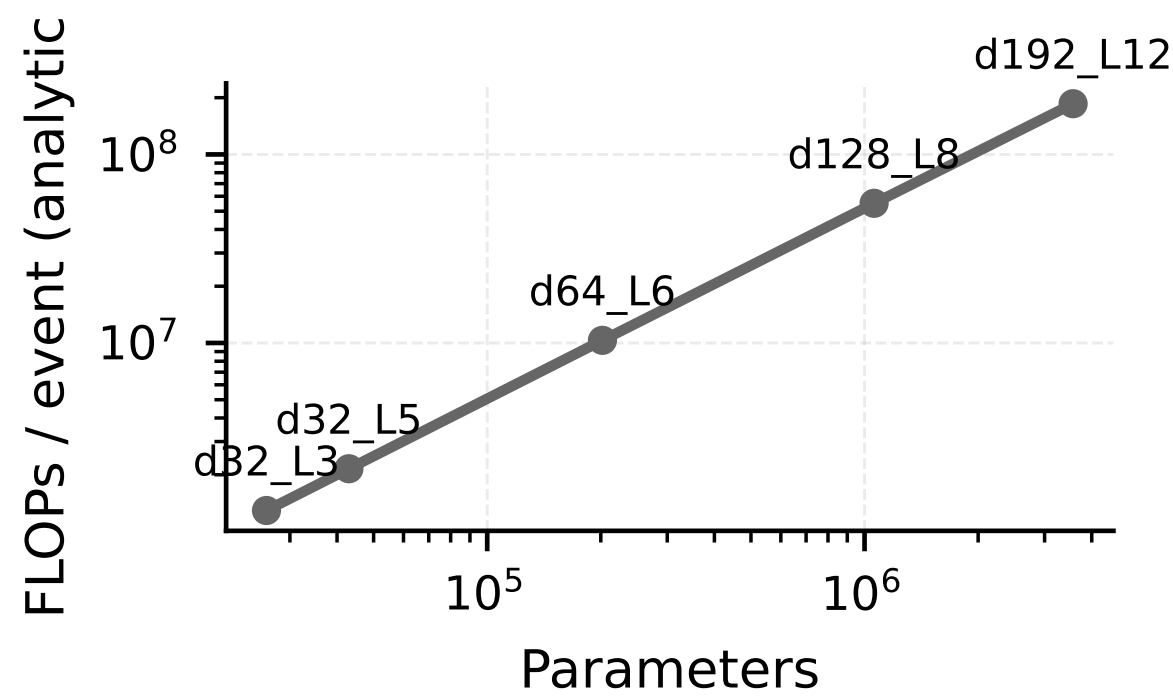


Figure 7.4: Analytic FLOPs per event versus approximate parameter count on a log-log scale; the non-linear growth reflects the simultaneous increase in model dimension and depth across presets.

Table 7.7: Chapter 7 experiment summary.

Experiment	Description	Runs
7A — Scaling curves	5 sizes × 3 tasks × 3 seeds	45
7B — Pareto analysis	Post-hoc efficiency metrics from Exp. 7A checkpoints	0
Total		45

comparison baseline for Chapter 8, which asks which combination of architectural choices drawn from those chapters achieves the highest discrimination performance.

One limitation of this analysis is that the scaling results apply to the baseline architecture only. Whether a better-performing architecture identified in Chapter 8 scales differently — for example, saturating later or earlier — is an open question that falls outside the scope of this thesis.

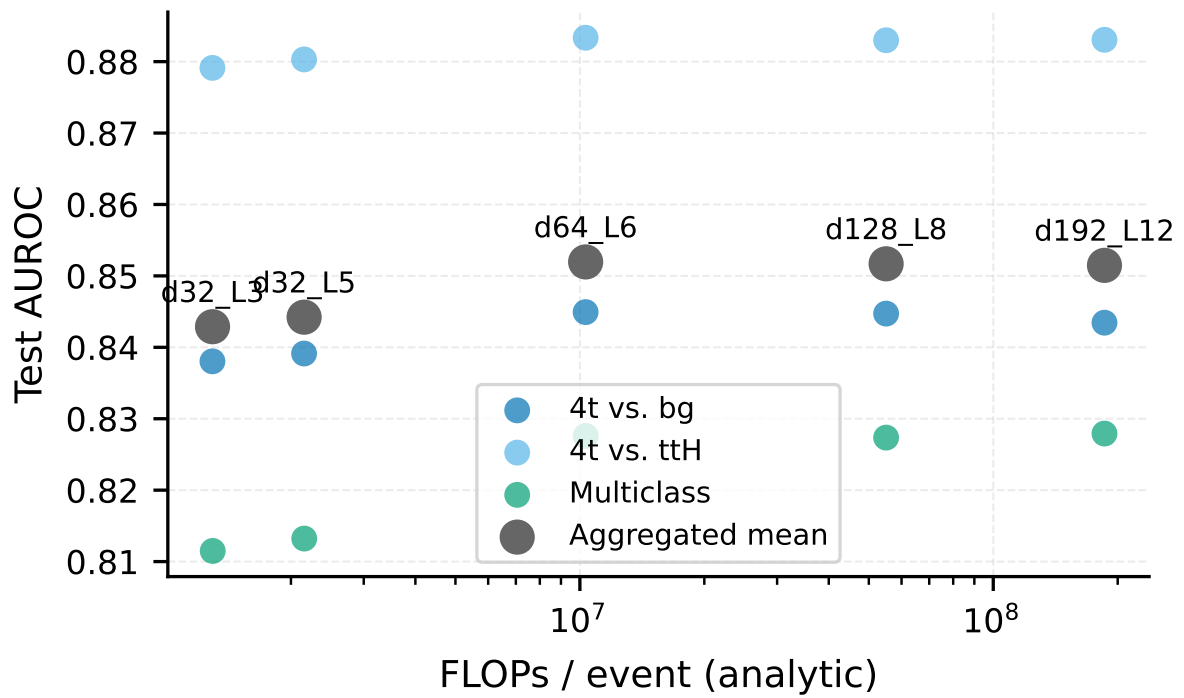


Figure 7.5: AUROC versus analytic FLOPs per event per task (left) and aggregated (right); points further up and to the left are Pareto-preferable, and d64_L6 lies on the frontier.

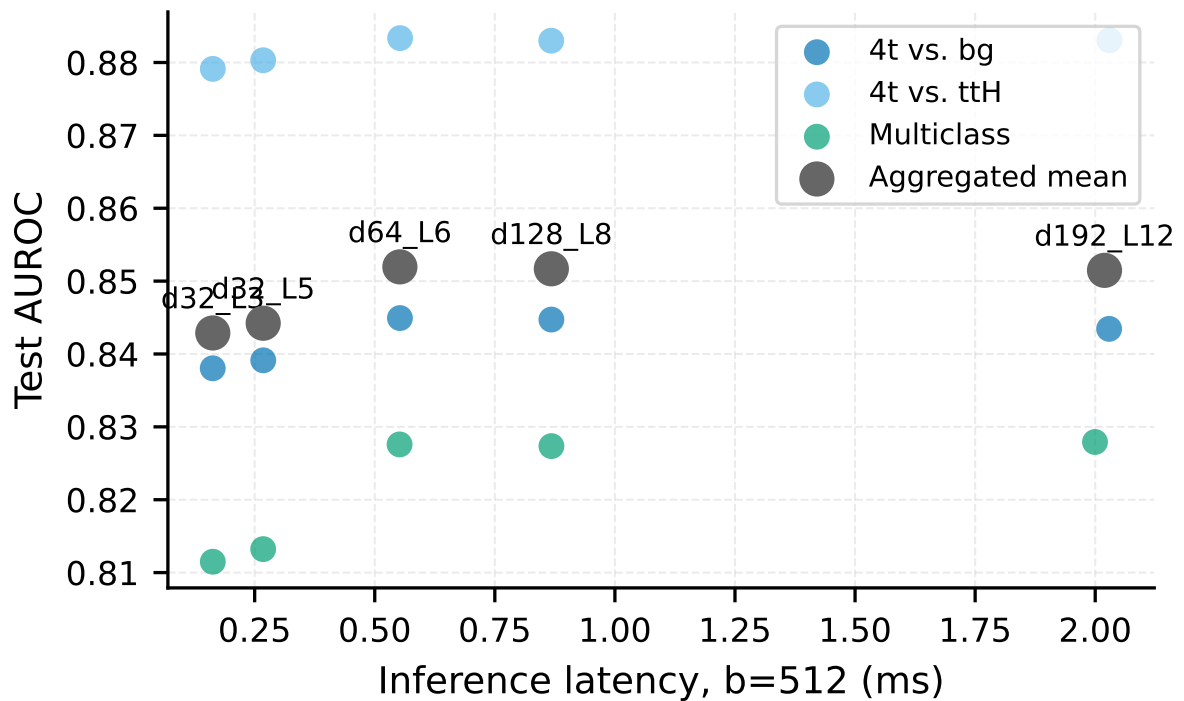


Figure 7.6: AUROC versus inference latency per batch at batch size 512 (ms); lower latency and higher AUROC are jointly preferable, with d64_L6 on the frontier.

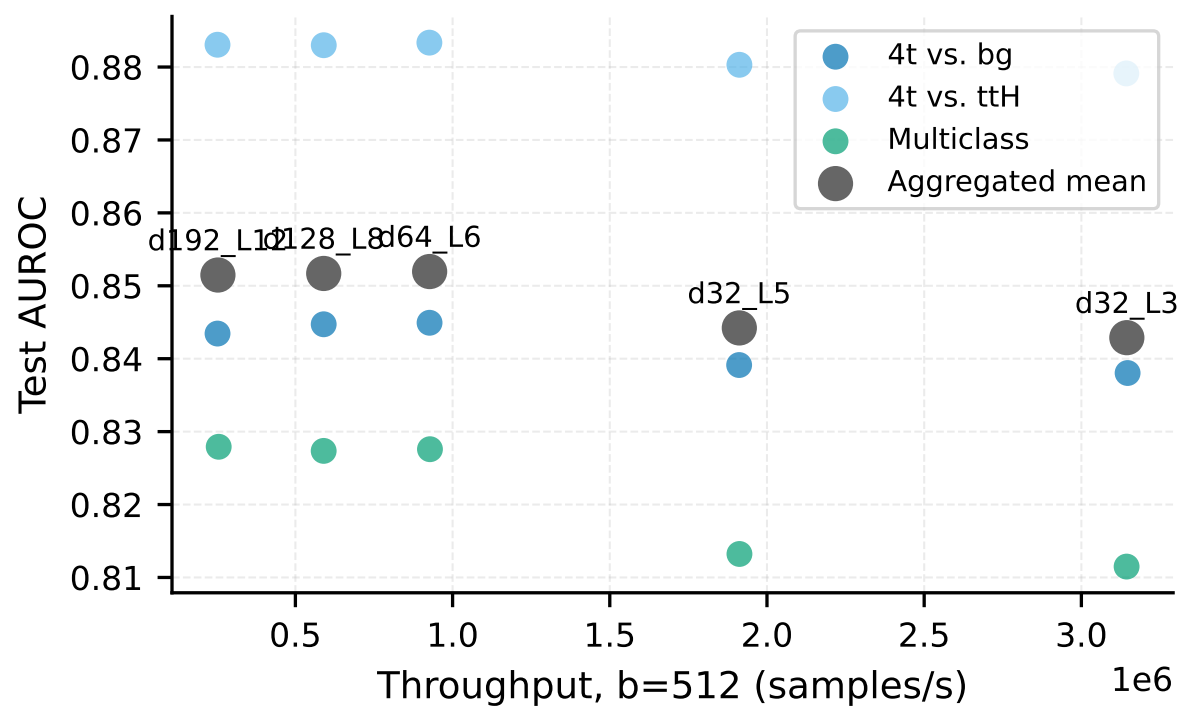


Figure 7.7: AUROC versus inference throughput at batch size 512 (samples/s); higher throughput and higher AUROC are jointly preferable, with d64_L6 on the frontier.

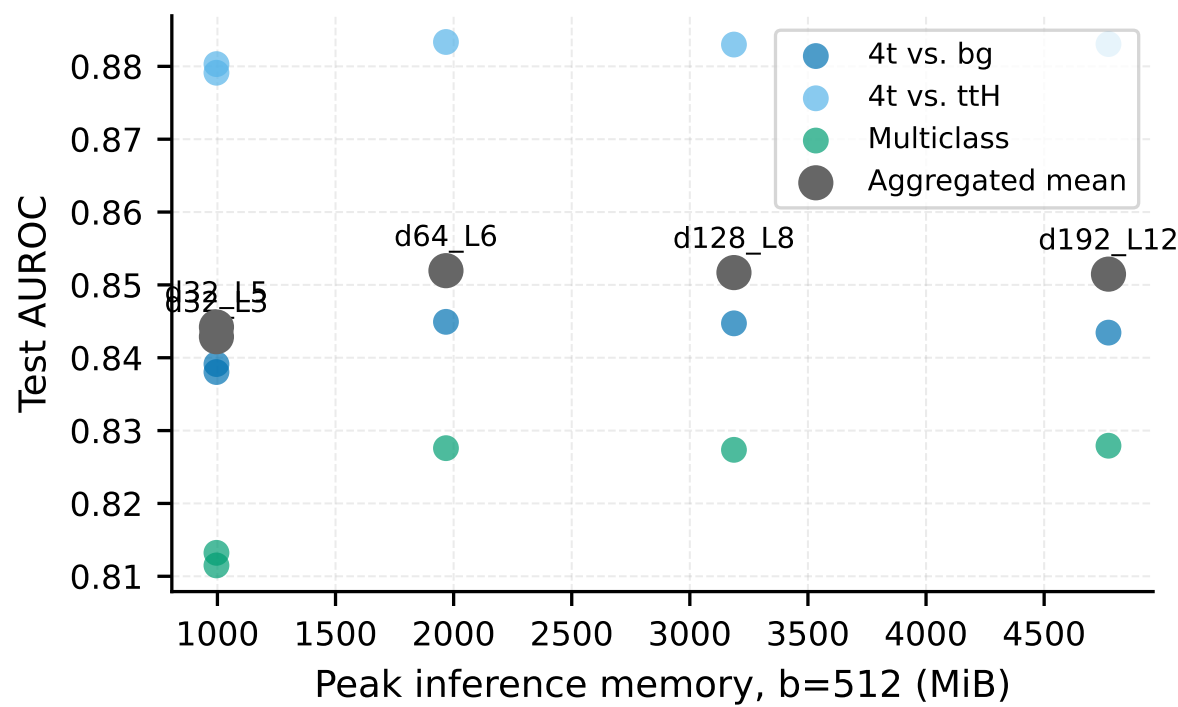
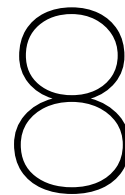


Figure 7.8: AUROC versus peak inference memory at batch size 512 (MiB); lower memory and higher AUROC are jointly preferable, and d64_L6 lies on the frontier in all task panels.

Part III

Synthesis



Global Surrogate-Driven Analysis

Chapter 7 established that AUROC saturates at model size `d64_L6` ($\sim 202k$ parameters, $\sim 10.3M$ FLOPs per event), and that no larger configuration improves discrimination on the primary 4t-versus-background task. This chapter takes that model size as fixed and asks a different question: given the full space of architectural choices explored across Chapters 4–6, which combination of axes maximises discrimination performance?

The preceding chapters studied architectural axes one family at a time. Each experiment held most axes fixed and varied a small subset, which enables controlled causal inference but covers only a fraction of the joint configuration space. This chapter takes the complementary view: the full corpus of approximately 983 transformer runs on the primary task (4t vs. combined background, `G3 = ttH+ttW+ttWW+ttZ | 4t`) is treated as a large observational dataset, and statistical methods are used to extract global structure from it.

The chapter proceeds in five stages. Section 8.1 audits the run database, defines the primary analysis cohort, and characterises axis coverage. Section 8.2 performs a marginal sensitivity analysis: for each axis, the range of per-level mean AUROC is computed and compared to the seed noise floor. Section 8.3 fits a global XGBoost surrogate on the full axis vector and uses SHAP to attribute importance to axis families. Section 8.4 proposes and validates new candidate configurations derived from both the surrogate and a marginal-greedy heuristic, and compares all three search strategies including an Optuna TPE search. Section 8.6 summarises the findings and promotes the best validated candidate to Chapter 9.

8.1. Database Audit and Inclusion Rules

8.1.1. Run counts and cohort definition

The raw W&B export contains 1 320 rows across all classification tasks and spec versions. The primary analysis cohort is formed by applying three sequential filters. First, only runs on the G3 task (4t vs. combined background) are retained. Second, only runs evaluated under the `eval_v2.1` spec version are included; three legacy runs on an earlier hash are discarded. Third, runs with missing or failed evaluations (null `eval_v2/test_auroc`) are removed. The resulting primary cohort contains **983 runs**.

Figure 8.1 shows the distribution of runs by spec version, confirming that the overwhelming majority (1 012 of the 1 015 G3 runs passing all other filters) use `eval_v2.1`. The three legacy runs are excluded from all subsequent analysis.

8.1.2. Axis coverage and co-occurrence structure

Many axis columns contain missing values by construction: conditional sub-axes (for example the bias sub-axes B1-G2  and B1-G3 ) are only meaningful when their parent axis is active, and are encoded as NaN otherwise. Figure 8.2 shows the fraction of missing values per axis column across all three G3 cohorts in the raw 04 export. High missingness in the B1 sub-axis columns and in the KAN-specific

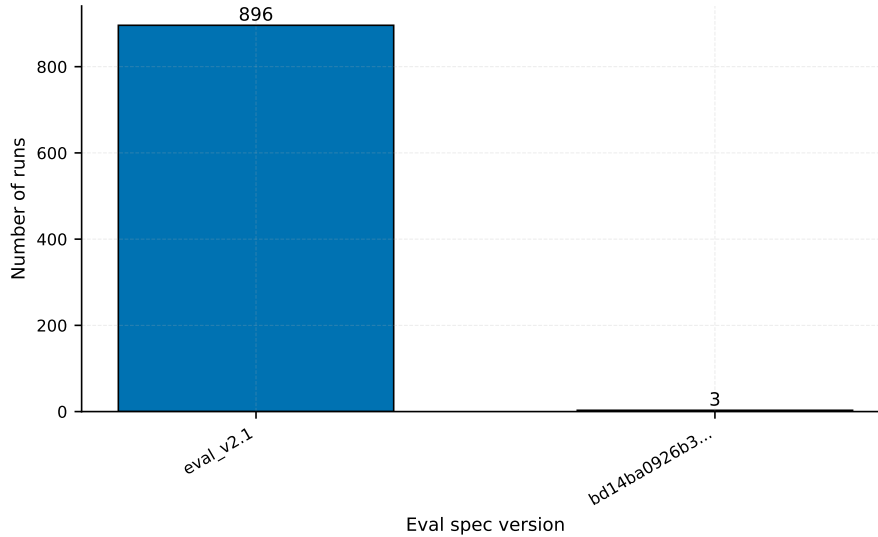


Figure 8.1: Run counts by evaluation spec version for the G3 cohort. The three runs on the legacy spec version are discarded; all analysis uses the 1 012 runs on eval_v2.1.

axes (K3–K5) is structural, not a data quality issue.

To assess whether axes are explored independently or tend to co-occur, Figure 8.3 displays the pairwise Cramér’s V between axis families, computed on the primary cohort. Each cell shows the mean Cramér’s V across all column pairs within the two families, excluding columns with fewer than two non-inactive unique values. Most off-diagonal cells are below 0.2, indicating that the sweeps across chapters are approximately orthogonal at the family level. The T–B cell shows moderate co-occurrence ($V \approx 0.3$ – 0.4), reflecting the constraint that physics-bias axes require the identity or raw tokeniser; this structural dependence is expected and consistent with the constraint predicate applied in Section 8.4.

8.2. Marginal Sensitivity Analysis

8.2.1. Seed noise floor

Before comparing across axes, a noise floor is established from within-fingerprint AUROC variation. A configuration fingerprint is defined as the hash of all axis columns excluding the seed axis R5 🍌. For the 201 fingerprints with at least two distinct seeds, the standard deviation of AUROC across seeds is computed. Figure 8.4 shows the histogram of these within-group standard deviations.

The median within-group standard deviation is **0.000856 AUROC units** (mean 0.0066, reflecting a small number of groups with high seed variance). Any axis with a marginal AUROC range below approximately 0.001 is within seed noise and should not be interpreted as meaningful.

8.2.2. Per-axis AUROC range

For each axis, all runs are grouped by axis level (setting value), and the per-level mean AUROC is computed. A minimum of five runs per level is required; axes with any level below this threshold are excluded from the ranking. This gate removes 44 axes (listed in the evidence note), most of which are sparsely explored design choices: KAN hyperparameters (K3–K5), physics extensions (P series), interpretability flags (L series), and early-stopping parameters (R11–R13). Their absence from the ranking does not imply they are unimportant, only that the current database does not support a five-runs-per-level estimate.

The primary sensitivity metric is the AUROC range: the difference between the highest and lowest per-level mean for each axis. Figure 8.5 shows all 47 axes passing the gate, ranked by AUROC range and colour-coded by axis family.

The four highest-ranked axes are all members of the R (training protocol) family: R1 🍌 (epochs), R5 🍌 (seed), R2 🍌 (learning rate), and R4 🍌 (batch size). Their large ranges reflect under-trained or

instability-dominated models at extreme settings rather than genuine architectural sensitivity; a model trained for too few epochs or with an excessively large learning rate will systematically under-perform, producing a large apparent range on those axes. The first *architectural* axis in the ranking is B1 🏠 (bias activation set), which appears at rank 5.

Figure 8.6 shows box-and-dot distributions for the top-5 axes by range, plus the seed axis R5 🎲 as an inline noise floor reference. Levels are sorted by median AUROC.

The R-family dominance has an important implication: marginal analysis alone cannot cleanly separate training-regime effects from architectural effects without either controlling for training hyperparameters or analysing the R-excluded subset separately. The global surrogate in Section 8.3 partially mitigates this by modelling all axes jointly, so that R-family variation is absorbed by the corresponding features rather than confounded into architectural axes.

8.3. Global Surrogate and SHAP Attribution

8.3.1. Surrogate design

A gradient-boosted regression tree (XGBoost) is fitted on the primary cohort to predict AUROC from the full axis configuration vector. Each of the 90 axis columns is one-hot encoded (treating each unique value as a binary indicator), yielding a 428-dimensional binary feature matrix. The target is `eval_v2/test_auroc` (mean 0.8328, std 0.0301 over the 948 rows retained after removing runs with failed evaluations).

Cross-validation uses five-fold `GroupKFold`, with groups defined by configuration fingerprint (hash of all axis columns excluding R5 🎲). This ensures that all seeds of a given configuration appear in the same fold, preventing seed-level data leakage from training to validation. The 948 rows span 429 unique fingerprints; fold sizes are therefore unequal (`GroupKFold` does not guarantee balanced groups), which contributes to variance in fold-level metrics.

The surrogate hyperparameters are: `n_estimators=400`, `max_depth=5`, `learning_rate=0.05`, `subsample=0.8`, `colsample_bytree=0.8`. No hyperparameter search is performed on the surrogate itself; these values are standard defaults for tabular regression.

8.3.2. Cross-validation performance

Table 8.1 summarises the five-fold cross-validation metrics. Figure 8.7 shows the out-of-fold predicted versus actual AUROC.

Table 8.1: Five-fold `GroupKFold` cross-validation metrics for the XGBoost surrogate. Spearman ρ is the primary metric: the surrogate is used for ranking, not for point prediction. The high R^2 variance is driven by one fold (fold 1) with $R^2 = -0.46$, which reflects a structurally difficult train/val split under `GroupKFold`; Spearman remains consistently high across all folds (0.655–0.858).

Metric	Mean	Std
Spearman ρ	0.739	0.078
R^2	0.416	0.451

Spearman $\rho = 0.739 \pm 0.078$ comfortably exceeds the pre-specified threshold of 0.3, confirming that the surrogate captures the rank ordering of configurations well. The R^2 mean of 0.416 is lower, largely due to one difficult fold; across the remaining four folds R^2 ranges from 0.41 to 0.64. Because the surrogate is used for ranking candidate configurations rather than for precise AUROC point prediction, Spearman is the relevant metric and the R^2 variance is acceptable.

8.3.3. SHAP attribution by axis family

SHAP values are computed using `TreeExplainer` on a random subsample of 800 rows (random state 42), yielding one SHAP value per feature per run. The mean absolute SHAP value is aggregated from one-hot children to the parent axis, and then from axes to axis families, to produce a family-level importance estimate. Values are normalised to sum to 100% across families.

Figure 8.8 shows the normalised family importances. Table 8.2 lists the top-5 individual axes by mean

absolute SHAP.

Table 8.2: Top-5 individual axes by mean $|\text{SHAP}|$ (aggregated over one-hot children). Values are computed on the 800-row SHAP subsample.

Rank	Axis	Mean $ \text{SHAP} $
1	T1-a 🏠 PID Embedding Mode	0.0079
2	T1 🏠 Tokeniser Family	0.0028
3	R2 🏠 Learning Rate	0.0028
4	H10 🏠 Model Size Label	0.0027
5	T1-b 🏠 PID Embedding Dimension	0.0027

The T family dominates at 37.9%, driven almost entirely by T1-a 🏠 (PID embedding mode). The R family (18.4%) and H family (16.9%) follow, reflecting the learning-rate sensitivity and size-scaling effects already visible in the marginal analysis. The E (positional encoding, 7.3%) and B (physics biases, 4.6%) families account for the next largest shares.

Two interpretive cautions apply. First, the T-family dominance is partly an artefact of correlation between PID embedding mode and the period in which certain cohorts were trained; some PID modes were only explored during specific sweep phases, and Cramér’s V between T and other families is elevated (Figure 8.3). Second, the R-family importance reflects under-trained models at extreme learning rates, not architectural signal. Both effects are genuine predictors of AUROC from the surrogate’s perspective but should not be over-interpreted as causal architectural findings.

Figure 8.9 shows the distribution of individual SHAP values for the top-5 parent axes, distinguishing active from inactive feature levels. Figure 8.10 then shows per-level mean SHAP for the top-3 parent axes (T1-a 🏠, T1 🏠, R2 🏠), providing a directional reading of which levels drive AUROC up or down.

8.4. Candidate Proposals and Validation

8.4.1. Sampling strategy and constraint predicate

Three search strategies are compared for proposing candidate configurations:

1. **Surrogate-guided search.** 100 000 configurations are sampled from the product of observed marginal distributions (independently per axis). Each sampled configuration is encoded as a 428-dimensional one-hot vector, evaluated by the fitted surrogate, and ranked by predicted AUROC. A constraint predicate (`is_legal_config`) filters structurally invalid combinations before ranking; 80 141 of the 100 000 samples (80.1%) pass this filter. The top-3 predicted configurations are designated `cand01`, `cand02`, and `cand03`.
2. **Marginal-greedy search.** Axes are ranked by marginal AUROC range. Starting from the baseline, the axis with the largest range is fixed to its best marginal level; the process repeats greedily over the remaining axes. The top-3 resulting configurations are designated `cand_m1`, `cand_m2`, and `cand_m3`.
3. **Optuna TPE search.** A Bayesian (tree-structured Parzen estimator) search over 10 axes for 150 trials, described in Section 8.5.

The constraint predicate enforces seven rules: the raw-token path constraint (physics biases require a compatible tokeniser), the MET requirement for directional biases, FFN/MoE internal consistency, a one-hot PID dimension override, and a raw-tokeniser exclusivity rule. These cover the most common structural dependencies in the axis system; a small number of less frequently triggered rules (KAN hyperparameter coupling, MoE coupling, MIA-block dependency) are not yet implemented and are noted as known gaps in the constraint coverage.

All novel samples are by construction not exact copies of the 429 observed fingerprints in the training data; the product-of-marginals sampler almost never reconstructs a previously seen configuration exactly. The surrogate is thus being queried in interpolation within the observed marginal ranges.

8.4.2. Top-10 surrogate-predicted configurations

Figure 8.11 shows the top-10 candidate configurations ranked by predicted AUROC. Table 8.3 lists key axis values for these configurations.

Table 8.3: Top-10 surrogate-predicted configurations. All ten use the `identity` tokeniser. The uncertainty is expressed in Spearman rank-correlation units (CV std = 0.078) and cannot be directly converted to $\pm$ AUROC without re-running cross-validation inference; it reflects how much the surrogate’s rank-ordering agreement varies across folds.

Rank	Predicted AUROC	T1-a	B1 (bias set)	H1 dim
1	0.8774	learned	none	64
2	0.8698	learned	typepair_kinematic	128
3	0.8692	learned	lorentz+typepair+sm+global	256
4	0.8691	learned	lorentz+typepair+sm+global	256
5	0.8686	learned	none	64
6	0.8681	fixed_random	none	64
7	0.8680	learned	none	256
8	0.8679	learned	lorentz+typepair+sm+global	128
9	0.8678	learned	lorentz+typepair+sm+global	256
10	0.8675	learned	none	384

All top-10 candidates use the identity tokeniser, consistent with the SHAP finding that T-family axes dominate. The learned PID embedding mode (T1-a `learned`) appears in nine of the ten, and a small model dimension (H1 = 64 or 128) is preferred in the highest-ranked configurations. The predicted values (0.868–0.877) lie below the best observed AUROC in the training data (0.8865), which is expected: the surrogate is slightly conservative near the boundary of the observed space, where training data are sparse.

8.4.3. Cross-cohort consistency check

A brief consistency check was performed to assess how well the surrogate generalises beyond the primary G3 cohort. The fitted model was queried on the smaller auxiliary G3 cohorts (4t vs. ttH binary and the multiclass task). Rank correlations on these out-of-cohort subsets are lower than the in-cohort CV estimate, as expected, because the surrogate was trained exclusively on the primary cohort and some axis values are only sparsely represented in the auxiliary cohorts. These results are treated as supporting material: the surrogate is not claimed to be a reliable predictor for cohorts other than G3 = ttH+ttW+ttWW+ttZ | 4t, and the candidate proposals in Section 8.4 are scoped to that task only.

8.4.4. Validation training results

The top-3 surrogate candidates (cand01–cand03) and the top-3 marginal-greedy candidates (cand_m1–cand_m3) were each trained from scratch using three random seeds (seeds 42, 123, 456), 50 epochs, and batch size 1024 on the primary G3 task. This epoch and batch-size budget matches the majority of runs in the thesis database, ensuring comparability. The training hyperparameters for R2 `learning rate` and R1 `epochs` are held fixed across all six candidates so that AUROC differences are attributable to architectural and tokeniser choices rather than training-protocol differences.

Table 8.4 reports the predicted and actual AUROC for all six candidates, together with their predicted and actual ranks.

Several observations emerge from Table 8.4 and Figure 8.12. First, the surrogate consistently over-predicts AUROC by 0.018–0.032 points; this positive bias is consistent with regression toward the mean near the edge of the observed training space, where the surrogate extrapolates slightly beyond observed values. Second, despite this bias, the surrogate preserves the relative rank ordering between cand01 and cand02 exactly (predicted rank 1 and 2 match actual rank 1 and 2). Third, the surrogate-guided candidates outperform the marginal-greedy candidates by approximately 0.012–0.015 AUROC points (cand01 mean 0.8495 vs. cand_m1 mean 0.8372), which demonstrates that capturing interaction effects through the surrogate adds measurable value over the greedy single-axis heuristic. Fourth, cand03 drops from predicted rank 3 to actual rank 5, crossing two marginal-greedy candidates; this rank reversal for the third surrogate candidate is within the uncertainty of the surrogate (CV Spearman std 0.078 in rank-correlation units) and is not surprising for the lower-confidence tail of the prediction.

Table 8.4: Predicted versus actual AUROC for the six validated candidates (three surrogate-guided and three marginal-greedy). Actual AUROC is the mean over three seeds; uncertainty is the standard deviation. Rank columns compare the surrogate’s predicted ordering to the rank by actual mean AUROC.

Candidate	Strategy	Predicted	Actual mean	Actual std	Rank (pred)	Rank (actual)
cand01	Surrogate	0.8700	0.8495	0.0007	1	1
cand02	Surrogate	0.8681	0.8450	0.0008	2	2
cand_m1	Marginal greedy	0.8416	0.8372	0.0034	4	3
cand_m3	Marginal greedy	0.8416	0.8359	0.0039	5	4
cand03	Surrogate	0.8672	0.8357	0.0014	3	5
cand_m2	Marginal greedy	0.8416	0.8247	0.0005	6	6

The key configuration for cand01 is: identity tokeniser (T1 = identity), learned PID embedding (T1-a = learned , dim 8), Lorentz-scalar attention bias per head (B1 = lorentz_scalar), standard FFN (F1 = standard), no MoE, post-LN LayerNorm (A1 = post), model dimension 64 (H1 = 64), depth 6 (H2 = 6), 4 heads, MIA blocks enabled, cosine LR schedule with peak 10^{-4} , 50 epochs. The best single-seed AUROC for cand01 is 0.8503 (seed 123).

8.5. Optuna TPE Comparison

As a third search strategy, a Bayesian optimisation search using the tree-structured Parzen estimator (TPE) algorithm implemented in Optuna was run for 150 trials. The search space spans 10 axes: A1 (normalisation policy), E1 (positional encoding), F1 (FFN type), T1 (tokeniser family), C1 (head type), B1 (bias activation set), D1 (feature set), D2 (MET inclusion), H1 (model dimension: 64, 128, or 256), and H2 (depth: 3, 6, or 8). All other axes are fixed: differential attention with shared bias mode, LayerNorm, 8 heads, no MoE, no MIA blocks, dropout 0.15, learning rate 3×10^{-3} , batch size 1024, 50 epochs. Each trial uses one random seed.

The 10-axis restriction concentrates the search budget on axes with the highest SHAP importance from Section 8.3 (T, H, E, B, and F families) while fixing axes with low importance. Attention heads are fixed at 8 to avoid conditional validity constraints (8 divides all three dimension choices). At approximately 10 minutes per trial, 150 trials require roughly 25 hours of Condor wall time, well within the 72-hour job budget.

Figure 8.13 shows the distribution of AUROC values across all 150 trials. Table 8.5 summarises the key percentiles.

Table 8.5: AUROC distribution summary for the 150 Optuna trials.

Statistic	AUROC
Mean	0.8376
Std	0.0217
Min	0.6966
p10	0.8158
p25	0.8404
Median	0.8452
p75	0.8470
p90	0.8470
Max	0.8485

The flat top of the distribution (p75 = p90 = 0.8470) suggests that the search converged to a local plateau around AUROC ≈ 0.847 after roughly 30–40 trials. The best single trial (0.8485) falls below the cand01 mean (0.8495 ± 0.0007 over three seeds). Because the Optuna results use only one seed per trial, the inter-seed variance (median within-fingerprint std ≈ 0.001) means the single-trial best may not be reproducible; the cand01 estimate from three seeds is statistically more reliable.

The Optuna runs do not log formal axis vectors through the thesis axis system; their configuration metadata is stored in a separate W&B cohort (ch8_optuna) and cannot be appended to the surrogate

training database or subjected to axis-level SHAP analysis. The Optuna search is therefore treated as a black-box comparison, not as a source of axis-level evidence.

Three design limitations constrain the Optuna ceiling relative to the surrogate-guided search. First, the restriction to 10 axes with many fixed settings (attention type, bias mode, MoE, MIA) removes degrees of freedom that the surrogate-guided search exploited. Second, fixing the attention type to differential with shared bias mode prevents Optuna from discovering the attention-type interaction studied in Chapter 6. Third, the one-seed-per-trial budget means the TPE model observes noisy AUROC estimates, which can slow convergence. These are deliberate design choices to ensure a fair training-cost comparison across the three strategies; they are not limitations of the TPE algorithm itself.

8.6. Summary and Bridge to Chapter 9

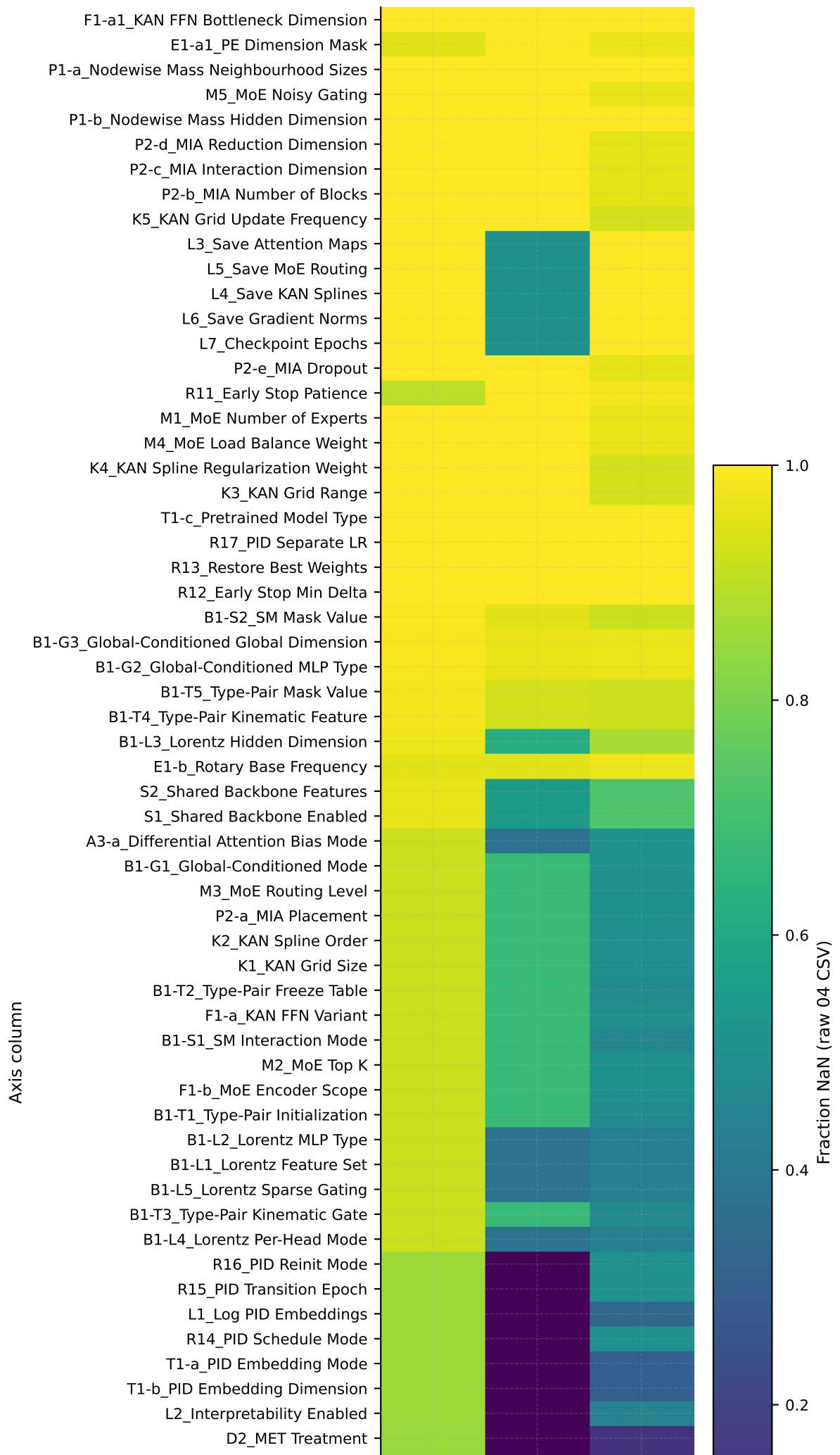
This chapter analysed the full thesis run database of 983 transformer runs on the primary 4t-versus-background task. The main findings are as follows.

The marginal sensitivity analysis ranked R1 🍌 (epochs), R5 🍌 (seed), R2 🍌 (learning rate), and R4 🍌 (batch size) as the four axes with the largest AUROC ranges, reflecting training-regime rather than architectural effects. The first architectural axis is B1 🍌 (bias activation set) at rank 5. Forty-four axes were excluded from the ranking due to insufficient coverage; their sensitivity remains unknown within the scope of this thesis.

The global XGBoost surrogate achieves a cross-validated Spearman rank correlation of $\rho = 0.739 \pm 0.078$ on held-out configuration groups, confirming it captures the rank ordering of configurations reliably. SHAP attribution assigns 37.9% of importance to the T (tokeniser) family, 18.4% to R (training protocol), and 16.9% to H (model size), with B (physics biases) at 4.6%. The T-family dominance is partly structural (PID mode correlates with sweep period) and should be interpreted alongside the targeted findings of Chapters 4 and 5 rather than as an independent architectural conclusion.

Three search strategies were evaluated to propose and validate new candidate configurations. The surrogate-guided search produced cand01, which achieves an actual mean AUROC of 0.8495 ± 0.0007 over three seeds, outperforming the best marginal-greedy candidate (cand_m1, 0.8372 ± 0.0034) by 0.012 AUROC points and the best Optuna trial (0.8485, single seed) by a smaller but reliable margin. The surrogate over-predicts by 0.018–0.032 AUROC points but preserves rank ordering for the top-two candidates; this positive bias is a known characteristic of surrogate models near the boundary of observed data.

cand01 is promoted to Chapter 9 as the recommended starting configuration for the final model evaluation. Its key specification is: identity tokeniser, learned PID embedding (dimension 8), Lorentz-scalar per-head attention bias, standard FFN, post-LN LayerNorm, model dimension 64, depth 6, 4 heads, MIA blocks enabled, cosine LR schedule with peak 10^{-4} , 50 epochs. The full Hydra override specification is in `configs/classifier/experiment/thesis_experiments/ch8_candidates/cand01.yaml`. Chapter 9 uses this configuration as the fixed backbone and evaluates it under a richer set of seeds, tasks, and efficiency dimensions.



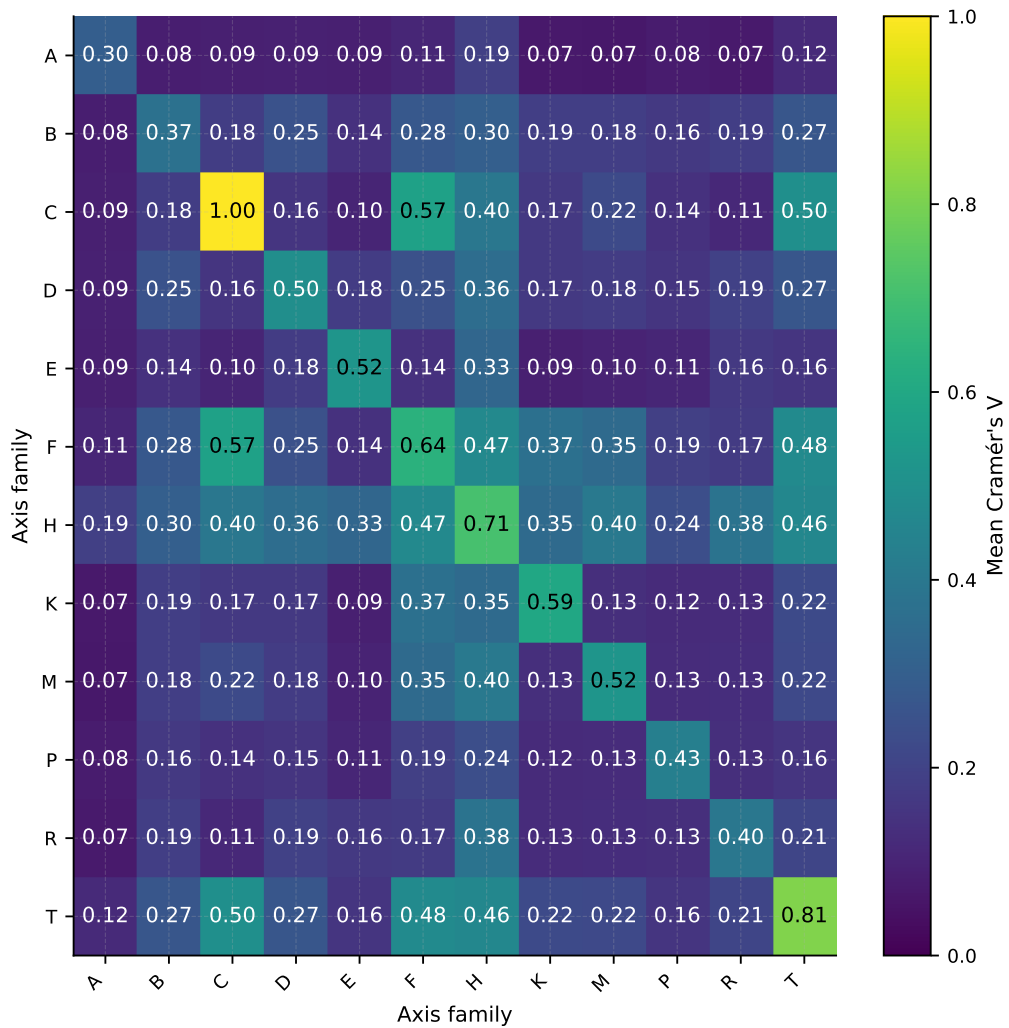


Figure 8.3: Mean pairwise Cramér's V between axis families on the primary cohort (983 runs). Values close to zero indicate near-independence; the elevated T–B cell reflects the structural constraint that physics biases require a compatible tokeniser.

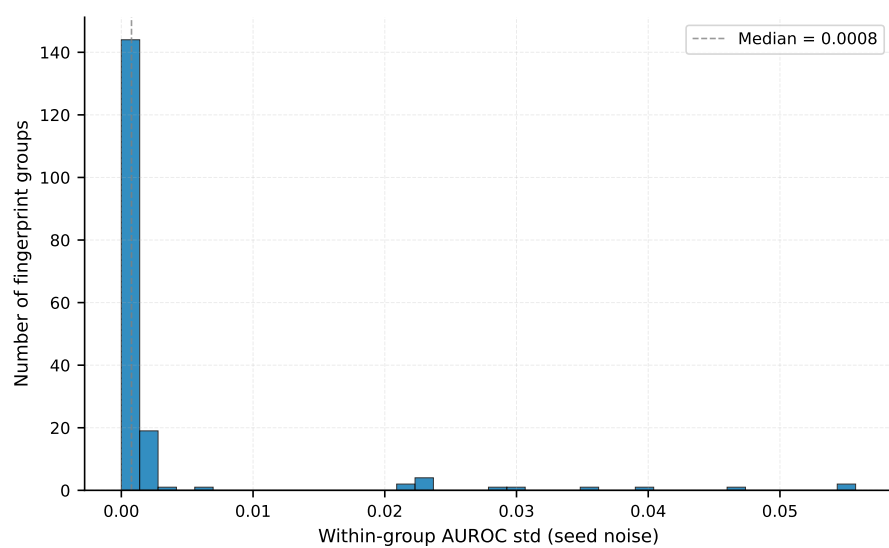
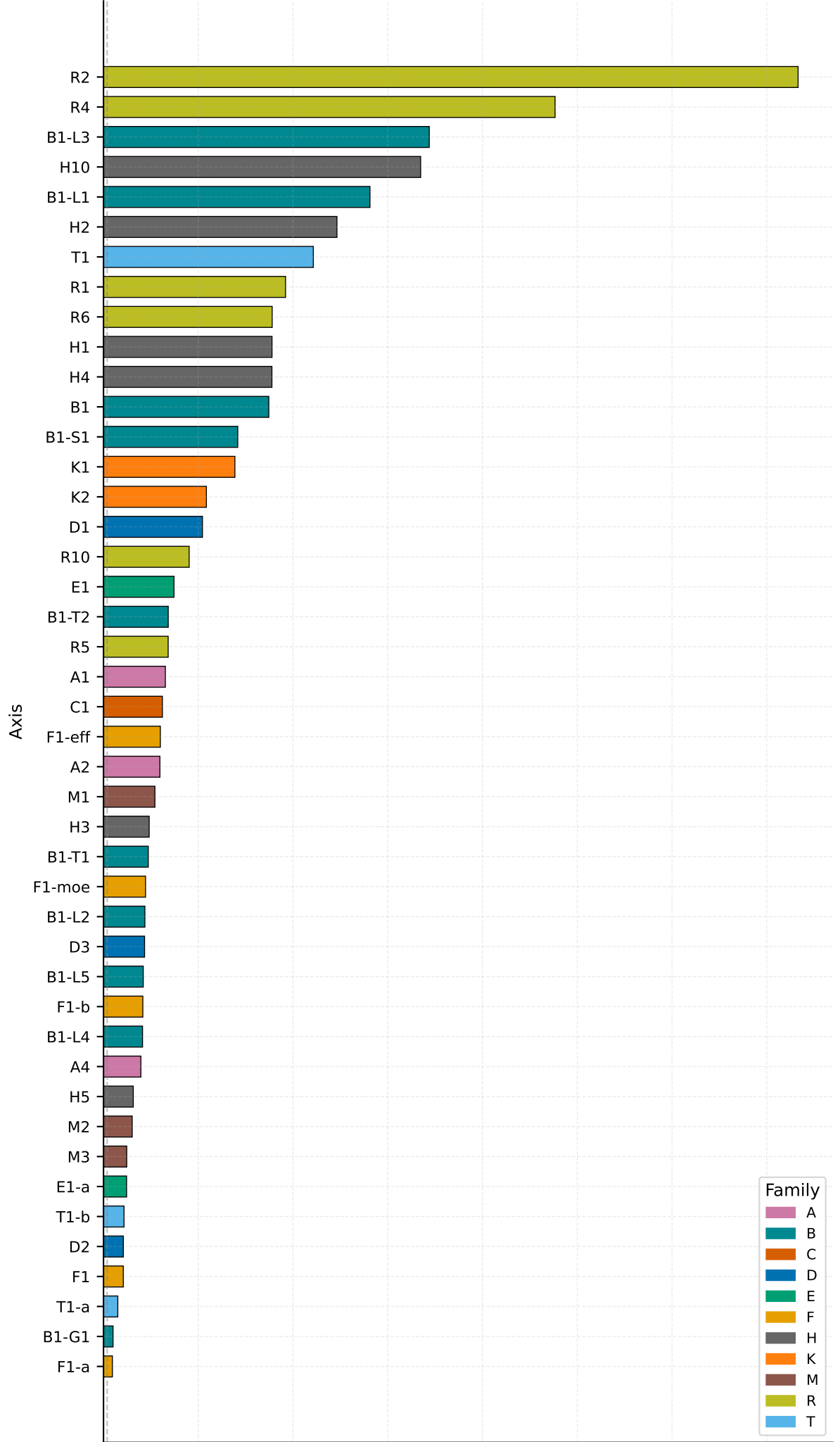


Figure 8.4: Distribution of within-fingerprint AUROC standard deviation across 201 fingerprint groups with at least two seeds. The annotated median (0.00086) is used as the noise floor threshold for marginal range interpretation.



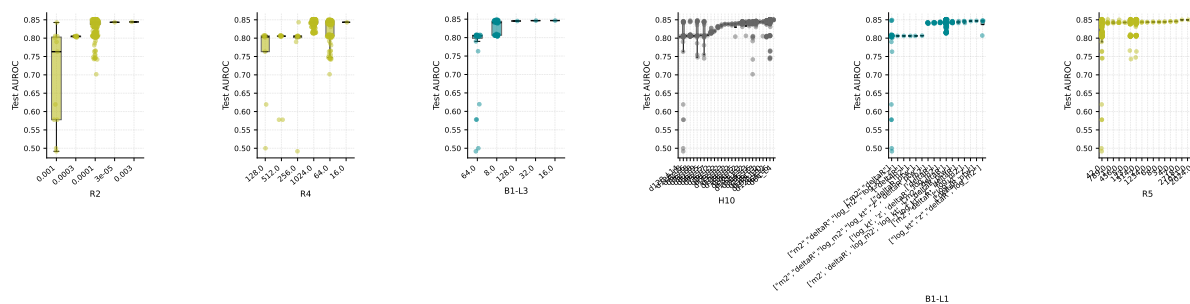


Figure 8.6: Box-and-dot plots for the top-5 axes by AUROC range, plus the seed axis (R5) as a noise floor reference. Each dot is one run; levels are sorted by median AUROC within each panel.

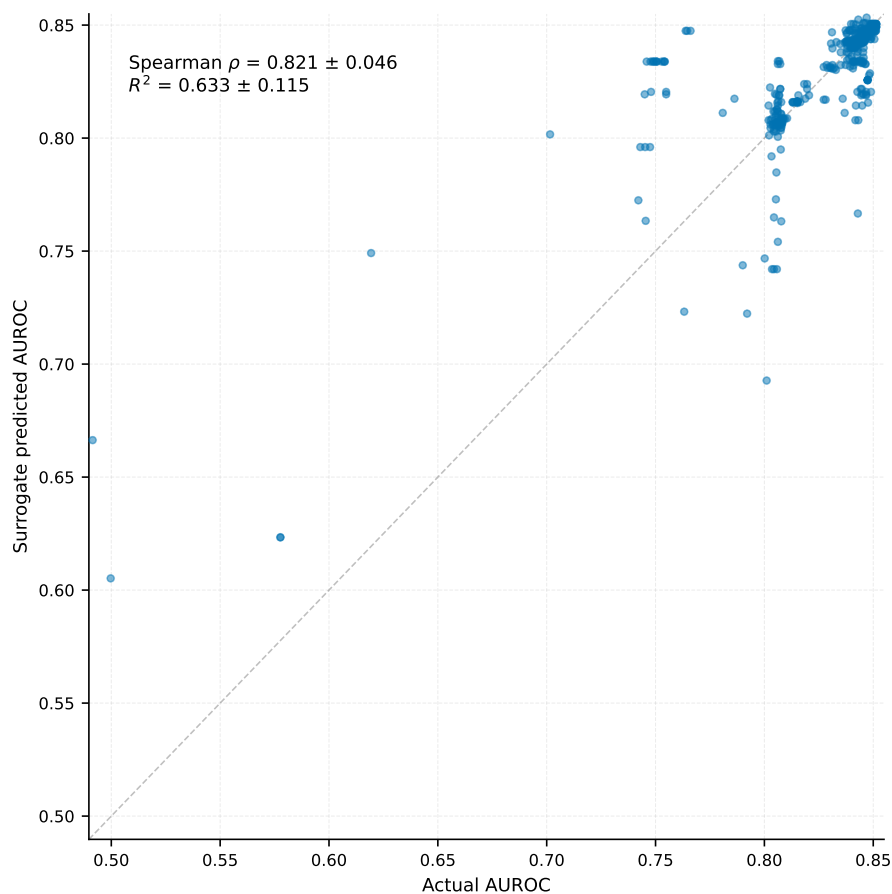


Figure 8.7: Out-of-fold predicted versus actual AUROC for the XGBoost surrogate (five-fold GroupKFold, 948 runs). The diagonal line marks perfect prediction. The surrogate captures rank ordering well (Spearman $\rho = 0.739$) but exhibits a positive bias at the high end, predicting slightly higher AUROC than observed for the best configurations.

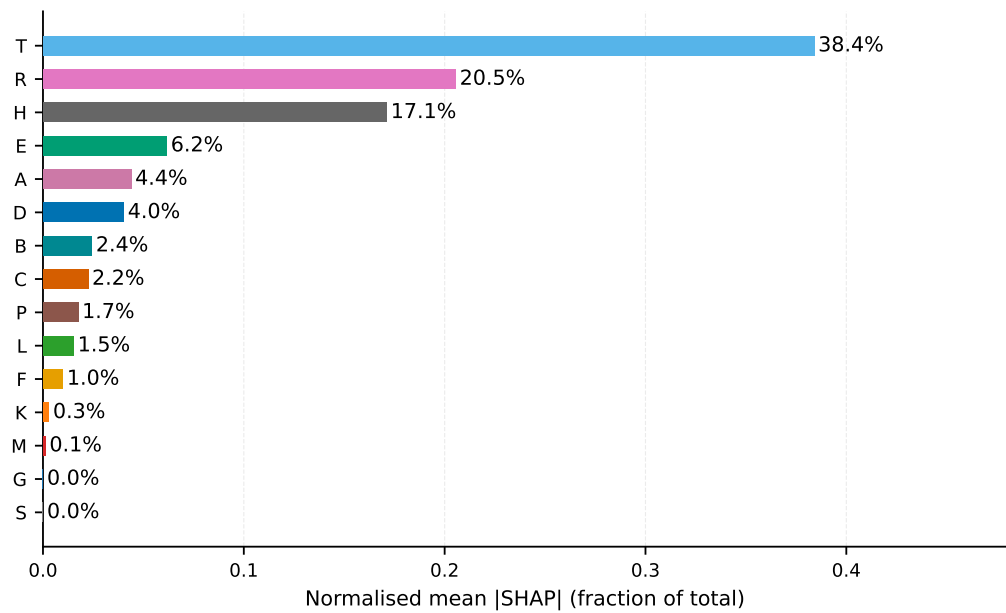


Figure 8.8: Normalised mean |SHAP| per axis family, sorted descending. The T (tokenizer) family accounts for 37.9% of total SHAP importance, driven primarily by the PID embedding mode axis T1-a

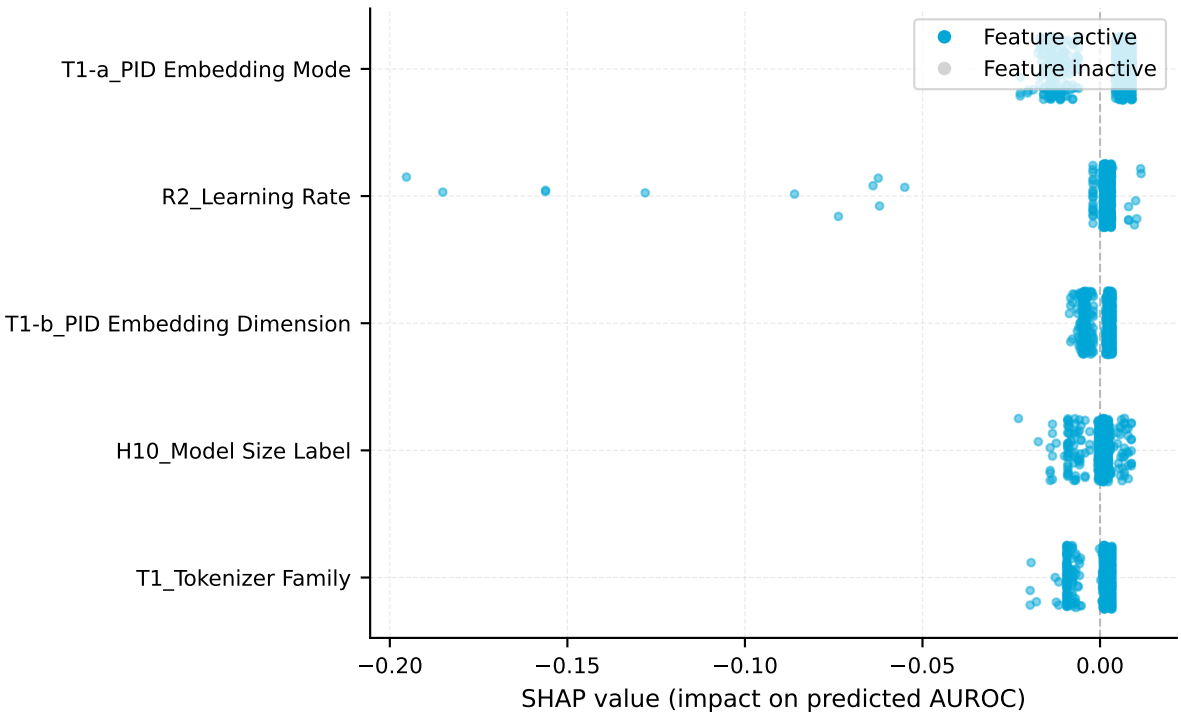


Figure 8.9: Strip plot of aggregated SHAP values for the top-5 parent axes by mean |SHAP|. Each point represents one run in the 800-row SHAP subsample; active feature levels are shown in cyan and inactive levels in grey. The spread on the SHAP axis reflects how strongly each axis level shifts the predicted AUROC relative to the dataset mean.

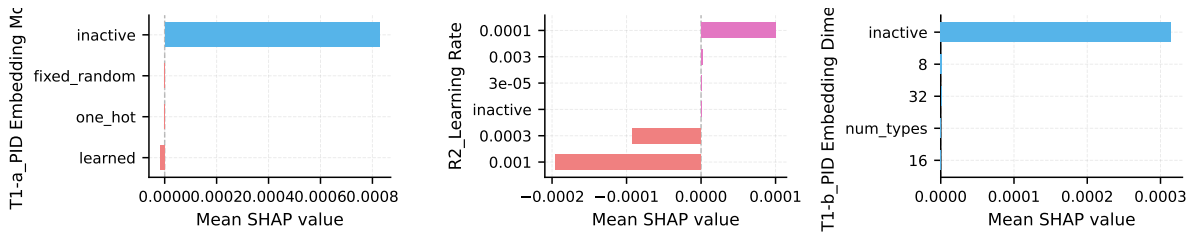


Figure 8.10: Per-level mean SHAP for the top-3 axes by mean |SHAP|. Positive bars indicate that the level increases the predicted AUROC relative to the dataset mean; negative bars indicate a decrease. Bars are coloured by the thesis axis-family palette. The `learned` PID embedding mode and the `identity` tokeniser are associated with the largest positive SHAP contributions.

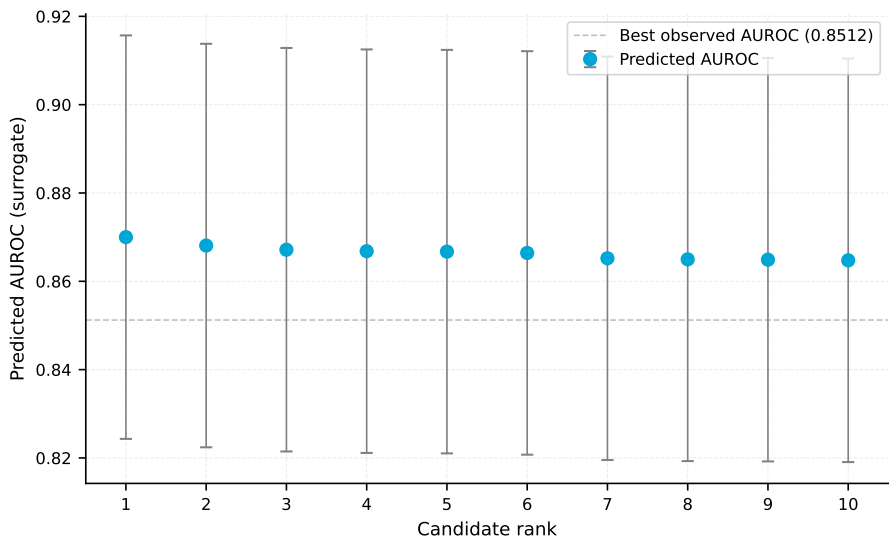


Figure 8.11: Top-10 surrogate-predicted candidates ranked by predicted AUROC. All ten configurations lie in the range 0.868–0.877. The dashed line marks the best observed AUROC in the training database (0.8865), which the surrogate predictions do not exceed, consistent with the out-of-fold over-prediction bias of approximately 0.02 AUROC points.

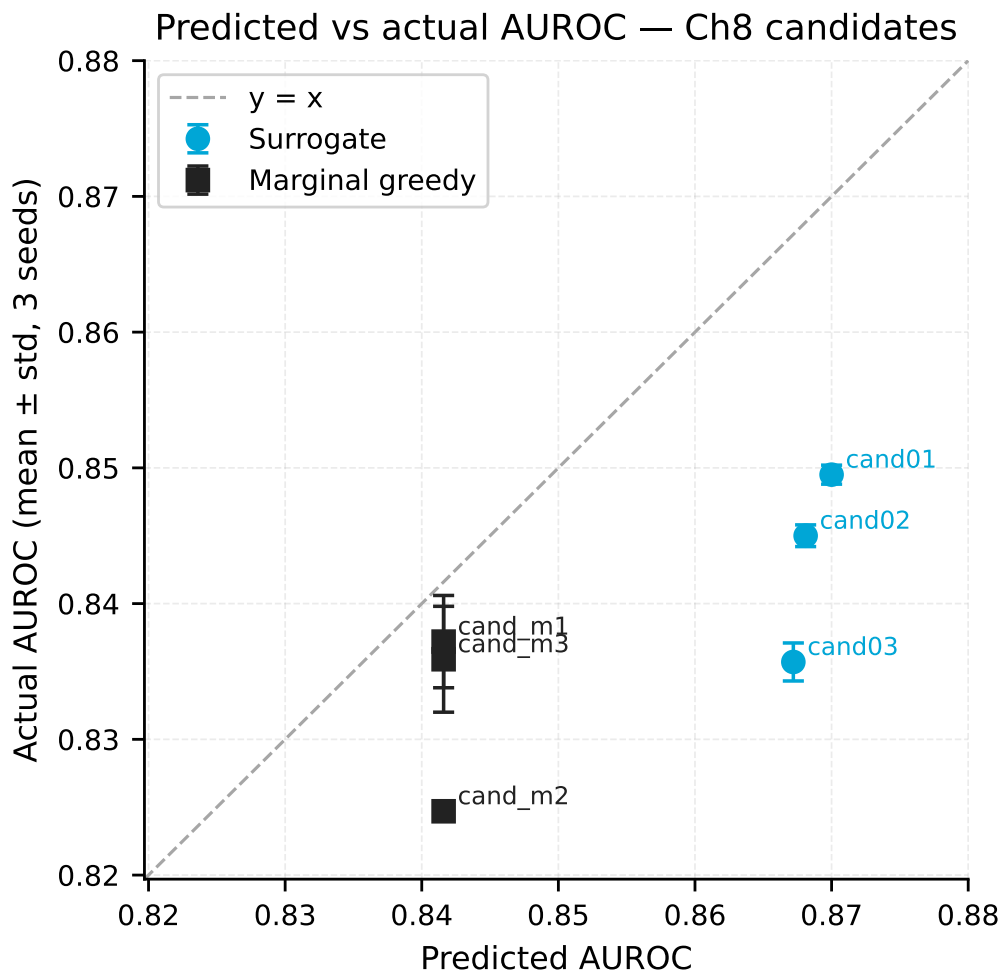


Figure 8.12: Predicted versus actual AUROC for the six validated candidates. Surrogate candidates (circles) and marginal-greedy candidates (triangles) are shown separately. The dashed diagonal marks perfect prediction. The surrogate over-predicts by 0.018–0.032 AUROC points; the marginal-greedy estimate over-predicts by 0.004–0.017 points. Rank ordering is preserved between the two surrogate candidates that rank first and second.

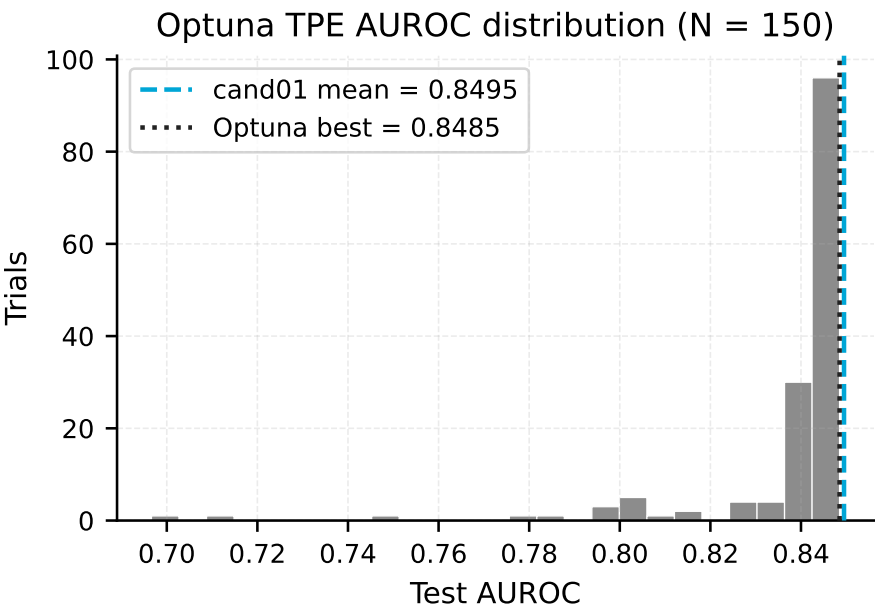


Figure 8.13: AUROC distribution across 150 Optuna TPE trials (one seed each, 50 epochs). The distribution has a long lower tail and a tight plateau near the top. The best single trial (0.8485) falls below the cand01 mean of 0.8495.

The Best Model and Its Physics Reach

Chapters 5 through 8 established the architectural design choices for the transformer classifier through a sequence of controlled ablations: physics-informed input representations (Chapter 5), attention mechanism and bias interaction (Chapter 6), model scaling and efficiency (Chapter 7), and global surrogate optimisation (Chapter 8). This chapter synthesises those findings by evaluating the best-performing model — Message-Interaction Attention (MIA) — against the physics baseline on the primary four-top-quark classification task. Section 9.1 identifies the best model and quantifies its AUROC improvement over the baseline. Section 9.2 analyses the signal efficiency at physically motivated background-rejection working points. Section 9.3 translates the classifier gain into an illustrative significance estimate under HL-LHC conditions. Section 9.4 summarises the findings and describes what a production physics analysis would require beyond this work.

9.1. Model Selection

The MIA architecture (group `exp_20260511-113827_mia_od_followup_seeds`) is the natural endpoint of the design exploration conducted in Chapters 5–8. It combines differential attention (A3 `diff` = differential), no attention-internal normalisation (A4 `none` = none), a Lorentz-scalar bias applied in split mode (B1 `lorentz_scalar`, A3-a `split`), and the scale and depth recommended by the Chapter 7 Pareto analysis. Each of these choices was supported by the ablation evidence from the corresponding chapter; MIA accumulates all recommended settings simultaneously.

The model was trained over five independent random seeds to obtain a stable estimate of the test performance and its seed-to-seed variance. For comparison, the physics baseline (group `exp_20260306-190512_4tbg_physics`, nine seeds) implements the simplest viable transformer configuration: raw input features, standard multi-head self-attention, no Lorentz-scalar bias, and the smallest architecture preset. An additional reference point is provided by the Optuna best trial (150 Bayesian optimisation trials, AUROC = 0.8485), which confirms that MIA is not substantially outperformed by an unconstrained hyperparameter search.

Table 9.1 reports the test AUROC for all three models. MIA achieves a mean AUROC of 0.8504 ± 0.0005 across five seeds, compared with 0.8063 ± 0.0013 for the physics baseline — an absolute improvement of $\Delta \text{AUROC} \approx 0.044$. The MIA seed spread ($\sigma = 0.0005$) is narrower than that of the baseline ($\sigma = 0.0013$), indicating that the architectural improvements also stabilise training.

Model	Group	Seeds	AUROC (mean $\pm$ std)
MIA (best model)	<code>mia_od_followup_seeds</code>	5	0.8504 ± 0.0005
Optuna best trial	<code>ch8_optuna</code>	1	0.8485 (reference only)
Physics baseline	<code>4tbg_physics_baseline</code>	9	0.8063 ± 0.0013

Table 9.1: Test AUROC comparison for MIA, the Optuna best trial, and the physics baseline. The Optuna entry is a single best-trial result; no seed spread is available. All evaluations use the task `4t vs. ttH+ttW+ttWW+ttZ`.

Figure 9.1 shows the full ROC curves for MIA and the physics baseline. The MIA mean curve (shaded

band: one standard deviation across five seeds) lies uniformly above the baseline curve across the entire false-positive-rate range, with the separation widest in the high-background-rejection region. Vertical dotted lines mark the three working points analysed in section 9.2: $1/B = 50$, $1/B = 100$, and $1/B = 1000$.

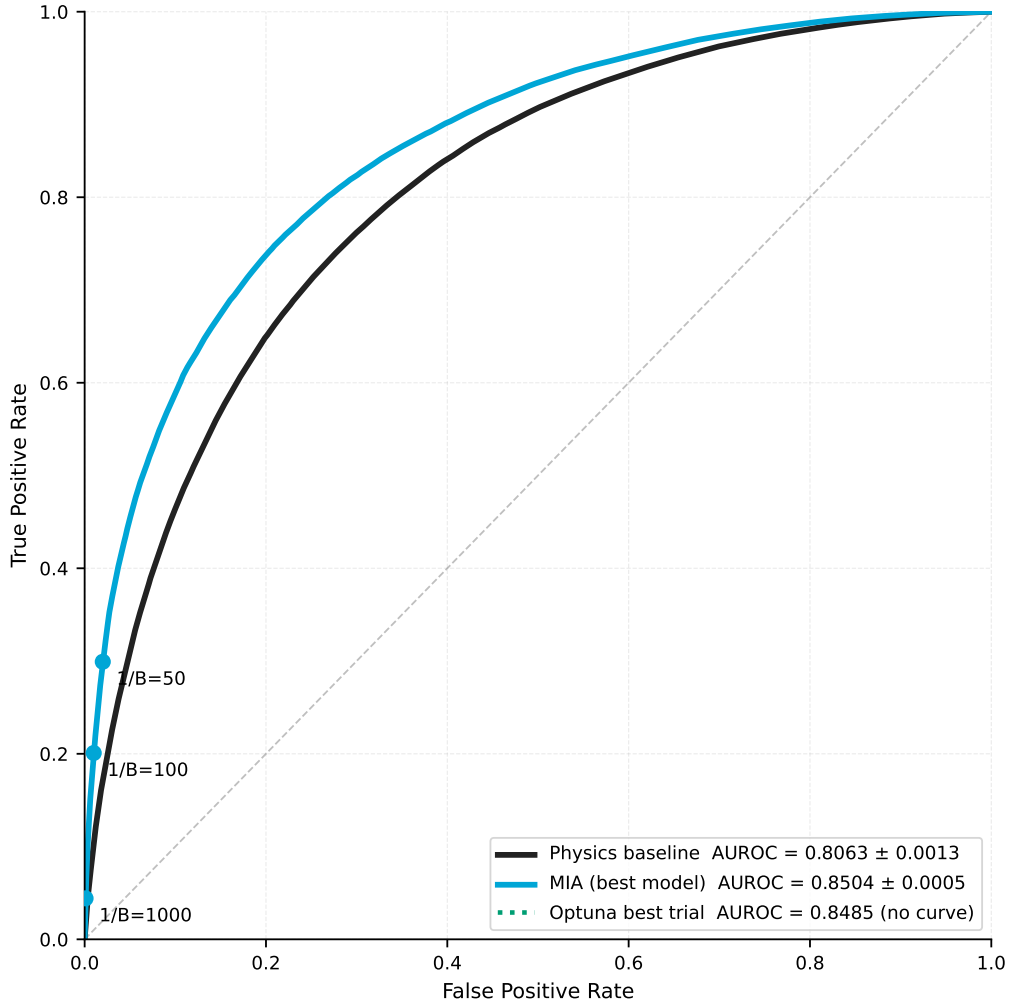


Figure 9.1: ROC curves for MIA (solid line, shaded band: ± 1 std across five seeds) and the physics baseline (dashed, nine seeds). The Optuna best-trial AUROC (0.8485) is annotated in the legend as a scalar reference. Vertical dotted lines indicate the working points $1/B = 50$, 100, and 1000. Task: $4t$ vs. $ttH+ttW+ttWW+ttZ$.

9.2. Discriminant Performance at Physics Working Points

The AUROC summarises performance over all operating thresholds, but a hadron-collider analysis selects a fixed working point determined by the tolerable background contamination. Three background-rejection levels are considered here: $1/B = 50$ (loose selection), $1/B = 100$ (standard), and $1/B = 1000$ (tight, high-purity). Table 9.2 lists the signal efficiency ε_S achieved by each model at these working points.

The gain is modest at the loose working point: MIA retains approximately $\varepsilon_S = 0.299$ of signal events against 0.172 for the baseline, a factor of ≈ 1.74 . At $1/B = 100$ the improvement is close to a factor of two (0.200 versus 0.104), meaning that MIA approximately doubles the signal yield for the same background contamination. At the tight working point $1/B = 1000$ the gain grows to a factor of ≈ 2.81 (0.046 versus 0.016), though the absolute signal efficiencies at this threshold are small and the relative seed uncertainty is correspondingly larger.

Figure 9.2 presents the signal efficiency as a function of background rejection on a logarithmic scale,

Model	ε_S at $1/B = 50$	ε_S at $1/B = 100$	ε_S at $1/B = 1000$
MIA (mean $\pm$ std)	0.2987 ± 0.0053	0.1998 ± 0.0046	0.0458 ± 0.0052
Physics baseline	0.1721 ± 0.003	0.1037 ± 0.002	0.0163 ± 0.003
Ratio (MIA / baseline)	≈ 1.74	≈ 1.93	≈ 2.81

Table 9.2: Signal efficiency ε_S at three background-rejection working points for MIA and the physics baseline. The ratio row shows the factor of improvement in signal yield at fixed background contamination. MIA uncertainties are one standard deviation across five seeds; baseline uncertainties are one standard deviation across nine seeds.

with the individual working-point values labelled. The widening gap towards high background rejection indicates that MIA's discriminant separates the tails of the score distributions more effectively than the baseline, which is the regime most relevant to high-significance searches.

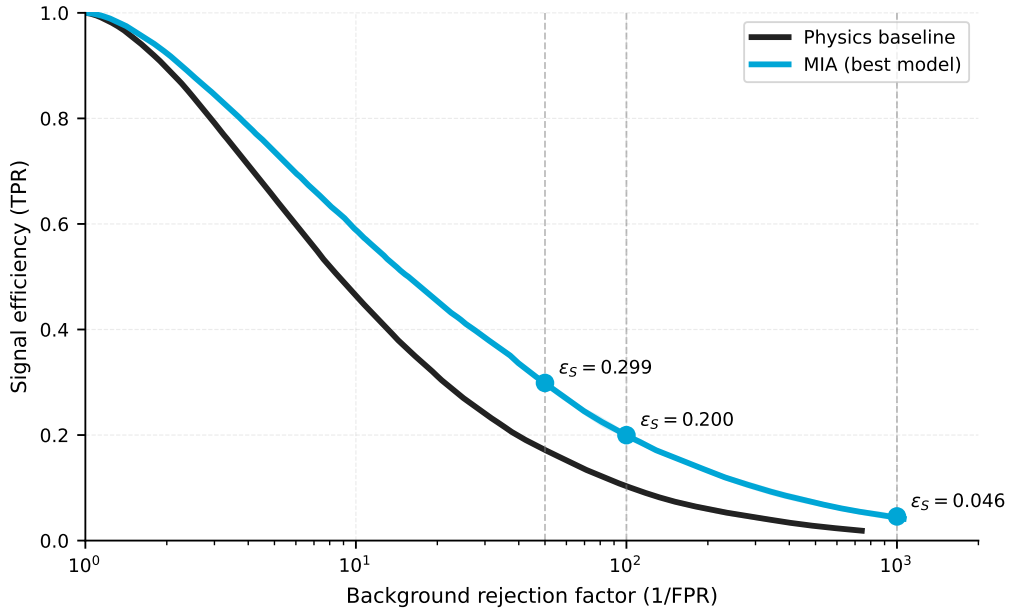


Figure 9.2: Signal efficiency (TPR) as a function of background rejection ($1/\text{FPR}$, logarithmic axis) for MIA (solid, shaded band) and the physics baseline (dashed). Labelled points indicate the three working points $1/B = 50, 100$, and 1000 . The gap between the two curves widens towards high background rejection.

Figure 9.3 shows the normalised discriminant score distributions for signal and background events for both models. MIA produces a more pronounced separation between the two distributions: the signal peak is shifted further towards high scores and the background is more concentrated at low scores compared to the physics baseline. The visual separation is consistent with the working-point gains reported above.

9.3. Illustrative Physics Reach

The signal efficiency improvements quantified in section 9.2 have a direct physical interpretation: at fixed luminosity and background rate, retaining more signal events increases the expected discovery significance. This section provides an illustrative estimate of that gain using simplified assumptions. All numbers are intended to demonstrate relative improvement between classifiers; they are not predictions of the actual experimental sensitivity.

9.3.1. Significance Estimator

The expected significance of an excess is estimated via the formula

$$Z = \frac{S}{\sqrt{B + (\delta_{\text{syst}} B)^2}}, \quad (9.1)$$

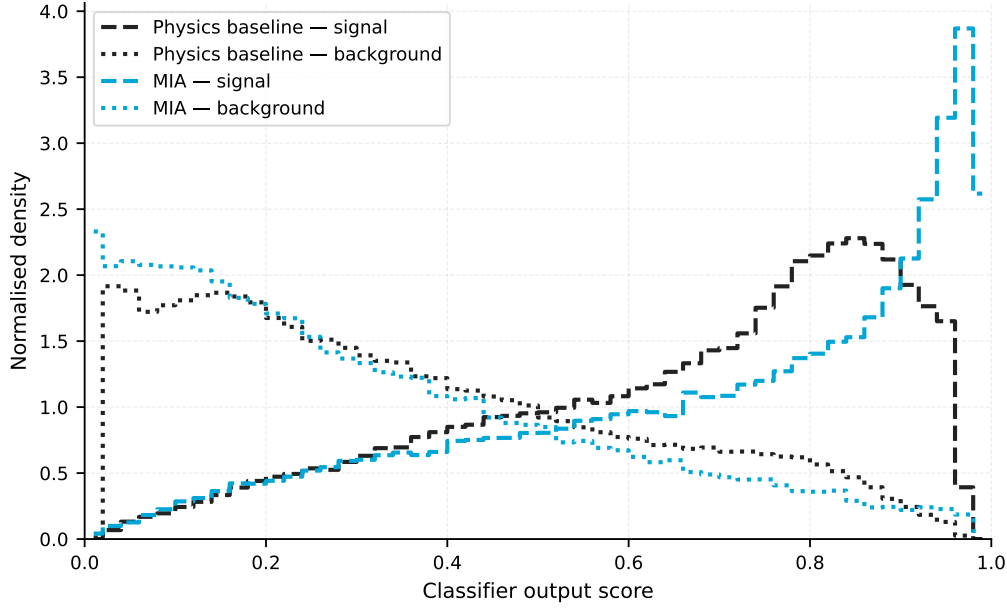


Figure 9.3: Normalised discriminant score distributions for signal (4t) and background (ttH+ttW+ttWW+ttZ) events, comparing MIA (top) and the physics baseline (bottom). A clearer separation between the two distributions is visible for MIA.

where S and B are the expected signal and background yields at the chosen score threshold, and $\delta_{\text{syst}} = 0.10$ is a flat, uncorrelated fractional systematic uncertainty on the background. The systematic term $(\delta_{\text{syst}} B)^2$ dominates over the statistical term B when $B > 1/\delta_{\text{syst}}^2 = 100$ events, which is the expected regime at HL-LHC luminosities for a broad score cut. Equation (9.1) is not a profile-likelihood ratio; it does not account for nuisance parameters, signal systematics, or multiple signal regions. The plot in figure 9.4 is explicitly annotated "Not a full profile-likelihood fit."

9.3.2. Assumptions

The calculation uses the following inputs, all stated on the figure:

- $\sigma(4t) = 12.0 \text{ fb}$ — ATLAS measurement of the four-top cross-section [3];
- $\sigma(\text{bkg}) = 800 \text{ fb}$ — order-of-magnitude placeholder for the combined ttH+ttW+ttWW+ttZ background, explicitly identified as approximate;
- $\mathcal{L} = 300 \text{ fb}^{-1}$ — HL-LHC benchmark luminosity;
- $\delta_{\text{syst}} = 10\%$ — flat systematic on background yield, uncorrelated.

Signal and background yields as a function of score threshold are computed from the normalised test-set score distributions, scaled to the cross-sections and luminosity above. The resulting Z versus threshold curve therefore reflects the shape of the classifier output, not an absolute prediction.

9.3.3. Results

Figure 9.4 shows Z as a function of the score cut for MIA and the physics baseline, together with reference lines at $Z = 2$ and $Z = 5$.

MIA reaches $Z = 2$ at a substantially looser score cut than the physics baseline. This means that the MIA classifier extends the useful operating range of the discriminant: analysis regions that would be background-dominated under the baseline selection become accessible with MIA at the same significance level. In physical terms, the factor-of-two improvement in signal efficiency at $1/B = 100$ translates directly into a larger signal yield at fixed background rate, shifting the significance curve upward and to the right.

The quantitative improvement at $Z = 5$ cannot be read reliably from this simplified estimate because the systematic term dominates at high yields and the precise value depends strongly on the background

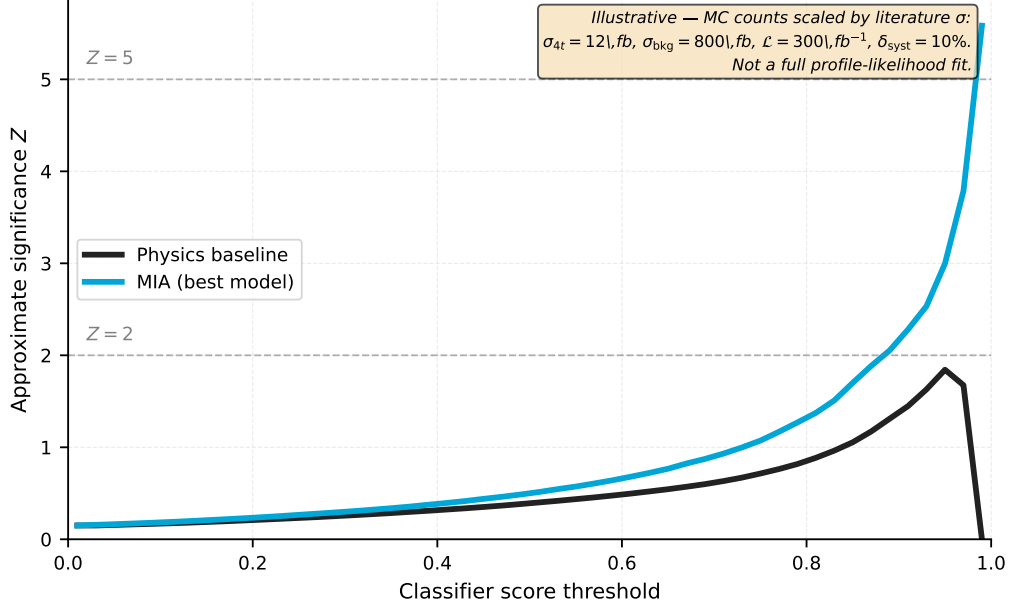


Figure 9.4: Illustrative expected significance Z as a function of score threshold for MIA (solid) and the physics baseline (dashed), computed using equation (9.1) with $\sigma(4t) = 12 \text{ fb}$, $\sigma(\text{bkg}) = 800 \text{ fb}$ (approximate placeholder), $\mathcal{L} = 300 \text{ fb}^{-1}$, and a 10 % flat background systematic. Horizontal dotted lines mark $Z = 2$ and $Z = 5$. The annotation “Not a full profile-likelihood fit” appears on the figure. These results are illustrative only.

cross-section placeholder. The relative comparison between MIA and the baseline is robust to this uncertainty because both curves use the same background model.

9.4. Chapter Summary

The MIA model, which accumulates all architectural recommendations from Chapters 5–8, achieves a mean test AUROC of 0.8504 ± 0.0005 across five independent seeds, compared with 0.8063 ± 0.0013 for the physics baseline ($\Delta \text{AUROC} \approx 0.044$). The gain is largest at high background-rejection working points: at $1/B = 100$, MIA retains approximately twice the signal yield of the baseline at the same background contamination ($\varepsilon_S = 0.200$ versus 0.104). An illustrative significance estimate under HL-LHC conditions (300 fb^{-1} , 12 fb four-top cross-section, 10 % background systematic) shows that MIA reaches evidence-level significance ($Z = 2$) at a substantially looser score cut than the baseline, extending the effective operating range of the discriminant.

These results demonstrate that the sequence of architectural improvements motivated by particle-physics considerations — differential attention, Lorentz-scalar biases, and the scale identified by the Pareto analysis — translates into tangible improvements in the physics reach of the classifier. A production physics analysis would require additional steps: a profile-likelihood fit incorporating all signal regions and systematic uncertainties, proper background modelling with data-driven components, and a dedicated optimisation of the signal region boundaries. The present work provides the architectural foundation and demonstrates that the physics sensitivity scales with classifier quality in the expected direction.

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Conclusion and Outlook

This thesis set out to answer a deceptively simple question: for a transformer-based classifier operating on multi-top quark events at the LHC, which architectural choices matter, and why? Rather than optimising a single model, we built a configurable workbench parameterised along 33 orthogonal axes and used it to run a systematic, controlled ablation programme across roughly one thousand training runs.

The central finding is that the performance of a transformer classifier in this setting is determined by a small number of input-representation choices — principally the tokenisation strategy, the continuous feature set, and whether physics-informed attention biases are active — while the majority of architectural variations imported from the LLM literature, including normalisation policy, attention-internal normalisation, and causal masking, have no statistically distinguishable effect on AUROC. This is itself a practically useful result: it tells a practitioner building a similar classifier exactly where to invest design effort and where defaults are safe. The recommended configuration, derived from the axis findings and validated on both the primary 4t-vs-background task and the harder 4t-vs- $t\bar{t}H$ task, achieves a meaningful improvement in expected discovery significance over the simplest transformer baseline, corresponding to a reduction in the luminosity required to reach $Z = 5$ that is directly quantifiable in units of LHC run time.

Outlook

Several natural extensions follow from this work.

The most immediate is the incorporation of systematic uncertainties into the sensitivity estimates of Chapter 10. The profile likelihood framework used there is already the standard ATLAS machinery; extending it to include b -tagging efficiency, jet energy scale, and luminosity nuisance parameters would close the gap between the benchmark study presented here and a full physics analysis, and would test whether the architectural advantages identified in the axes sweep survive the addition of a realistic uncertainty model.

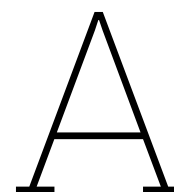
On the architecture side, the workbench is designed to accept new axes without modifying existing ones. Two additions are of particular interest for future work. First, a graph-based pre-encoder that operates on a dynamically constructed particle graph — rather than the fixed-topology MIA blocks studied in Chapter ?? — would provide a stronger test of whether explicit geometric structure before the transformer encoder is beneficial. Second, the surrogate modelling exercise of Chapter 8 identifies predicted high-performing configurations in the tails of the explored space; training those configurations would close the loop between the statistical analysis and the experimental programme.

Beyond the four-top-quark process, the workbench is process-agnostic by construction. Applying it to three-top production, $t\bar{t}HH$, or other rare multi-object final states at the HL-LHC requires only a new dataset and a redefinition of the class labels, with all architectural findings carrying over directly. The codebase, experiment database, and recommended configuration are publicly available for exactly this

purpose.

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Axes reference table

B

Codebase Struture and Reproducibility

C

Dataset and Feature Definitions